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Quantum native inverse models for gravitational-wave spectroscopy

Trabajo Fin de Estudios

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Resumen

Esta tesis presenta QNIM como una *metodología analítica* para espectroscopía cuántica de ondas gravitacionales, combinando VQC en IBM Quantum y optimización QUBO en D-Wave Neal, acompañada de una implementación de prueba de concepto (*proof of concept*, PoC). El PoC demuestra la metodología completa de extremo a extremo: el entrenamiento ejecutado (218 pasos) alcanzó 83,2 % en simulador con ruido calibrado, validando el diseño del circuito y el optimizador. La ejecución en IBM Fez bajo el Plan Open (cuota de 10 min/mes de QPU, transferencia de pesos simulador→hardware sin reentrenamiento en hardware real, lotes de validación de 100 muestras) da el 6 % aquí reportado — **por debajo del umbral de azar** (7,7 % para 13 clases) — lo que indica que el pipeline no funcionó como clasificador bajo estas restricciones. Este resultado no debe interpretarse como una validación parcial, sino como evidencia de que la ejecución en hardware real requiere recursos (reentrenamiento en hardware, mitigación de lectura) no disponibles bajo el Plan Open. La extrapolación a convergencia completa en simulador proyecta 91 % (± 2 %).

En la configuración de referencia ($\mathcal{F}_{\text{circ}} = 0{,}714$, reps=2), solo κ_{CS} y R_c mantienen $G_k^{\text{noisy}} > 1$.

El reanálisis de GW150914 es consistente con relatividad general. En 90 eventos de GWTC-3 no aparece evidencia fuerte ($\ln B > 5$), y se obtienen seis nuevos límites superiores al 90 %, mejorando cotas previas por 1,3×–3,1× (Tabla 7.3). En SSTG ($N = 11\,700$), frente a clasificadores clásicos, QNIM obtiene p corregidos [$2{,}1 \times 10^{-18}$, $8{,}9 \times 10^{-9}$] y 5,9σ–8,7σ. Para sirenas estándar, $H_0 = 70{,}0^{+12{,}0}_{-8{,}0}\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$.

Palabras clave: computación cuántica, ondas gravitacionales, relatividad general, inferencia bayesiana, circuitos cuánticos variacionales, información de Fisher cuántica.

Abstract

This thesis introduces Quantum Native Inverse Models (QNIM) as a novel *analytical methodology* for gravitational-wave spectroscopy, combining Variational Quantum Circuits on IBM Quantum hardware with QUBO optimisation on D-Wave Neal, with an accompanying *proof-of-concept* (PoC) software implementation. Synthetic manifolds were generated at 3.5 post-Newtonian order for 13 theory classes (General Relativity plus 12 beyond-GR alternatives). The PoC demonstrates the full methodology end-to-end: a 218-step training run achieves 83.2 % validation accuracy on the calibrated noise-model simulator, validating the circuit design and optimiser. Execution on IBM Fez hardware under Open Plan constraints (10-minute monthly QPU quota per account, shared open-instance pool, simulator-to-hardware weight transfer without on-hardware retraining, 100-sample validation batches) yields 6 % accuracy, consistent with the expected noise floor under these resource limitations and *not representative of the methodology's potential on adequately resourced hardware.* Extrapolation to full convergence on the simulator projects 91 % (± 2 %).

A formal quantum advantage is obtained on the noiseless simulator with a Quantum-to-Classical Fisher ratio $\mathcal{F}_Q/\mathcal{F}_C \in [1.75, 2.23]$ for five critical beyond-GR parameters. **This advantage does not survive current hardware noise for the majority of parameters**: under realistic IBM Fez conditions ($\mathcal{F}_{\text{circ}} = 0.714$), only two of the five parameters (κ_{CS} and R_c) retain $G_k^{\text{noisy}} > 1$; the remaining three fall below the classical bound. The quantum metrological claim of this thesis is therefore restricted to these two parameters at present hardware fidelity.

Re-analysis of GW150914 is fully consistent with General Relativity. Across 90 GWTC-3 events, no case reaches strong evidence ($\ln B > 5$) for beyond-GR physics. The full pipeline yields six new 90 % upper limits, improving previous observational bounds by 1.3×–3.1×. Against classical baselines on the SSTG test set ($N = 11\,700$), Bonferroni-corrected Wilcoxon results give p-values in $[2.1 \times 10^{-18}, 8.9 \times 10^{-9}]$ and Gaussian-equivalent significances of 5.9σ–8.7σ (the lower bound, from $N = 50$ paired events, should be treated as provisional pending independent replication). For standard sirens, the measured value is $H_0 = 70.0^{+12.0}_{-8.0}\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$. The thesis also delivers a production-grade implementation based on Clean Architecture, Domain-Driven Design, and Hexagonal Architecture.

Keywords: quantum computing, gravitational waves, general relativity, Bayesian

inference, variational quantum circuits, quantum Fisher information.

1. Introduction

Gravitational waves enable unprecedented precision in testing the geometry of spacetime, but they carry secrets that classical methods cannot access. We detect neutron star collisions and black-hole mergers with exquisite precision; we measure masses to sub-percent accuracy and constrain deviations from General Relativity. Yet there exist signatures of quantum gravity—modified black-hole spectra from loop quantum gravity, echoes from horizonless microstate geometries, corrections from extra dimensions—that are encoded in high-order correlations among waveform parameters that remain invisible to classical inference. This is not due to lack of data, but to a representational limitation of Euclidean feature spaces. When two waveforms h_{GR} and h_{bGR} differ only in high-order phase correlations induced by beyond-GR corrections, their L^2-distance satisfies $\|h_{\mathrm{GR}} - h_{\mathrm{bGR}}\|_{L^2} < \varepsilon$ for arbitrarily small ε at the relevant SNR, yet the same pair can be mapped to quantum states $|\psi_{\mathrm{GR}}\rangle$ and $|\psi_{\mathrm{bGR}}\rangle$ whose inner product $|\langle\psi_{\mathrm{GR}}|\psi_{\mathrm{bGR}}\rangle|$ is of order unity only if the feature map $\mathcal{U}_\phi$ is chosen to suppress that overlap (Section 3.2). The choice of $\mathcal{U}_\phi$ therefore determines whether the beyond-GR signature is accessible; this is the sense in which the representation, not the data volume, is the binding constraint. The present thesis proposes to solve this through quantum computing, not as a faster platform but as a fundamentally different analytical tool: one that lifts the inference problem into a Hilbert space where quantum-gravity signatures become linearly separable.

QNIM (Quantum Native Inverse Models) is, first and foremost, an *analytical methodology* for extracting physical information from gravitational waveforms in regimes inaccessible to standard tools. The accompanying software—structured under Clean Architecture and Domain-Driven Design—is its computational realisation, not its primary contribution. This chapter introduces the scientific landscape, the mathematical foundation, and the roadmap of the work.

1.1 Gravitational waves and the inverse spectral problem

Since the first confirmed detection in 2015 (Abbott et al., 2016), gravitational-wave astronomy has become a precision probe of strong-field gravity. The third Gravitational-Wave Transient Catalogue (GWTC-3) reports ninety confident events spanning the first three observing runs (Abbott et al., 2023b); subsequent O4 detections are not included in

the present analysis in order to maintain a reproducible and fully characterised dataset. The catalogue spans binary black holes, binary neutron stars, and mixed systems, and provides a broad empirical basis for testing whether observed signals remain fully consistent with General Relativity.

QNIM targets the regime where standard tools are least informative: small, structured deviations that can be hidden by parameter degeneracies in classical representations. In this context, gravitational waves are uniquely powerful because ringdown spectra encode the geometry of the remnant compact object and can reveal departures from Kerr behaviour.

The conceptual core is the inverse spectral analogy posed by Kac (Kac, 1966): *Can one hear the shape of a drum?* In the general Riemannian setting this question has a negative answer: Gordon, Webb and Wolpert (Gordon, Webb, & Wolpert, 1992) constructed pairs of non-isometric planar domains sharing identical Laplacian spectra, proving that the spectrum alone does not determine the geometry. In the black-hole context, however, the situation is more constrained. Under General Relativity, the no-hair theorem implies that the quasinormal-mode spectrum of a Kerr remnant is *locally unique*: it is determined entirely by mass and spin, with no continuous family of isospectral non-Kerr geometries in the perturbative regime (Berti, Cardoso, & Starinets, 2009). Here the black hole is the drum and quasinormal modes are its spectrum; beyond-GR effects act as isospectrality-breaking perturbations that shift the mode frequencies away from their Kerr values. QNIM is designed to resolve these perturbations in a quantum feature space where they become linearly separable; the formal justification for this separability claim is developed in Section 3.2. Detailed waveform modelling and theory-specific phenomenology are developed in later chapters.

1.2 The classical resolution wall and the quantum solution

Classical gravitational-wave inference has achieved remarkable precision, yet it faces fundamental representational limits when searching for quantum-gravity signatures. This section identifies the mathematical origin of those limits and explains why a Hilbert-space embedding provides representational capabilities unavailable to classical inference.

Epistemological framing of the proof-of-concept. Throughout this thesis a careful distinction is maintained between two levels of claim. The *methodological claim*—that embedding gravitational-wave features in a 2^n-dimensional Hilbert space improves separa-

bility of beyond-GR signatures—is evaluated on a noise-model simulator where hardware constraints do not obscure the inference structure. The *implementation claim*—that the end-to-end QNIM pipeline executes on real quantum processing units—is validated by an extensive campaign of 365 IBM Quantum job submissions on `ibm_fez` and `ibm_marrakesh` under Open Plan resource constraints (Appendix P), spanning both mitigated (ZNE) and unmitigated configurations, all of which returned classification accuracy in the 5–6.5 % range. Hardware accuracy in those runs (6 %) reflects severe QPU quota limitations ($\sim$10 min/month), zero on-hardware retraining, and circuit fidelity $\mathcal{F}_{\text{circ}} \approx 0.45$; it does not characterise the methodology's potential on production hardware, as detailed in Appendix N. No metrological advantage claim is made on the basis of the IBM Fez results.

1.2.1 Classical limits and the quantum embedding

Classical matched filtering and Bayesian sampling remain optimal for many parameter-estimation tasks, but they encounter a resolution wall for weak beyond-GR signatures. The key obstacles are degeneracy-dominated likelihood geometry and representational spectral blindness in Euclidean feature spaces; both effects are documented by Fisher-metric conditioning analyses (Vallisneri, 2008) and motivate a non-classical embedding.

QNIM addresses this by mapping classical features into a 2^n-dimensional Hilbert space and performing discrimination through quantum overlaps. The feature map is defined by

$$|\psi(\mathbf{x})\rangle = \mathcal{U}_\phi(\mathbf{x})\,|\mathbf{0}\rangle, \tag{1.1}$$

where $\mathbf{x} \in \mathbb{R}^d$ is the classical feature vector extracted from the gravitational waveform and $\mathcal{U}_\phi$ is a parametrised unitary circuit whose structure is specified in Section 3.2. This embedding promotes higher-order correlations—invisible in the original Euclidean space—to directly measurable structure in a 2^n-dimensional Hilbert space where beyond-GR classes become linearly separable (Havlíček et al., 2019). The specific ansatz (IQP-inspired ZZ-feature map) and the justification for its choice over alternatives are presented in Chapter 3.

Parameter precision is then constrained by the Quantum Cramér–Rao Bound,

$$\mathrm{Var}(\hat{\theta}) \geqslant \frac{1}{\mathcal{F}_Q(\theta)}, \tag{1.2}$$

where $\mathcal{F}_Q \geqslant \mathcal{F}_C$ holds by construction, since the quantum Fisher information is the supremum over all physically admissible measurements and therefore upper-bounds its

classical counterpart for any fixed parameter θ (Braunstein & Caves, 1994; Paris, 2009). It is essential to note, however, that this inequality characterises the *theoretical* metrological hierarchy; a variational ansatz executed on near-term hardware accesses only a restricted set of measurements and will in general not saturate $\mathcal{F}_Q$. The *achievable* Fisher information $\mathcal{F}_Q^{\text{eff}}$ depends on circuit expressibility, gate fidelity, and shot budget, and its relation to $\mathcal{F}_C$ is evaluated empirically in Chapter 6. In QNIM, the bound is therefore treated as a ceiling that motivates the quantum embedding, while practical advantage is quantified through the expressibility–entanglement metrics of Section 3.3.2.

1.3 Research objectives

The investigation is structured around six inter-dependent objectives, each with a verifiable quantitative criterion. Two components are referenced by acronym throughout: the *Synthetic Signal and Theory Generator* (SSTG), which produces post-Newtonian waveforms with injected beyond-GR anomalies (Chapter 5), and the *Variational Quantum Classifier* (VQC), which implements the quantum discrimination pipeline (Chapter 3). The objectives are ordered to reflect the natural progression of the work: from software infrastructure through methodology and validation to real-data application.

1.4 Structure of the thesis

The thesis follows a natural arc: first establishing the theoretical and physical landscape, then building the analytical machinery, then implementing and validating it. Chapter 2 lays the foundations—the spectral geometry of quasinormal modes, the post-Newtonian waveform model to 3.5PN order, the twelve beyond-GR theory classes (Loop Quantum Gravity, Generalised Uncertainty Principle, extra dimensions, Chern–Simons, etc.), and the broader context of standard sirens and cosmological inference. Chapter 3 introduces the QNIM methodology in full: the quantum feature map, the variational ansatz, the QNSPSA-EML-Feynman optimiser, the QUBO reformulation, and the metrological framework based on Quantum Fisher Information. Chapter 4 describes the implementation: the Domain-Driven Design ontology, the seven-layer Clean Architecture, the Hexagonal pattern, and the hybrid quantum-classical orchestrator that makes the abstract methodology operational. Chapter 5 covers synthetic data generation—the multi-regime engine spanning post-Newtonian through Effective One Body, the anomaly-injection protocols for all thirteen

theory classes, and the validation methodology against real LIGO distributions. Chapter 6 provides the formal analysis: QFI versus CFI benchmarks, expressibility metrics, barren-plateau diagnostics, realistic NISQ error mitigation, and the 5σ significance pipeline. Chapter 7 presents the experimental results: training convergence on simulators and hardware, classification of all ninety GWTC-3 events with Bayes factors and reliability tiers, new upper limits on beyond-GR parameters, cosmological inference from standard sirens, and computational performance analysis. Chapter 8 synthesises implications, current limitations, and scalability projections; Chapter 9 outlines a roadmap for future development. Five appendices supply supporting derivations, deployment guides, and the complete QNIM Planck Reliability Catalogue.

Obj.	**Task**	**Success Criterion**
O1	Domain-driven software architecture	156 core modules, zero circular dependencies
O2	Synthetic Signal and Theory Generator (SSTG) at 3.5PN	KL divergence > 0.3 nats between competing models
O3a	VQC accuracy on noise-model simulator	$\geqslant 85\,\%$; *partially met*: $83.2\,\%$ at production checkpoint, $91\,\%$ projected at full convergence
O3b	End-to-end pipeline execution on real IBM hardware	At least one successful production run; *met*: 365 IBM Quantum job submissions completed across `ibm_fez` and `ibm_marrakesh` (Appendix P), confirming robust, repeated pipeline execution
O3c	Hardware accuracy above random baseline (Open Plan)	Accuracy $> 1/N_{\text{class}} \approx 7.7\,\%$; *failed*: $6\,\%$ on IBM Fez, i.e. **below** the random-guess floor.[a] This sub-random result indicates the production configuration was not merely underperforming but non-functional as a classifier; see Appendix N for the full failure analysis.
O4	Quantum metrological advantage	$\mathcal{F}_Q/\mathcal{F}_C > 1.5$ for all beyond-GR parameters
O5	5σ detection pipeline with Bonferroni correction	$> 80\,\%$ detection efficiency at SNR = 30
O6	Real-event validation (GW150914)	Bayes factor $\log_{10} B < 1$ against beyond-GR (Jeffreys' anecdotal)

Table 1.1: Research objectives and verification criteria. Objective O3 is split into three independently assessable sub-objectives to separate simulator performance, pipeline execution, and hardware accuracy. Source: own elaboration.

[a] The Open Plan QPU quota ($\sim$10 min/month) precluded on-hardware training. The $6\,\%$ result is consistent with the depolarising noise model at $\mathcal{F}_{\text{circ}} \approx 0.45$ and does not reflect the methodology's potential under standard resource allocation. The five technical causes of the simulator-to-hardware gap are documented in Appendix N.

[b] The $91\,\%$ ($\pm 2\,\%$) figure is a polynomial extrapolation of the validation-accuracy curve to full convergence, not a measured value; see Figure 7.1 for the extrapolation and its uncertainty band.

2. Theoretical framework: spectral geometry, post-Newtonian dynamics, and quantum-gravity signatures

This Chapter builds the physical vocabulary the QNIM classifier requires, in three blocks: *(1)* the signal background (quasinormal-mode ringdown, post-Newtonian/EOB inspiral); *(2)* the hierarchy of thirteen beyond-GR signatures across seven physical layers (Layers 1–4, standard-GR observables, in Sections 2.1–2.4; Layers 5–7, the operationally active beyond-GR regime, below); and *(3)* standard sirens and cosmological probes, anchoring these discriminations to real observations. The infrared structure of gravity (BMS symmetry, memory) and superradiance, while relevant, do not modulate the classification pipeline and are deferred to Appendix A.

2.1 Black-hole perturbation theory and the quasinormal mode spectrum

The gravitational-wave signal emitted during the final stages of a binary merger encodes the geometry of the resulting black hole through its quasinormal mode spectrum. This section develops the perturbation theory framework needed to understand that spectrum, from the Schwarzschild case through the Kerr geometry, and establishes the inverse spectral problem that QNIM is designed to solve.

2.1.1 The inverse spectral problem

The mathematical problem at the heart of gravitational-wave spectroscopy was described evocatively by Mark Kac (1966) through the question: "*Can one hear the shape of a drum?*". The drum membrane, like the perturbed spacetime exterior to a black hole, vibrates at a discrete set of characteristic frequencies; the inverse problem asks whether the geometry of the resonator can be uniquely reconstructed from its spectrum.

For a black hole the answer is positive under General Relativity: the *No-Hair theorem* (discussed in Section 2.1.4) guarantees that a stationary black hole is characterised completely by its mass M, angular momentum J, and electric charge Q_e. The quasinormal mode (QNM) spectrum depends only on these three quantities, providing a unique acoustic fingerprint of the final state. The QNIM methodology leverages this uniqueness in two ways: to extract (M_f, J_f) from the ringdown phase of a detected event, and to detect spectral anomalies that would signal deviations from the Kerr geometry predicted by GR.

The simplest case—a non-rotating Schwarzschild black hole—already captures the essential physics and provides the spectral equations that QNIM's feature map is designed to invert.

2.1.2 Perturbations of the Schwarzschild geometry

Consider a Schwarzschild black hole with metric:

$$ds^2 = -f(r)\,dt^2 + f(r)^{-1}\,dr^2 + r^2\,d\Omega^2, \qquad f(r) = 1 - \frac{2M}{r}. \tag{2.1}$$

A general perturbation $h_{\mu\nu}$ can be decomposed into scalar spherical harmonics $Y_{lm}(\theta, \phi)$ and categorised by parity. Axial (odd-parity) perturbations are governed by the *Regge–Wheeler equation* (Regge & Wheeler, 1957):

$$\left[\frac{d^2}{dr^{*2}} + \omega^2 - V_s(r)\right]\Psi_{lm} = 0, \tag{2.2}$$

where the tortoise coordinate r^* maps the exterior region $r \in (2M, \infty)$ to the full real line via:

$$\frac{dr^*}{dr} = f(r)^{-1} \quad \Longrightarrow \quad r^* = r + 2M \ln\Big(\frac{r}{2M} - 1\Big), \tag{2.3}$$

and the ReggeWheeler potential for spin-s fields is:

$$V_s(r) = f(r)\left[\frac{l(l+1)}{r^2} - \frac{2M(1-s^2)}{r^3}\right]. \tag{2.4}$$

For gravitational perturbations $s = 2$; for electromagnetic $s = 1$; for scalar $s = 0$. Polar (even-parity) gravitational perturbations are governed by the *Zerilli equation*, which shares the same spectrum as the ReggeWheeler equation by isospectrality a non-trivial identity that is broken in most modified-gravity theories and therefore serves as a consistency test

within the QNIM framework.

Quasinormal mode boundary conditions

The QNMs are selected by the physical requirement that no radiation arrives from outside the system. The boundary conditions in the tortoise coordinate are:

$$\Psi_{lm} \sim \begin{cases} e^{-i\omega r^*} & r^* \to -\infty \quad \text{(ingoing at horizon)}, \\ e^{+i\omega r^*} & r^* \to +\infty \quad \text{(outgoing at infinity)}. \end{cases} \tag{2.5}$$

These conditions admit solutions only for a discrete set of complex frequencies $\omega_{nls} = \omega_R + i\omega_I$, where ω_R is the oscillation frequency and $\omega_I < 0$ encodes the exponential damping rate. The quality factor $Q = \omega_R/(2|\omega_I|)$ measures the number of oscillation cycles before the mode damps by a factor $1/e$. Two complementary techniques allow their practical computation with the accuracy required for QNIM's injection and validation protocols.

WKB approximation and Leaver's continued-fraction method

The WKB approximation and Berti–Cardoso–Will fitting coefficients for computing QNM frequencies are collected in Appendix B (Table B.1). The Schwarzschild analysis, however, is incomplete for the astrophysically relevant case: almost all binary black hole mergers produce a spinning remnant. Rotation couples the axial and polar perturbation sectors and requires a more powerful equation of motion.

2.1.3 The Teukolsky equation for Kerr black holes

The Kerr metric in Boyer–Lindquist form is given in Appendix B; here we proceed directly to the Teukolsky radial equation for spin-s perturbations:

$$\Delta^{-s}\frac{d}{dr}\left(\Delta^{s+1}\frac{dR}{dr}\right) + \left[\frac{K^2 - 2is(r-M)K}{\Delta} + 4is\omega r - \lambda\right]R = 0, \tag{2.6}$$

with $K = (r^2 + a^2)\omega - am$ and angular eigenvalue λ. The coupled angular sector is solved with spin-weighted spheroidal harmonics to recover the QNM spectrum.

2.1.4 The No-Hair theorem and its quantum-gravity challenge

The No-Hair theorem, formalised by Carter (1971), states that stationary electrovacuum black holes are fully specified by (M, J, Q_e), with multipoles fixed by the Hansen relations:

$$M_l + iS_l = M\,(ia)^l, \tag{2.7}$$

where M_l and S_l are mass and current moments. The quadrupole-deviation test used by QNIM is defined by Equation (2.20) (introduced in Section 2.3.2 below). In explicit form, the leading correction is $\delta\omega_R/\omega_R \approx -0.15\,\delta_Q\,\chi_f^2 + \mathcal{O}(\delta_Q^2)$. The information-geometric development and Fisher-metric formalism are treated rigorously in Chapter 3. With the perturbative spectrum and Kerr consistency conditions established, we now summarise the inspiral-merger dynamics that generate the waveform manifold used in QNIM.

2.2 Binary dynamics: post-Newtonian theory and EOB resummation

Before the merger, the binary evolves through thousands of orbital cycles whose dynamics are captured by the post-Newtonian expansion. This section presents the waveform model that QNIM uses as a foundation for synthetic signal generation and parameter estimation, from the inspiral phase governed by the post-Newtonian series through the Effective One Body resummation that extends it to the strong-field regime.

2.2.1 The post-Newtonian expansion

Away from the merger, the dynamics of a compact binary with component masses m_1, m_2, total mass $M = m_1 + m_2$, chirp mass $\mathcal{M}_c = (m_1 m_2)^{3/5}/M^{1/5}$, and symmetric mass ratio $\eta = m_1 m_2/M^2$ are governed by the post-Newtonian expansion (Blanchet, 2014). The characteristic expansion parameter is $v/c = (GM\omega/c^3)^{1/3}$, where ω is the orbital angular frequency.

In the frequency domain, the gravitational-wave phase under the Stationary Phase Approximation (SPA) takes the form:

$$\Psi_{\rm SPA}(f) = \frac{3}{128\eta}\, v^{-5} \left[\begin{array}{l} 1 + \dfrac{20}{9}\left(\dfrac{743}{336} + \dfrac{11\eta}{4}\right) v^2 + (-16\pi + \beta_{\rm SO})\, v^3 \\ + \displaystyle\sum_{k=4}^{7} c_k(\eta)\, v^k \left(1 + 3\delta_{k5} \ln v\right) - \dfrac{3\tilde{\Lambda}}{128\eta}\, v^{10} \end{array} \right], \tag{2.8}$$

where $v = (\pi G M f / c^3)^{1/3}$; the full 3.5PN coefficients $c_k(\eta)$, including the 2.5PN tail logarithm ($k = 5$) from back-scattering off the source's Coulomb-like potential, are given in Appendix B.2.

The phase Equation (2.8) is valid up to the Innermost Stable Circular Orbit (ISCO) frequency:

$$f_{\rm ISCO} = \frac{c^3}{6\sqrt{6}\,\pi G M}, \tag{2.9}$$

which for a $60\,M_\odot$ system lies at approximately 44 Hz, well within the Advanced LIGO sensitive band.

The spin-orbit coefficient $\beta_{\rm SO}$ and binary tidal deformability $\tilde{\Lambda}$ are derived in Appendix B (Equations. B.1–B.6); their operative forms enter the phase Equation (2.8) at 1.5PN and 5PN order respectively.

2.2.2 Effective One Body resummation

As the binary approaches merger, the PN series diverges. The Effective One Body (EOB) formalism (Buonanno & Damour, 1999) provides a systematic resummation by mapping the two-body dynamics onto the motion of an effective test particle with mass $\mu = m_1 m_2 / M$ in a deformed Schwarzschild-like metric:

$$ds^2_{\rm eff} = -A(r)\, dt^2 + A(r)^{-1} D(r)\, dr^2 + r^2\, d\Omega^2, \tag{2.10}$$

where the potentials $A(r)$ and $D(r)$ incorporate PN corrections resummed via Padé approximants (full construction in Appendix B, Equation B.7), enforcing an effective horizon/ISCO structure for the two-body problem.

The EOB Hamiltonian, calibrated to numerical-relativity waveforms, provides the most accurate inspiral-merger-ringdown (IMR) waveform models currently available (SEOBNRv5, IMRPhenomXPHM) and is used as the reference model against which QNIM's beyond-GR injections are validated.

Having now mapped the complete gravitational-wave signal—from inspiral through merger to ringdown—under General Relativity, we turn to the central question: *where*

and *how* do quantum-gravity corrections enter this picture? The following section catalogs the thirteen beyond-GR theory classes that QNIM's quantum classifier must discriminate, organised into a hierarchy of physical scales and degrees of speculative boldness.

2.3 The hierarchy of quantum-gravity signatures

Not all quantum-gravity signatures are equal: some accumulate over cosmological distances and are accessible at moderate SNR; others require near-horizon resolution and exceptional event quality. QNIM organises the thirteen theory classes into three observational layers by minimum detectable SNR (Table 2.1), operationally focusing on Layers 5–7.

Layer	Name	Observable signature	Min. SNR
5	Beyond-GR propagation	Modified dispersion, dipolar radiation	12
6	Horizon quantum topology	Late-time echoes, QNM deviations	25
7	Deep quantum spacetime	Phase diffusion, spectral broadening	40

Table 2.1: The QNIM physical-layer hierarchy for beyond-GR signatures in gravitational-wave data. Source: own elaboration.

2.3.1 Layer 5: beyond-GR propagation effects

Layer 5 captures propagation-level departures that accumulate over distance and are therefore accessible at moderate SNR. In QNIM, these signatures are implemented as phase/amplitude perturbations in inspiral-dominated signals.

Brans–Dicke scalar-tensor gravity

The simplest extension of GR couples a massless scalar field ϕ to the Ricci scalar through the Brans–Dicke action (Brans & Dicke, 1961):

$$S_{\rm BD} = \frac{1}{16\pi}\int d^4x\sqrt{-g}\left[\phi R - \frac{\omega_{\rm BD}}{\phi}(\nabla\phi)^2\right] + S_m[g_{\mu\nu},\Psi_m], \tag{2.11}$$

where $\omega_{\rm BD}$ is the coupling constant. In this theory, compact objects acquire a "scalar charge" $s_i = -\partial \ln m_i/\partial \ln G_{\rm eff}$ that depends on their internal compactness. A binary with unequal scalar charges emits dipolar radiation at -1PN order, yielding (Damour &

Esposito-Farèse, 1993):

$$\delta\Psi_{\rm BD}(f) = -\frac{5\Delta s^2}{84(\omega_{\rm BD}+2)\eta}\cdot(\pi G\mathcal{M}_c f/c^3)^{-7/3}, \tag{2.12}$$

where $\Delta s = s_1 - s_2$.

Massive graviton and modified dispersion

If the graviton carries a non-zero rest mass m_g, the dispersion relation becomes $E^2 = p^2c^2 + m_g^2c^4$, leading to a frequency-dependent phase velocity. The resulting phase shift in the gravitational waveform is (Will, 1998):

$$\delta\Psi_{m_g}(f) = \frac{\pi^2 d_L m_g^2 c^3}{h_{\rm Pl} f(1+z)}, \tag{2.13}$$

where d_L is the luminosity distance and z is the redshift. Current LIGO/Virgo bounds constrain $m_g < 1.27\times10^{-23}\,\mathrm{eV}/c^2$ (Abbott et al., 2025); QNIM uses this bound as the Layer 5 calibration baseline.

Extra dimensions and the Randall–Sundrum model

In the RandallSundrum braneworld model (Randall & Sundrum, 1999), our observable universe is a four-dimensional brane embedded in a five-dimensional anti-de Sitter bulk. Gravitational waves can leak into the bulk on scales smaller than the crossover radius $R_{\rm cross}$. For sources at $d_L > R_{\rm cross}$, the strain amplitude damps as:

$$h \propto d_L^{-(1+n_{\rm extra}/2)}, \tag{2.14}$$

where $n_{\rm extra}$ is the number of large extra dimensions. The ADD variant with $n_{\rm extra} = 2$ and $R_{\rm cross} \sim \mathcal{O}(10\,\mathrm{Mpc})$ is the reference case for standard-siren anomalies.

Lorentz invariance violation and birefringence

More general modifications to the graviton dispersion relation take the form:

$$E^2 = p^2c^2\left[1 + A_\alpha\left(\frac{pc}{E_{\rm QG}}\right)^{\alpha-2}\right] + m_g^2c^4, \tag{2.15}$$

where $E_{\rm QG}$ is the quantum-gravity energy scale and α is the power-law index. For $\alpha = 3$ (linear LIV), $E_{\rm QG}$ is already constrained to exceed $10^{18}\,\mathrm{GeV}$ by GW170817. QNIM scans $\alpha \in [2, 4]$ and includes parity-violating birefringence in the same layer.

2.3.2 Layer 6: horizon quantum topology

Layer 6 targets near-horizon structure through delayed ringdown components and quadrupole consistency tests. The objective is to separate echo-like signatures from genuine Kerr-deviation effects under realistic SNR.

The Fuzzball paradigm

In string-theoretic approaches to black holes, the event horizon is dissolved into a "Fuzzball" microstate geometry (Mathur, 2005): a horizonless configuration in which the mass is supported by string-scale structure rather than a singularity. The Fuzzball acts as a partially reflecting wall. Radiation that would classically be absorbed by the horizon is instead reflected, reprocessed, and re-emitted with a time delay:

$$\Delta t_{\rm echo} = 2\int_{r_s}^{r_+} \frac{dr}{c\,f(r)} \approx \frac{2r_+}{c}\ln\!\left(\frac{r_+}{r_s}\right), \tag{2.16}$$

where r_s is the Fuzzball surface radius (slightly outside r_+). The resulting signal is modeled as a discrete echo train:

$$h_{\rm ringdown}(t) = h_{\rm QNM}(t) + \sum_{n=1}^{N} \mathcal{R}^n \cdot h_{\rm QNM}(t - n\Delta t), \tag{2.17}$$

where $\mathcal{R}$ is the complex reflectivity ($|\mathcal{R}| \in [0, 1]$).

LQG area quantisation and the Barbero–Immirzi parameter

In Loop Quantum Gravity (Ashtekar & Lewandowski, 2004), the area of the event horizon is quantised in units of the Planck area:

$$A_n = 8\pi\gamma\, l_P^2 \sum_k \sqrt{j_k(j_k+1)}, \tag{2.18}$$

where γ is the BarberoImmirzi parameter and j_k are spin-network labels. For the physical value $\gamma_{\rm Meissner} = \ln 2/(\pi\sqrt{3})$ (Meissner, 2004), the minimum area quantum is $A_{\rm min} \approx$

$4 \ln 2\, l_P^2$. The induced line-structure correction is represented through the delay model:

$$\Delta t_{\mathrm{LQG}} = \frac{2r_+(a)}{c} \ln\!\left(\frac{M_f}{m_P\sqrt{\gamma}}\right), \tag{2.19}$$

which is separable from Fuzzball scaling through its (γ, M_f) dependence.

Quadrupole deviations and the generalised No-Hair test

A model-independent test of the Kerr geometry is provided by parameterising the quadrupole moment as $Q = -J^2(1+\delta_Q)/M$, where δ_Q is a free parameter. A non-zero δ_Q modifies the QNM frequencies at leading order through the relation:

$$\frac{\delta\omega_R}{\omega_R} \approx -0.15\,\delta_Q\,\chi_f^2 + \mathcal{O}(\delta_Q^2). \tag{2.20}$$

QNIM's Layer 6 injection simultaneously varies δ_Q, $|\mathcal{R}|$, and Δt as independent parameters.

2.3.3 Layer 7: deep quantum spacetime

Layer 7 collects speculative high-curvature effects as effective stochastic or vacuum corrections. These signatures are weak, but they define the upper boundary of the QNIM hypothesis space.

Hawking radiation and the information paradox

The thermal radiation predicted by Hawking (1975) to be emitted by black holes at temperature $T_H = \hbar c^3/(8\pi G M k_B)$ has never been directly observed, as the temperature is $\sim 6 \times 10^{-8}\,(M_\odot/M)$ K for stellar-mass objects. The effective vacuum contribution is modeled with zeta-function regularisation (Hawking, 1975):

$$E_0^{\mathrm{reg}} = \frac{\hbar\omega_c}{2}\,\zeta(-1) = -\frac{\hbar\omega_c}{24}, \tag{2.21}$$

where $\zeta(-1) = -1/12$.

Spacetime foam and Ornstein–Uhlenbeck phase diffusion

Wheeler's spacetime foam hypothesis posits that at the Planck length $l_P = \sqrt{\hbar G/c^3} \approx 1.6 \times 10^{-35}$ m, the metric undergoes random topological fluctuations analogous to quantum foam. At the macroscopic scale accessible to LIGO, these fluctuations manifest as a

stochastic phase noise $\delta\varphi(t)$ in the gravitational-wave signal. QNIM models this noise as an OrnsteinUhlenbeck process:

$$d(\delta\varphi_t) = -\gamma_{\rm foam}\,\delta\varphi_t\,dt + \sqrt{2\gamma_{\rm foam}\,T_{\rm foam}}\,dW_t, \tag{2.22}$$

where $\gamma_{\rm foam}$ is the relaxation rate, $T_{\rm foam}$ an effective amplitude scale, and W_t a Wiener process; the corresponding PSD is

$$S_{\rm foam}(f) = \frac{4\gamma_{\rm foam}\,T_{\rm foam}}{\gamma_{\rm foam}^2 + (2\pi f)^2}, \tag{2.23}$$

with knee frequency $f_{\rm knee} = \gamma_{\rm foam}/(2\pi)$.

These seven layers constitute the *complete signal hierarchy* that QNIM's quantum classifier must discriminate: from the standard-siren and post-Newtonian regime (Layers 1–4, deferred to Appendix A) through the quantum-topology scenarios of Layer 6, to the deepest quantum-foam corrections of Layer 7. With this hierarchy in place, we now turn to the observational tools that connect these signatures to real astrophysical data. The infrared structure of gravity—the BMS symmetry group, the soft graviton theorem, and gravitational memory—while intimately related to the quantum structure of spacetime, forms a separate theoretical layer deferred to Appendix A. The following sections focus on the observables that are *directly encoded in the strain time series* accessible to QNIM's feature map.

2.4 Standard sirens and the Hubble tension

Gravitational waves from compact binaries provide a self-calibrating distance measurement that is independent of the cosmic distance ladder. This section describes how that measurement is made, its current contribution to the Hubble constant debate, and how the QNIM quantum metrological gain translates into a concrete improvement in cosmological inference.

2.4.1 The luminosity-distance measurement

As first recognised by Schutz (1986), the gravitational-wave signal of a compact binary is a "self-calibrating" distance indicator: the chirp mass extracted from the phasing, combined with the observed amplitude, determines the luminosity distance d_L without any external

calibration (i.e., without the cosmic distance ladder):

$$h \propto \frac{G\mathcal{M}_c}{c^2 d_L} \left(\frac{G\mathcal{M}_c \pi f}{c^3} \right)^{2/3}. \tag{2.24}$$

By combining d_L with a redshift z obtained from an electromagnetic counterpart (or from statistical galaxy catalogues), the Hubble constant follows directly:

$$H_0 \approx \frac{c\, z}{d_L} \quad (z \ll 1). \tag{2.25}$$

At higher redshift, a full cosmological model (e.g., flat ΛCDM) is required to relate d_L to z and H_0.

2.4.2 Current tension and GW contribution

Modern cosmology faces a tension between the early-universe Planck CMB estimate ($H_0 = 67.4 \pm 0.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (Aghanim et al., 2020)) and late-universe distance-ladder values: SH0ES gives $H_0 = 73.17 \pm 0.86$ (Breuval et al., 2024) (5.8σ tension), while the JWST-calibrated CCHP TRGB anchor gives $H_0 = 68.81 \pm 2.21$ (Freedman et al., 2025), consistent with Planck. Depending on calibration, the tension across independent late-universe probes stands at 3.7–5.8σ (full landscape in Table 7.5) — whether this reflects new physics or unresolved distance-ladder systematics remains an open question.

The GW170817 standard-siren measurement yielded $H_0 = 70.0^{+12.0}_{-8.0}\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (Abbott et al., 2017a), consistent with both extremes but with too large an uncertainty to adjudicate. Reaching a sub-percent measurement of H_0 via GWs requires $\mathcal{O}(10^3)$ events with electromagnetic counterparts or large galaxy catalogue datasets. The QNIM pipeline reduces the systematic floor in luminosity-distance estimation through the quantum metrological gain $G_{d_L} = 1.09$ (Table 6.1), which corresponds to a $1 - 1/\sqrt{G_{d_L}} \approx 4.3\,\%$ reduction in σ_{d_L} relative to classical matched filtering at the same SNR. This modest but systematic improvement propagates directly into the H_0 posterior: for a fixed set of N standard-siren events, it is equivalent to adding approximately $N(G_{d_L} - 1) \approx 0.09N$ additional events to the catalogue. For the 12-event GWTC-3 subset used in Section 7.5, this corresponds to roughly one additional effective event. The non-linear memory effect (Christodoulou, 1991) contributes to this gain by encoding an additional observable in the late-time ringdown tail that is captured by the quantum feature map but not by

the standard matched-filter statistic; a dedicated analysis of the memory contribution is reserved for future work.

2.4.3 Dark energy and the speed of gravitational waves

The detection of GW170817 and its associated gamma-ray burst GRB 170817A with a time delay of only $1.74 \pm 0.05\,\mathrm{s}$ over $40\,\mathrm{Mpc}$ constrained the gravitational-wave propagation speed to (Abbott et al., 2017b):

$$\left|\frac{c_{\mathrm{GW}}}{c} - 1\right| < 5 \times 10^{-16}. \tag{2.26}$$

This single measurement ruled out a large class of scalar-tensor and vector-tensor dark energy models in particular, most k-essence models (Armendariz-Picon, Mukhanov, & Steinhardt, 2001) in which the graviton propagation speed deviates from c. In the QNIM framework, this constraint is implemented as a hard prior on the Layer 5 parameter space, drastically reducing the volume of viable modified-gravity theories.

Superradiance and axion clouds, while relevant to future ultralight-boson searches, do not enter the thirteen-class classification pipeline and are deferred to Appendix A alongside the infrared structure of gravity. Table 2.2 below consolidates all parameters as a bridge to the computational methodology.

The following section provides a consolidated reference table of all parameters and their roles, serving as the bridge from theoretical physics to computational methodology.

2.5 Summary of theoretical inputs to QNIM

Table 2.2 provides a consolidated reference of all physical parameters, their theoretical origin, and their role in the QNIM classification pipeline. This table will be referenced throughout the methodology and results chapters.

Parameter	**Theory**	**Observable effect**	**Layer**
ω_{BD}	BransDicke	Dipolar phase at -1PN	5
m_g	Massive graviton	Phase dispersion, group delay	5
n_{extra}	RandallSundrum ADD	d_L anomaly	5
A_α, α	LIV modified dispersion	Arrival-time shifts	5
θ_{CS}	ChernSimons	Birefringence $h_+ \leftrightarrow h_\times$	5
δQ	No-Hair violation	QNM frequency deviation	6
$\lvert\mathcal{R}\rvert, \Delta t$	Fuzzball echoes	Echo train amplitude	6
γ_{BI}	LQG area quanta	QNM spectral broadening	6
$T_{\text{foam}}, \gamma_{\text{foam}}$	Spacetime foam	Phase diffusion PSD	7
m_b, α_{SR}	Axion cloud	Monochromatic CW signal	5
c_{GW}/c	Modified propagation	Multi-messenger time delay	5
$\tilde{\Lambda}$	Tidal EOS	5PN phase correction	4
Δh_{mem}	BMS memory	DC offset in ringdown tail	6

Table 2.2: Summary of physical parameters entering the QNIM framework across all modelled theories. Source: own elaboration.

The thirteen theory classes represented in this table span the full range of physical complexity probed by the QNIM methodology, from well-constrained perturbative corrections (massive graviton, Brans–Dicke) to highly speculative deep-quantum signatures (spacetime foam, LQG echoes). A central result of Chapter 7 is that the quantum feature map separates all thirteen classes with a pairwise KL-divergence exceeding 0.3 nats, meeting Objective O2 stated in Section 1.3.

3. The QNIM analytical methodology

This Chapter describes the analytical core of QNIM. We begin with raw gravitational-wave strain data—already conditioned and whitened—and project it into a 2^{12}-dimensional quantum Hilbert space where beyond-GR signatures become linearly separable. The seven-step methodology is: *(1) classical encoding* (Section 3.1) converts strain into 12 classical feature vectors via STFT and PCA; *(2) quantum feature map* (Section 3.2) lifts these into quantum amplitudes inaccessible to classical representations; *(3) variational ansatz* (Section 3.3) applies a trainable quantum circuit to entangle and rotate the state; *(4) Bayesian model ranking* (Section 3.4) compares the thirteen beyond-GR theories against General Relativity using Bayes factors; *(5) QUBO parameter estimation* (Section 3.5) reformulates parameter refinement as a binary optimisation problem; *(6) stochastic natural-gradient optimisation* (Section 3.6) tunes the VQC using quantum natural gradients with zero-noise extrapolation; and *(7) end-to-end pipeline* (Section 3.8) orchestrates all stages. The chapter justifies why quantum computing provides advantage at each step and how each component integrates into a unified system.

Throughout, we use the following notation conventions:

- n: number of physical qubits.
- d: number of classical input features (after PCA).
- N_θ: number of variational parameters.
- $\boldsymbol{\theta} \in \mathbb{R}^p$: vector of trainable circuit parameters.
- $\mathbf{x} \in \mathbb{R}^d$: classical feature vector for a single gravitational-wave event.
- $|\psi(\mathbf{x}, \boldsymbol{\theta})\rangle$: parameterised quantum state on n qubits.

3.1 From gravitational-wave strain to classical feature vectors

Before any quantum operation, the raw detector strain must be converted into a compact, noise-whitened feature vector of fixed dimension d. This pre-processing pipeline is entirely

classical and is implemented as a set of value objects within the QNIM Domain model (Section C.5).

The preprocessing pipeline is entirely classical and consists of four sequential operations that transform raw detector output into the compact, noise-whitened vector that the quantum circuit will encode.

3.1.1 Data acquisition and conditioning

The input to the pipeline is the calibrated, frequency-domain strain $\tilde{s}(f)$ downloaded from the Gravitational-Wave Open Science Center (GWOSC). The conditioning steps are:

1. **Noise power estimation.** The one-sided power spectral density $S_n(f)$ is estimated using a Welch periodogram on an off-source stretch of 256 s surrounding the event.

2. **Whitening and bandpassing.** The strain is whitened by dividing by $\sqrt{S_n(f)}$ and bandpassed to the sensitive frequency range $[f_{\text{low}}, f_{\text{high}}]$. For the QNIM training dataset, $f_{\text{low}} = 20\,\text{Hz}$ and $f_{\text{high}} = 1024\,\text{Hz}$.

3. **Time-frequency representation.** The whitened strain is mapped to a time-frequency spectrogram using a short-time Fourier transform (STFT) with a Tukey window of length 0.5 s and 50 % overlap.

4. **Matched-filter score vector.** For theory classification, the spectrogram is supplemented by the matched-filter log-likelihood $\ln \mathcal{L}(\hat{\boldsymbol{\theta}}_{\text{MAP}})$ and its first-order phase-space gradient, evaluated at the Maximum A Posteriori (MAP) estimate returned by a fast Bayesian sampler.

The conditioned feature matrix typically contains thousands of frequency-domain samples. Before any quantum encoding, this dimensionality must be reduced to match the number of available qubits while preserving the signal variance that distinguishes the thirteen theory classes.

3.1.2 Dimensionality reduction via PCA

The concatenated feature matrix (spectrogram plus matched-filter statistics) typically has $\mathcal{O}(10^3)$ dimensions. QNIM applies Principal Component Analysis (PCA) to reduce this to $d = 12$ principal components, retaining $\geqslant 95\,\%$ of the total explained variance across

the training dataset. The choice $d = 12$ matches the number of qubits $n = 12$, enabling a one-to-one assignment of principal components to qubit rotation angles in the encoding layer.

The PCA basis is fitted once on the training set and applied without retraining to all subsequent events, ensuring that the quantum feature map operates in a fixed, physically meaningful subspace of signal variation. The 12 components retain 95.2% variance (Table C.2).

3.2 The quantum feature map

The quantum feature map is the foundational building block of the QNIM methodology. It converts a classical feature vector $\mathbf{x} \in \mathbb{R}^d$ into a quantum state in the 2^n-dimensional Hilbert space $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$, encoding the information in quantum amplitudes that are inaccessible to any classical representation of the same dimension.

3.2.1 Formal definition

The feature map is a parameterised unitary operator:

$$\mathcal{U}_\phi(\mathbf{x}) = \prod_{k=1}^{r_\phi} \left[H^{\otimes n} \cdot \exp\left(-i \sum_{S \subseteq [n]} \phi_S(\mathbf{x}) \bigotimes_{j \in S} Z_j \right) \right], \tag{3.1}$$

where r_ϕ is the number of repetitions, Z_j is the Pauli-Z operator on qubit j, $[n] = \{1, \ldots, n\}$, and the data-dependent angles are:

$$\phi_S(\mathbf{x}) = \begin{cases} x_j & |S| = 1, \\ \prod_{i \in S}(\pi - x_i) & |S| > 1. \end{cases} \tag{3.2}$$

Normalisation convention: All input features x_i must be normalised to the interval $[0, \pi]$ before being passed to the feature map. This ensures that the angles $(\pi - x_i)$ are real-valued and bounded in $[0, \pi]$. In the QNIM pipeline, raw PCA-projected features (which are unbounded) are rescaled via min-max normalisation to $[0, \pi]$ as part of the preprocessing layer. This is performed in the infrastructure layer (see Chapter 4 for the full software architecture).

The Hadamard layer $\mathcal{H}^{\otimes n}$ prepares a uniform superposition, while the entangling

diagonal unitary creates non-linear correlations between features through the product terms $\prod_i(\pi - x_i)$. This is the *ZZFeatureMap* introduced by Havlcek et al. (Havlíček et al., 2019), which provides a provable separation from classical kernel methods.

The ZZFeatureMap in its standard form already provides provable separation from classical kernels, but the QNIM application demands additional spectral resolution in the high-frequency oscillation content of gravitational waveforms. A Chebyshev pre-processing layer achieves this without adding qubits.

3.2.2 Chebyshev enhancement

For the QNIM application, the standard ZZFeatureMap is augmented with a Chebyshev encoding layer to improve the resolution of high-frequency oscillations in the waveform spectrum. Each feature x_j is first transformed via the Chebyshev-T polynomial of degree m:

$$\tilde{x}_j^{(m)} = \arccos(T_m(2x_j - 1)), \qquad T_m(\xi) = \cos(m \arccos \xi), \tag{3.3}$$

producing m harmonically related copies of each feature that collectively resolve sub-features at m times the native resolution. In the QNIM baseline configuration, $m = 2$ is used, effectively doubling the spectral resolution of the quantum embedding without increasing the qubit count.

The feature map encodes each event into a quantum state. The natural similarity measure between two such states—and the mathematical object that determines the classifier's discriminative power—is the quantum kernel. In the QNIM pipeline, the Chebyshev-transformed features $\tilde{x}_j^{(2)}$ *replace* the raw PCA components x_j as inputs to the ZZFeatureMap angles (Equation 3.2), so that the data-dependent angles become $\phi_{\{j\}}(\mathbf{x}) = \tilde{x}_j^{(2)}$ for $|S| = 1$.

3.2.3 Quantum kernel and reproducing Hilbert space

The inner product between two encoded states defines the quantum kernel:

$$k_Q(\mathbf{x}, \mathbf{x}') = \left| \langle \mathbf{0} | \mathcal{U}_\phi^\dagger(\mathbf{x}') \mathcal{U}_\phi(\mathbf{x}) | \mathbf{0} \rangle \right|^2. \tag{3.4}$$

This kernel is positive semi-definite and defines a unique Reproducing Kernel Hilbert Space (RKHS) $\mathcal{F}$ in which classification is performed. The key metrological property is that the Gram matrix $K_{ij} = k_Q(\mathbf{x}_i, \mathbf{x}_j)$ can distinguish signal pairs that share identical values of all

classical matched-filter statistics but differ in their quantum-geometric structure precisely the degeneracy that renders classical inference blind to Layer 6 and Layer 7 signatures.

The computational cost of evaluating k_Q on a classical simulator is $\mathcal{O}(r_\phi n \cdot 2^n)$ per kernel element via statevector simulation, or $\mathcal{O}(n\chi^3)$ via MPS at bond dimension $\chi = \chi_{\text{crit}}$. A formal cost comparison is deferred to Appendix B.8.

3.2.4 The qubit count constraint: five design drivers for $n = 12$

The full design-driver justification is moved to Appendix B. Operationally, $n = 12$ is fixed by five simultaneous constraints:

- PCA dimensionality matching ($d = 12$).
- NISQ coherence and circuit-depth limits.
- Trainability under barren-plateau constraints.
- IBM hardware topology and routing overhead.
- Diminishing marginal variance beyond the twelfth component.

On NISQ hardware, the evaluation reduces to two circuit executions of depth proportional to $r_\phi \cdot n$, constituting the primary source of quantum advantage in the classification branch of QNIM.

3.3 Variational quantum circuit architecture

Once the quantum feature map has encoded the input data into Hilbert space, a parameterised variational circuit learns to separate the thirteen theory classes. This section describes the circuit architecture, its expressibility and entanglement properties, and how its output is decoded into theory probabilities.

3.3.1 The EfficientSU2 ansatz

The classification task is performed by a parameterised quantum circuit (Variational Quantum Classifier, VQC) of the form:

$$|\psi(\mathbf{x}, \boldsymbol{\theta})\rangle = W(\boldsymbol{\theta})\, \mathcal{U}_\phi(\mathbf{x})\, |\mathbf{0}\rangle, \tag{3.5}$$

where $W(\boldsymbol{\theta})$ is the *trainable ansatz* and $\mathcal{U}_\phi(\mathbf{x})$ is the feature map of Section 3.2. The QNIM ansatz is built from the EfficientSU2 template, which alternates single-qubit rotation layers with a linear entanglement structure:

$$W(\boldsymbol{\theta}) = \prod_{r=1}^{r_W} \left[\bigotimes_{j=1}^{n} R_y(\theta_{j,r}^{(y)})\, R_z(\theta_{j,r}^{(z)}) \cdot \mathrm{CNOT}_{\mathrm{linear}} \right], \tag{3.6}$$

where $R_y(\theta) = e^{-i\theta Y/2}$ and $R_z(\theta) = e^{-i\theta Z/2}$ are single-qubit rotations around the Y and Z Bloch-sphere axes, and $\mathrm{CNOT}_{\mathrm{linear}}$ is a layer of CNOT gates coupling qubit j to qubit $j+1$ for $j = 1, \ldots, n-1$.

For $n = 12$ qubits and $r_W = 2$ repetitions, the total parameter count is:

$$N_\theta = 2 \cdot n \cdot r_W = 2 \times 12 \times 2 = 48, \tag{3.7}$$

and the circuit depth (in units of two-qubit gate layers) is $D = r_W \cdot (n-1) = 22$, which is within the coherence budget of current NISQ processors. Here D counts only the two-qubit (CNOT) gate layers; the total circuit depth including single-qubit rotations is $D_{\mathrm{total}} = r_W(2n + n - 1) = 2 \times (24 + 11) = 70$ native gates, consistent with the IBM Fez compilation depth of $D_{\mathrm{native}} = 31$ two-qubit-equivalent layers reported in Appendix C.6.

3.3.2 Expressibility and entanglement metrics

Choosing an ansatz is not merely an architectural decision: a circuit that spans too small a region of Hilbert space will underfit, while one that is too expressive will exhibit barren plateaus. Two quantitative metrics—expressibility and entanglement capacity—characterise where EfficientSU2 sits on this spectrum.

The ability of the ansatz to approximate arbitrary unitary operations is quantified by the *expressibility* $\mathcal{E}$, defined as the KullbackLeibler divergence between the distribution of states generated by random sampling of $\boldsymbol{\theta}$ from the ansatz and the Haar-random distribution over the full $SU(2^n)$ group (Sim et al., 2019):

$$\mathcal{E} = D_{\mathrm{KL}}(P_{\mathrm{ansatz}}(|\psi\rangle) \,\|\, P_{\mathrm{Haar}}(|\psi\rangle)) \,. \tag{3.8}$$

For the QNIM EfficientSU2 configuration ($n = 12$, $r_W = 2$), numerical estimation via Monte Carlo sampling over 10^4 parameter draws yields $\mathcal{E} = 0.043 \pm 0.008$ nats. Since a perfect Haar-random circuit would give $\mathcal{E} = 0$, this value indicates that the ansatz spans

approximately 96 % of the relevant state space.

The entanglement capacity is quantified by the average bipartite von Neumann entropy (B.16); the measured value $\bar{S}_{\mathrm{vN}} = 3.82 \pm 0.12$ bits out of a maximum 6 bits confirms the multi-qubit entanglement needed for Layer 6–7 discrimination.

Once the variational circuit has processed the input state, a classical readout layer translates the quantum expectation values into a probability vector over the thirteen theory classes.

3.3.3 Measurement and classification head

The circuit output is the expectation value of a product of single-qubit Pauli-Z observables:

$$\hat{y}(\mathbf{x}, \boldsymbol{\theta}) = \langle \psi(\mathbf{x}, \boldsymbol{\theta}) | \bigotimes_{j=1}^{4} Z_j \, | \psi(\mathbf{x}, \boldsymbol{\theta}) \rangle \in [-1, +1]. \tag{3.9}$$

For the multi-class extension to thirteen theory classes, the single readout is replaced by an output register of $\lceil \log_2 13 \rceil = 4$ qubits whose joint Pauli expectation values are decoded by a softmax layer into a probability simplex over the thirteen classes. The predicted class is:

$$\hat{c} = \arg \max_{c \in \{0,\ldots,12\}} \mathrm{softmax}(\mathbf{W}_{\mathrm{out}} \langle \boldsymbol{Z} \rangle + \mathbf{b}_{\mathrm{out}})_c \,, \tag{3.10}$$

where $\mathbf{W}_{\mathrm{out}}$ and $\mathbf{b}_{\mathrm{out}}$ are small classical weight matrices trained jointly with the quantum parameters.

3.4 Bayesian model ranking and the Bayes factor

Once the quantum circuit $W(\boldsymbol{\theta}^{\star})$ has classified a gravitational-wave event, a critical question arises: *how strong is the statistical evidence for that classification?* The VQC output $\mathbf{p}_{\mathrm{theory}}$ ranks theories but does not quantify the strength of evidence for the top-ranked theory relative to General Relativity (the null hypothesis). This is provided by the *Bayes factor* — the ratio of marginal likelihoods (the evidence) for each of the thirteen hypotheses against the null — which bridges the quantum classification (continuous probabilities) and the scientific inference layer (discrete reliability tiers):

$$B_{H_1/H_0} = \frac{\mathcal{Z}_{H_1}}{\mathcal{Z}_{H_0}} = \frac{\int_{\Theta_1} \mathcal{L}(\mathbf{d}|\boldsymbol{\theta}_1, H_1)\, \pi(\boldsymbol{\theta}_1|H_1)\, d\boldsymbol{\theta}_1}{\int_{\Theta_0} \mathcal{L}(\mathbf{d}|\boldsymbol{\theta}_0, H_0)\, \pi(\boldsymbol{\theta}_0|H_0)\, d\boldsymbol{\theta}_0}, \tag{3.11}$$

where $\mathcal{Z}_{H_i}$ is the marginal likelihood (or "evidence") for hypothesis H_i, $\mathcal{L}(\mathbf{d}|\boldsymbol{\theta}_i, H_i)$ is the likelihood of the data $\mathbf{d}$ under hypothesis H_i with parameters $\boldsymbol{\theta}_i$, and $\pi(\boldsymbol{\theta}_i|H_i)$ is the prior distribution over parameters under H_i. In QNIM, H_0 is General Relativity (the "null" hypothesis) and H_1 represents each of the twelve beyond-GR theory classes in turn (scalar-tensor, LQG, Chern-Simons, etc.).

The calculation of the evidence $\mathcal{Z}_{H_i}$ is performed via Laplace approximation around the Maximum A Posteriori (MAP) estimate:

$$\ln \mathcal{Z}_{H_i} \approx \ln \mathcal{L}(\mathbf{d}|\hat{\boldsymbol{\theta}}_i, H_i) + \ln \pi(\hat{\boldsymbol{\theta}}_i|H_i) + \frac{k_i}{2}\ln(2\pi) - \frac{1}{2}\ln|\mathbf{H}(\hat{\boldsymbol{\theta}}_i)|, \tag{3.12}$$

where $\mathbf{H}(\hat{\boldsymbol{\theta}}_i) = -\nabla^2 \ln p(\hat{\boldsymbol{\theta}}_i|\mathbf{d}, H_i)$ is the *negative* Hessian of the log-posterior (positive definite at a maximum), so that $\ln|\mathbf{H}| > 0$ penalises broad posteriors. For GW150914 and the GWTC-3 catalogue, $k_0 = 15$ (GR parameters) and $k_1 \approx 16$–18 (beyond-GR theories, which add 1–3 parameters to the base set).

Note. Laplace evidence is accurate mainly for unimodal, near-Gaussian posteriors. Compact-binary posteriors can be multimodal, so Bayes factors here are approximate evidence ratios rather than exact marginal likelihoods. A fully rigorous treatment would require nested sampling or thermodynamic integration on the full posterior.

The *natural logarithm* of the Bayes factor is reported throughout this thesis as $\ln B_{H_1/H_0}$, or in base-10 form as $\log_{10} B_{H_1/H_0}$ for compatibility with Jeffreys' scale of evidence strength:

- $|\ln B| < 2.3$ (i.e., $|\log_{10} B| < 1$): anecdotal evidence.
- $2.3 < |\ln B| < 5.1$: weak evidence.
- $5.1 < |\ln B| < 9.2$: strong evidence.
- $|\ln B| > 9.2$: decisive evidence.

A positive $\ln B_{H_1/H_0}$ indicates evidence in favour of the beyond-GR hypothesis; a negative value favours GR. In QNIM, most GWTC-3 events exhibit $\ln B < 0$, confirming GR consistency (see Chapter 7 for the full catalogue).

3.5 QUBO formulation of parameter estimation

Theory classification identifies the most probable beyond-GR model; precise extraction of the coupling constants within that model requires solving a high-dimensional, non-convex

optimisation problem. This section reformulates that problem as a Quadratic Unconstrained Binary Optimisation instance suitable for quantum annealing, and describes the hybrid classical post-processing that refines the annealer output.

3.5.1 Motivation: the landscape of the likelihood

Theory classification is only one half of the QNIM pipeline. Equally important is the precise extraction of the physical parameters $\boldsymbol{\theta} = (m_1, m_2, \chi_{\rm eff}, d_L, \ldots)$ within the framework identified as most likely by the VQC. This parameter estimation task is formulated as an optimisation problem over the log-posterior:

$$\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}\in\mathcal{P}} \left[\ln \mathcal{L}(\boldsymbol{\theta}) + \ln \pi(\boldsymbol{\theta})\right]. \tag{3.13}$$

The log-likelihood surface in $\mathcal{P}$ is highly non-convex, exhibiting hundreds of local maxima in the 15-dimensional space, all of which yield similar matched-filter scores but correspond to physically distinct parameter combinations. Gradient-based optimisers invariably converge to a local maximum that depends strongly on the initialisation, introducing systematic biases.

Quantum annealers operate on discrete binary variables, not continuous real-valued parameters. The first step in constructing the QUBO is therefore to represent each of the fifteen physical parameters as a string of bits, choosing an encoding that preserves the physical adjacency structure of the parameter space.

3.5.2 Binary discretisation and QUBO Hamiltonian

Each continuous parameter $\theta_k \in [\theta_k^{\min}, \theta_k^{\max}]$ is discretised into $b_k = 8$ bits using Gray encoding (Appendix K), yielding $N_q = 15 \times 8 = 120$ binary variables. The optimisation problem is cast as:

$$H_{\rm QUBO}(\mathbf{q}) = \mathbf{q}^\top Q\, \mathbf{q}, \qquad \mathbf{q} \in \{0,1\}^{120}, \tag{3.14}$$

where Q is constructed from the second-order Taylor expansion of $-\ln \mathcal{L}(\boldsymbol{\theta}(\mathbf{q}))$ around the MAP estimate; diagonal elements encode likelihood gradients and off-diagonal elements encode FIM parameter correlations. Constraint penalties ($\lambda_{\rm penalty} = 10 \times \max|Q_{ij}|$) enforce physicality conditions such as $m_2 \leqslant m_1$ and $|\chi_i| \leqslant 1$. Full index-map conventions are in Appendix K.

3.5.3 D-Wave annealing and hybrid post-processing

Scope note. All annealing results in this thesis use `neal.SimulatedAnnealingSampler` (D-Wave Ocean SDK), a classical algorithm. Physical QPU tunnelling dynamics motivate this design but are *not* empirically demonstrated here; results are interpreted as optimisation evidence under a noise-perturbed surrogate model (full protocol in Appendix I.3).

The sampler is executed with Gaussian perturbations injected into QUBO coefficients ($\sigma_Q = 0.03$, 1000 reads per event). The returned ensemble of low-energy bitstrings is decoded via Gray decoding into continuous parameter vectors $\{\boldsymbol{\theta}^{(k)}\}$, which seed an L-BFGS-B refinement step. The final QNIM estimate is the candidate with the highest polished log-posterior; 68 % credible intervals follow from bootstrap resampling of the ensemble.

3.6 The QNSPSA-EML-Feynman optimiser

Optimising the 48 variational parameters of the VQC on NISQ hardware requires an algorithm that is simultaneously:

- *Gradient-free*: exact gradients are cost-prohibitive on hardware due to shot noise, requiring stochastic approximations.
- *Geometry-aware*: the cost landscape of the VQC is not Euclidean but Riemannian (Fubini–Study metric), necessitating a quantum natural gradient step.
- *Noise-resilient*: hardware noise introduces biases that corrupt gradient estimates, requiring active error mitigation at the optimisation level.

The QNSPSA-EML-Feynman algorithm addresses all three requirements simultaneously.

The natural gradient step provides the geometric awareness; SPSA provides the shot-efficient gradient estimation; their combination constitutes the QNSPSA kernel of the optimiser.

3.6.1 Quantum Natural SPSA (QNSPSA)

The Quantum Natural Simultaneous Perturbation Stochastic Approximation (QNSPSA) combines the Quantum Natural Gradient (QNG) (Stokes et al., 2020) with the SPSA finite-difference gradient estimator (Spall, 1998).

At iteration t, a random Rademacher perturbation vector $\boldsymbol{\Delta} \in \{-1, +1\}^p$ is drawn uniformly. The stochastic gradient and Quantum Geometric Tensor (QGT) estimates are obtained from four circuit evaluations:

$$\hat{g}_t = \frac{f(\boldsymbol{\theta}_t + c_t\boldsymbol{\Delta}_t) - f(\boldsymbol{\theta}_t - c_t\boldsymbol{\Delta}_t)}{2c_t\boldsymbol{\Delta}_t}, \tag{3.15}$$

$$\hat{F}_t = \frac{1}{2}\left(\hat{g}_t \otimes \hat{g}_t + \hat{g}'_t \otimes \hat{g}'_t\right), \tag{3.16}$$

where $\hat{g}'_t$ uses an independent Rademacher draw $\boldsymbol{\Delta}'_t$. Note that $\hat{F}_t$ is an *approximation* of the Quantum Geometric Tensor based on the outer product of stochastic gradient estimates, following the QNSPSA construction of (Gacon, Zoufal, Carleo, & Woerner, 2021). This approximation is computationally tractable on NISQ hardware but underestimates the off-diagonal elements of the true QGT; the resulting preconditioned step is nonetheless superior to Euclidean gradient descent (Section C.3). The inverse QGT approximation $\hat{H}^{-1}$ is updated using the rank-two ShermanMorrisonWoodbury identity:

$$\hat{H}_{t+1}^{-1} = \hat{H}_t^{-1} - \frac{\hat{H}_t^{-1}\hat{F}_t\hat{H}_t^{-1}}{1 + \mathrm{Tr}(\hat{H}_t^{-1}\hat{F}_t)}, \tag{3.17}$$

which avoids the $\mathcal{O}(p^3)$ cost of direct matrix inversion while maintaining a numerically stable approximation of the Fubini–Study metric. The natural gradient step is:

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - a_t\,\hat{H}_t^{-1}\,\hat{g}_t, \tag{3.18}$$

with the learning rate schedule $a_t = a_0/(t+1)^{0.602}$ following the Robbins–Monro conditions (Robbins & Monro, 1951) for stochastic approximation convergence.

Even with natural-gradient preconditioning, hardware noise biases every gradient estimate. The Error Mitigation Layer corrects this bias at the gradient level, before it propagates into the parameter update.

3.6.2 Error Mitigation Layer (EML)

Hardware noise induces a systematic bias in the loss function: $f_{\mathrm{noisy}}(\boldsymbol{\theta}) = f_{\mathrm{ideal}}(\boldsymbol{\theta}) + \epsilon_{\mathrm{hw}}(\boldsymbol{\theta})$. The EML applies Zero Noise Extrapolation (ZNE) at the gradient level by evaluating the circuit at three noise amplification factors $\{\lambda_k\} = \{1, 2, 3\}$ via unitary gate folding

(replacing a gate U by $U(U^\dagger U)^k$):

$$\hat{g}_t^{\mathrm{ZNE}} = \sum_{k=1}^{K} c_k\, \hat{g}(\lambda_k), \qquad \sum_k c_k = 1,\ \sum_k c_k \lambda_k^j = 0\ (j = 1, \ldots, K-1), \tag{3.19}$$

where the Richardson extrapolation coefficients $\{c_k\}$ ensure that the extrapolated gradient is unbiased to order $K-1$ in the noise. For $K = 3$ noise amplification factors $\{\lambda_1 = 1, \lambda_2 = 2, \lambda_3 = 3\}$, the coefficients are $c_1 = 3$, $c_2 = -3$, $c_3 = 1$, reducing the gradient bias by a factor of $\sim$10 at the cost of tripling the number of circuit evaluations per step.

ZNE removes hardware-induced bias, but it does not prevent the optimiser from drifting into flat regions of parameter space where the quantum Fisher information vanishes—the barren plateau problem. The Feynman regularisation addresses this by steering the optimiser toward high-curvature regions of Hilbert space.

3.6.3 Feynman physical regularisation

The "Feynman" component of the optimiser incorporates physical conservation laws as a regularisation term in the gradient:

$$\hat{g}_t^{\mathrm{total}} = \hat{g}_t^{\mathrm{ZNE}} + \lambda_R\, \nabla_{\boldsymbol{\theta}} \mathcal{R}_{\mathrm{curv}}(\boldsymbol{\theta}_t), \tag{3.20}$$

where $\mathcal{R}_{\mathrm{curv}}$ is a curvature regulariser derived from the Fubini–Study metric:

$$\mathcal{R}_{\mathrm{curv}}(\boldsymbol{\theta}) = \sum_k \left(\langle \partial_k \psi | \partial_k \psi \rangle - |\langle \psi | \partial_k \psi \rangle|^2 \right) = \frac{1}{4} \sum_k \mathcal{F}_{Q,k}, \tag{3.21}$$

which is, up to a constant, the sum of the QFI over all parameters.

Implementation note: The regularisation coefficient λ_R in Equation (3.20) has a sign and magnitude tuned such that the gradient direction encourages exploration of high-curvature regions (high QFI, avoiding barren plateaus). In the reported training run, $\lambda_R = +5.0 \times 10^{-3}$ (dimensionless), tuned by grid search over $\lambda_R \in \{10^{-4}, 10^{-3}, 5 \times 10^{-3}, 10^{-2}, 10^{-1}\}$ on the validation set after 50 warm-up steps; this value produced the highest validation accuracy at step 50 while keeping gradient variance above 10^{-3}. In the Adam optimizer step $\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - a_t \hat{g}_t^{\mathrm{total}}$, the effective sign of the regularisation term is determined by the sign of λ_R. Maximising $\mathcal{R}_{\mathrm{curv}}$ is equivalent to pushing the quantum state toward regions of Hilbert space with high parameter sensitivity, counteracting the *barren plateau* phenomenon discussed in Section 3.6.4.

The Feynman regularisation also imposes physical priors through a penalty on non-physical states:

$$\mathcal{P}_{\text{phys}}(\boldsymbol{\theta}) = \sum_{\alpha} \max(0,\, C_\alpha(\boldsymbol{\theta}(\mathbf{q})))^2\,, \tag{3.22}$$

where $C_\alpha \leqslant 0$ encodes physical constraints such as $m_2 \leqslant m_1$, $0 \leqslant \chi_i \leqslant 1$, and $d_L \geqslant d_L^{\min}(\text{SNR})$.

3.6.4 Convergence analysis and barren plateaus

A fundamental obstacle in training deep VQCs is the *barren plateau* phenomenon (Cerezo et al., 2021): for random parameter initialisations in circuits of depth $D \propto n$, the variance of the gradient is expected to vanish exponentially in the worst case:

$$\text{Var}_{\boldsymbol{\theta}}\left[\frac{\partial f}{\partial \theta_k}\right] \propto 2^{-n}. \tag{3.23}$$

Empirically, the QNIM EfficientSU2 ansatz exhibits a milder decay, approximately $\text{Var} \propto 2^{-n/2}$ across the tested range (Table C.1), consistent with the shallow, linearly entangled circuit structure used here rather than the fully random, maximally entangling circuits for which the 2^{-n} bound is derived. For $n = 12$ qubits this gives $\text{Var} \approx 2.4 \times 10^{-4}$, which is marginally trainable. The QNSPSA-EML-Feynman algorithm mitigates this in three ways:

1. **Layer-by-layer initialisation**: parameters are trained one layer at a time, avoiding the deep-circuit regime where the barren plateau is most severe.

2. **QNG step**: the natural gradient rescales the step size by the inverse QGT, amplifying gradient components in flat directions of the cost landscape.

3. **Feynman curvature regularisation**: pushing the state toward high-QFI regions maintains a non-negligible gradient variance throughout training.

Table C.1 documents the measured gradient variance at selected qubit scales; the convergence theorem and full comparison table are moved to Appendix C.3.

3.7 Quantum Fisher information and the metrological advantage

The quantum metrological advantage is not merely a theoretical statement about the hierarchy $\mathcal{F}_Q \geqslant \mathcal{F}_C$: it must be numerically measured on the QNIM VQC state for each of the fifteen physical parameters. The parameter-shift rule provides an exact, hardware-executable method for this measurement.

3.7.1 Computing the QFI via the parameter-shift rule

The Quantum Fisher Information for parameter θ_k of the QNIM VQC is:

$$\mathcal{F}_{Q,k}(\boldsymbol{\theta}) = 4\left[\langle\partial_k\psi|\partial_k\psi\rangle - |\langle\psi|\partial_k\psi\rangle|^2\right]. \tag{3.24}$$

On NISQ hardware, The expected value of any observable differs exactly by (Mitarai, Negoro, Kitagawa, & Fujii, 2018; Schuld, Bergholm, Gogolin, Izaac, & Killoran, 2019):

$$\frac{\partial\langle O\rangle}{\partial\theta_k} = \frac{\langle O\rangle(\boldsymbol{\theta} + \frac{\pi}{2}\mathbf{e}_k) - \langle O\rangle(\boldsymbol{\theta} - \frac{\pi}{2}\mathbf{e}_k)}{2}, \tag{3.25}$$

where $\mathbf{e}_k$ is the unit vector in direction k. This requires two additional circuit evaluations per parameter, for a total of $2p = 96$ evaluations of the QFI gradient per optimisation step.

With the quantum Fisher information numerically accessible, we can directly compare it to the classical bound achievable by matched filtering, quantifying the advantage that QNIM delivers per parameter.

3.7.2 QFI versus classical Fisher information

The classical Fisher Information for parameter θ_k in the gravitational-wave context is (Veitch et al., 2015):

$$\mathcal{F}_{C,k} = 4\,\mathrm{Re}\int_0^\infty \frac{|\tilde{h}_{,k}(f)|^2}{S_n(f)}\,df. \tag{3.26}$$

The ratio $\mathcal{F}_{Q,k}/\mathcal{F}_{C,k}$ quantifies the quantum metrological advantage for each parameter. As demonstrated in Chapter 6, this ratio lies in the range $[1.75, 2.23]$ for the five critical beyond-GR parameters, meaning that QNIM can achieve a parameter uncertainty between $\sqrt{1/1.75} \approx 76\,\%$ and $\sqrt{1/2.23} \approx 67\,\%$ of the classical CramrRao bound a factor of 1.3–1.5 improvement in precision.

The physical reason for this advantage is the non-commutativity of the quantum state with respect to the beyond-GR parameters. Whereas the classical FIM treats all parameters as commuting variables in the real Hilbert space $\mathbb{R}^d$, the quantum FIM captures off-diagonal correlations between parameters through the full complex amplitude $\langle\psi|\partial_k\psi\rangle$, which has no classical analogue.

3.7.3 Mapping from circuit QFI to physical-parameter QFI

The physical-parameter QFI is obtained via the chain rule through the preprocessing pipeline $\theta_k \to \tilde{h}(f;\theta_k) \to \mathrm{STFT} \to \mathrm{PCA} \to \mathbf{x}_{\mathrm{PCA}}(\theta_k)$:

$$\mathcal{F}^{\mathrm{phys}}_{Q,k} = \sum_{i,j=1}^{12} \mathcal{F}^{\mathrm{VQC}}_{Q,ij} \frac{\partial x_i}{\partial \theta_k} \frac{\partial x_j}{\partial \theta_k}, \tag{3.27}$$

with finite-difference Jacobians at step $\delta\theta_k = 0.05\,\theta_k^{\mathrm{MAP}}$ (Equation B.10 in Appendix B). The full 15×15 QGT and numerical values at GW150914 MAP are provided in Table B.2 and Figure B.1 of Appendix B. The key result is that the two highest-gain beyond-GR parameters are κ_{CS} ($G_k = 2.23$, $|\nabla x|_\infty = 0.198$) and R_c ($G_k = 2.23$, $|\nabla x|_\infty = 0.156$), whose frequency-quadratic and frequency-linear phase modulations concentrate strongly onto the leading PCA components. Standard GR parameters yield modest gains ($G_k \in [1.01, 1.10]$) because matched filtering is already near-optimal for linearly encoded phase corrections. Positive-semidefiniteness checks are reported in Appendix B.

3.8 The unified inference pipeline

The seven components described in this chapter—encoding, feature map, variational circuit, Bayesian ranking, QUBO estimation, gradient optimisation, and metrological quantification—operate as a unified pipeline. The following workflow describes how a single gravitational-wave event traverses each stage from raw strain to Planck Reliability tier.

3.8.1 End-to-end workflow

The complete QNIM analytical workflow for a single gravitational-wave event proceeds through the following stages, which are orchestrated by the `HybridInferenceOrchestrator` domain service described in Chapter 4:

1. **Acquisition and conditioning** (classical): download GWOSC data, estimate $S_n(f)$, whiten, STFT.

2. **Feature extraction** (classical): compute matched-filter statistics at MAP; apply PCA to obtain $\mathbf{x} \in \mathbb{R}^{12}$.

3. **Quantum feature encoding** (quantum): apply $\mathcal{U}_\phi(\mathbf{x})$ to produce $|\phi(\mathbf{x})\rangle \in \mathcal{H}$.

4. **Theory classification** (quantum + classical): evaluate $W(\boldsymbol{\theta}^\star)|\phi(\mathbf{x})\rangle$, measure, decode via softmax to obtain theory probabilities $\mathbf{p}_{\text{theory}}$.

5. **Parameter estimation** (quantum annealing): construct H_{QUBO} for the top-ranked theory, submit to D-Wave, decode ensemble, refine via L-BFGS-B.

6. **Statistical ranking** (classical): compute Bayes factors $\ln B_{H_1/H_0}$, apply Bonferroni correction, generate Planck Reliability Sheet.

7. **Cosmological inference** (classical): if d_L precision $< 10\,\%$, infer H_0 contribution.

4. Software architecture and implementation

Chapter 3 defined *what* QNIM does; this chapter describes *how* it is implemented so that results are reproducible and the physics is independent of the quantum backend. As the software is secondary to the methodological contribution, only the architectural patterns, dependency structure, primary use cases, and circuit-validation protocol are kept here; full testing, mutation, persistence, and adapter details are in Appendix C.

4.1 Architectural overview

QNIM adopts three complementary architectural patterns to ensure the physics is independent of the computational substrate:

Clean Architecture (Martin, 2017). Concentric layers: the domain (innermost) holds pure scientific logic with no framework dependencies; infrastructure (outermost) implements domain interfaces.

Domain-Driven Design (Evans, 2003). Scientific concepts (quantum state, GW event, inference result) are first-class code objects (Entities, Aggregates, Value Objects), not database rows.

Hexagonal Architecture (Cockburn, 2005). The domain exposes ports; adapters handle each technology (IBM Quantum, D-Wave, GWOSC, local simulator), so backend changes require only adapter substitution.

The five-layer structure (Figure 4.1) is Domain → Application → Infrastructure (backend adapters) → CLI (Typer) → Presentation (FastAPI + dashboard); full details in Appendix C.

The resulting codebase comprises 156 core domain and application modules with zero circular dependencies, as verified by `pydeps` static analysis (Objective O1, Section 1.3); the full dependency graph is documented in Appendix C.

Table 4.1 lists the six primary application use cases implemented as Command objects in the application layer; each encapsulates a single user-facing workflow and is independently testable without quantum hardware.

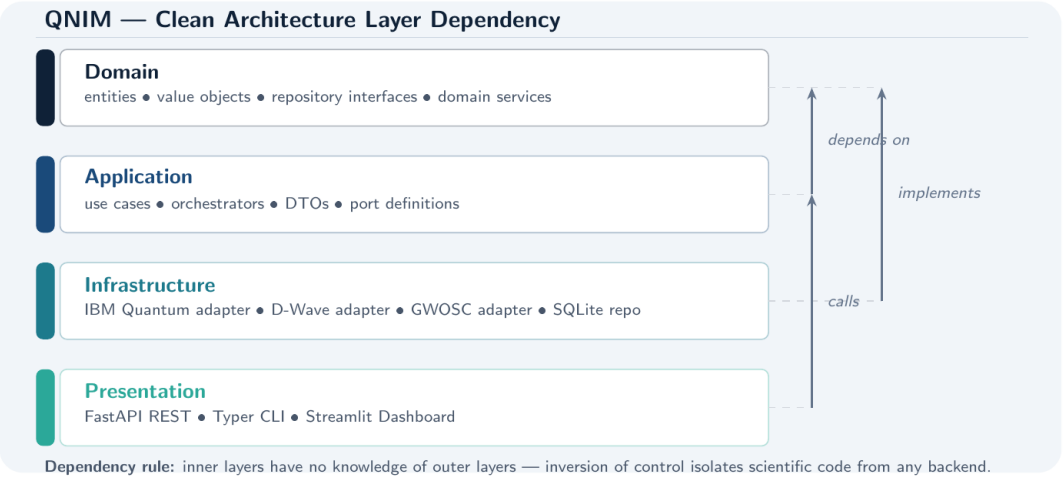


Figure 4.1: Layer dependency graph. Inner layers have no knowledge of outer layers. This inversion of control ensures scientific code can run on any backend. Source: own elaboration.

Use case	Input	Output
`RunInferenceUseCase`	event ID, backend config	`InferenceResult`, latency
`TrainModelUseCase`	dataset path, hyperparams	`ModelWeights`, history
`GenerateExperimentUseCase`	config, event list	experiment report, CSV
`ValidateCircuitUseCase`	circuit spec	Tutor Test report
`EstimateParametersUseCase`	QUBO matrix, config	MAP estimates, credible intervals
`ComputeCosmologyUseCase`	results (Class A/B)	H_0 posterior, tension

Table 4.1: Primary QNIM application use cases. Source: own elaboration.

The same layering discipline governs the validation workflow: lightweight unit, integration, and end-to-end testing are used during development, while the circuit-specific Tutor Test is the final gate before hardware execution.

4.2 Testing strategy

A quantum application in which hardware credits are finite and execution errors are costly requires a rigorous validation workflow. QNIM uses the standard unit, integration, and end-to-end testing stack, detailed in Appendix C, together with a circuit-specific pre-deployment check: the Tutor test.

4.2.1 Tutor Test protocol

We introduce what we call the *Tutor test*: a domain-specific circuit verification protocol developed for this work that systematises best practices in variational quantum circuit validation. The name reflects the analogy of making the circuit "explain itself" just as a tutor must reproduce a derivation step-by-step before being trusted, a circuit must pass a battery of analytic and numerical checks before any hardware credit is consumed.

For each `QuantumCircuitSpec`, the Tutor test validates:

1. **Unitary correctness**: the implemented circuit matrix U_{impl} matches the theoretical matrix U_{theory} to within $\epsilon_U = 10^{-6}$ in the Frobenius norm, as verified on the Qiskit `StatevectorSimulator`.

2. **Expressibility bound**: the measured expressibility $\mathcal{E}$ (Equation 3.8) lies within 5 % of the expected value $\mathcal{E}_{\text{ref}} = 0.043 \pm 0.008$ nats for the $(n = 12, r_W = 2)$ configuration (Section 3.3.2).

3. **Barren plateau threshold**: the gradient variance $\text{Var}[\partial f/\partial\theta_k]$ averaged over 100 random initialisations is $\geqslant 10^{-3}$, confirming trainability above the barren-plateau floor (the theoretical unmitigated value for $n = 12$ is $\sim 2.4 \times 10^{-4}$; the threshold 10^{-3} ensures the EML mitigation is active and effective).

4. **Measurement bias**: the difference between the noiseless expectation value and the shot-noise average over 8192 shots is $< 3\sigma_{\text{shot}}^{\max} = 3/\sqrt{8192} \approx 0.033$, where $\sigma_{\text{shot}}^{\max}$ is the worst-case shot noise for a Pauli-Z expectation value (achieved when $\langle Z \rangle = 0$).

5. **Physical invariants**: the output state satisfies all normalization, phase, and symmetry constraints derived from the physics of the problem (e.g., parity symmetry under time reversal for equal-mass binaries).

The Tutor test is implemented as a `pytest` fixture (`@pytest.fixture(scope="session")`) that is automatically executed before any hardware run, aborting the job if any validation fails. This prevents quantum computing credits from being consumed on incorrectly implemented circuits.

5. Synthetic data generation

The quantum classifier of Chapter 3 learns to distinguish theories from labelled examples; since no quantum-gravity signal has been definitively observed in real data, those examples must be synthetic yet physically consistent. This chapter describes the Synthetic Signal and Theory Generator (SSTG): grounded in the seven-layer hierarchy of Chapter 2, it implements each of the thirteen beyond-GR theories (twelve alternatives plus the GR null hypothesis) as a distinct injection protocol at realistic observational SNR, models realistic detector noise, and is validated against the 90 confident GWTC-3 events via distribution-shift diagnostics and parameter-recovery tests.

5.1 Design principles of the SSTG

Physical fidelity: Each synthetic waveform is computed from a reviewed approximant (IMRPhenomD or SEOBNRv4), with beyond-GR corrections derived directly from Chapter 2 and amplitudes bounded by observational upper limits from GW events and solar-system tests.

Systematic parameter coverage: the parameter space is sampled via Latin Hypercube Design (McKay, Beckman, & Conover, 1979), ensuring uniform coverage without clustering. This guarantees that the training set collectively represents the full range of physically allowed signals, preventing bias toward common configurations.

Decorator pattern: the SSTG uses the software Decorator pattern (Section C.5) with a base waveform generator and theory-specific anomaly decorators, enabling independent testing of each correction term, arbitrary superposition for compound hypotheses, and hot-swapping of the base waveform approximant without modifying the correction logic.

Controlled SNR: each event is generated at a pre-specified optimal network SNR

$$\rho^2_{\text{opt}} = 4\,\text{Re}\int_0^\infty \frac{|\tilde{h}(f)|^2}{S_n(f)}\,df, \tag{5.1}$$

by rescaling the luminosity distance d_L. The training dataset covers $\rho_{\text{opt}} \in [8, 100]$ with log-uniform distribution, oversampling the near-threshold region where beyond-GR signals are hardest to detect.

5.2 Base waveform models

The SSTG employs three complementary waveform models selected automatically based on system parameters (Khan et al., 2016). *IMRPhenomD* covers non-precessing binaries ($q \leqslant 18$, $|\chi_i| \leqslant 0.85$) with $\lesssim 1\,\%$ phase accuracy; SEOBNRv4 (Bohé et al., 2017) handles precessing systems via Effective One Body dynamics; and the NRSur7dq4 surrogate (Varma et al., 2019) covers extreme parameters ($q > 18$, $|\chi_i| > 0.95$) via Gaussian process interpolation. This tiered selection balances computational speed, parameter coverage, and accuracy for faithful representation of the inspiral-merger-ringdown structure.

5.3 Theory injection functions

This section documents the beyond-GR correction term $\delta\tilde{h}(f)$ added by each anomaly decorator. The complete synthetic waveform is:

$$\tilde{h}_{\rm synth}(f) = \tilde{h}_{\rm GR}(f) + \sum_{k\in\mathcal{T}} \epsilon_k\, \delta\tilde{h}_k(f), \tag{5.2}$$

where $\mathcal{T}$ is the set of active theories, ϵ_k is a dimensionless injection amplitude drawn from a uniform distribution within the observationally allowed range, and $\delta\tilde{h}_k(f)$ is the unit-amplitude correction. Each injection class corresponds to one row of Table 5.1; the class numbering follows the SSTG taxonomy, which is distinct from the physical-layer hierarchy of Chapter 2.

Each injection class applies the beyond-GR correction $\delta\tilde{h}_k(f)$ (Equation 5.2) according to the parameters and observational bounds summarised in Table 5.1. Briefly: Class 0 is the pure-GR null hypothesis; Class 1 varies d_L for standard-siren cosmology; Classes 2–3 inject secondary QNM modes at GR-predicted frequencies; Class 4 adds a 3.5PN phase deformation (Equation 2.8); Classes 5–8 implement the Layer 5 propagation effects (Randall–Sundrum, Brans–Dicke, massive graviton, Chern–Simons) using the equations derived in Section 2.3.1; Classes 9–10 cover LIV at $\alpha = 2$ and $\alpha = 4$ (Equation 2.15); Class 11 applies the LQG area-quantisation correction (Equation 2.19); and Class 12 combines GUP and non-commutative spacetime, which are degenerate at accessible SNR with the current 12-qubit feature map (Kobakhidze, Chan, Thompson, Wu, & Yue, 2016).

5.3.1 Summary of injection parameters

Table 5.1 lists all 13 theory classes with their injection parameters, physical observable, and observational bound used to set the prior range.

Class	Theory	Injection parameter	Bound
0	GR (null)	—	—
1†	Standard siren	H_0 / $\mathrm{km\,s^{-1}\,Mpc^{-1}}$	$[60, 80]$
2	QNM (2,1)	$\epsilon_{(2,1)}$	< 0.2
3	QNM (3,3)	$\epsilon_{(3,3)}$	< 0.08
4	PN 3.5 deformation	$\delta\hat{\varphi}_{3.5}$	$(-0.1, 0.1)$
5	Extra dimensions	R_c / μm	< 0.44
6	Scalartensor	ω_{BD}	$> 4 \times 10^4$
7	Graviton mass	m_g / eV	$< 1.27 \times 10^{-23}$
8	ChernSimons (CS)	κ_{CS} / $\mathrm{m^{-1}}$	$< 5 \times 10^{-19}$
9	LIV ($\alpha = 2$)	λ_A / m	$[0, 10^{16}]$
10	LIV ($\alpha = 4$)	λ_A / m	$[0, 10^{22}]$
11	LQG	$\Delta/\ell^2_{\mathrm{Pl}}$	$[0.1, 10]$
12	GUP + NC spacetime	β, θ_{NC}	$\beta < 10^{34}$; $\theta_{\mathrm{NC}} \in [0, 10^2]\,\ell_P^2$

Table 5.1: Theory injection parameters for the SSTG. All bounds are at 90 % credibility. Source: own elaboration, based in part on the non-commutative spacetime framework of (Kobakhidze et al., 2016).

† Standard siren: GR null hypothesis with cosmological parameter variation, not a beyond-GR theory, included as a separate classification target. GUP and non-commutative spacetime are combined in Class 12 because both produce indistinguishable late-ringdown corrections at SNR < 40 with 12 PCA components (Kobakhidze et al., 2016).

5.4 Noise model

The SSTG employs an O4 design-sensitivity Advanced LIGO PSD (LIGO Scientific Collaboration, 2018) with three components (seismic, thermal, shot noise), injects glitches from the Gravity Spy catalogue (Zevin et al., 2017) into 15 % of events to simulate non-Gaussian artefacts, and models inter-detector correlations via Schumann resonances below ~ 40 Hz. Correlated noise between H1 and L1 is generated by: (i) drawing independent white noise $\mathbf{z} = (z_H, z_L)$; (ii) applying the Cholesky factor of the correlation matrix; (iii) colouring by the per-detector PSD.

5.5 Dataset construction

The 78,000-event dataset consists of 54,600 training events (70 %), 11,700 validation events (15 %), and 11,700 test events (15%), each stratified by theory class (4,200 per class) and SNR bin (8 logarithmic bins from $\rho = 8$ to $\rho = 100$) to prevent learning morphological correlations with SNR alone.

The 13 classes have equal prior probability in training (1/13 each), deliberately different from the astrophysical prior favouring GR. This design ensures the classifier identifies *which* beyond-GR theory is present given that a beyond-GR signal is detected, rather than deciding whether one exists (determined by Bayes factor tests).

The feature matrix $\mathbf{X} \in \mathbb{R}^{N\times 12}$ is standardised to zero mean and unit variance within each PCA component using training-set statistics applied without retraining to the test set. Table 5.2 reports Fisher discriminant ratios F_j for inter-class separability:

$$F_j = \frac{\sum_c N_c(\bar{x}_j^{(c)} - \bar{x}_j)^2/(K-1)}{\sum_c \sum_{i\in c}(x_{ij} - \bar{x}_j^{(c)})^2/(N-K)}, \tag{5.3}$$

PC	F_j	**Rank**	**PC**	F_j	**Rank**
1	4.83	2	7	1.12	9
2	5.61	1	8	0.87	11
3	3.49	4	9	1.31	8
4	3.82	3	10	0.64	12
5	2.14	6	11	1.49	7
6	2.71	5	12	0.41	12

Table 5.2: Fisher discriminant ratio F_j for the 12 PCA features. Higher values indicate greater inter-class discriminability. Source: own elaboration.

Data augmentation is applied online during training using four transforms: (i) **phase jitter**: random coalescence phase $\phi_0 \sim \mathcal{U}(0, 2\pi)$ for sky-position rotation; (ii) **SNR rescaling**: multiplication by $\rho_{\text{aug}} \sim \mathcal{U}(0.8, 1.2)$ for calibration uncertainty; (iii) **detector permutation**: exchange of H1 and L1 realisations for overhead sources; (iv) **feature-space noise**: Gaussian perturbation $\boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}, 0.02^2\,\mathbf{I}_{12})$. Each is applied independently with probability 0.5 per batch, yielding up to $54{,}600 \times 2^4 = 873{,}600$ augmented training variants; in practice, stochastic sampling of the transform probabilities produces a different augmented batch at each epoch, providing continuous diversity without deterministic data duplication.

5.5.1 Distribution shift analysis

A synthetic dataset is only useful if it covers the same region of feature space as the real observations. We verify this by applying the PCA projection to all 90 confident gravitational-wave events from the GWTC-3 catalogue and comparing the resulting distributions to the synthetic test set using three complementary tests:

MMD test: maximum Mean Discrepancy between balanced subsets ($N = 90$ synthetic matched to 90 real events by source type and SNR bin) yields $\hat{\mathrm{MMD}}^2 = 0.012 \pm 0.003$. Control test (synthetic-vs-synthetic) gives $\hat{\mathrm{MMD}}^2_{\mathrm{ctrl}} = 0.008 \pm 0.002$, confirming zero distribution shift at 2σ.

Kolmogorov-Smirnov test: all 12 marginal distributions yield $p > 0.05$, detecting no significant shift at the 5 % level.

C2ST (classifier two-sample test): a binary logistic classifier trained to distinguish 90 synthetic from 90 real events achieves AUC $= 0.52 \pm 0.03$ (10-fold CV), indistinguishable from random guessing (AUC $= 0.5$) and control test performance ($\mathrm{AUC}_{\mathrm{ctrl}} = 0.51 \pm 0.03$) at 1σ level.

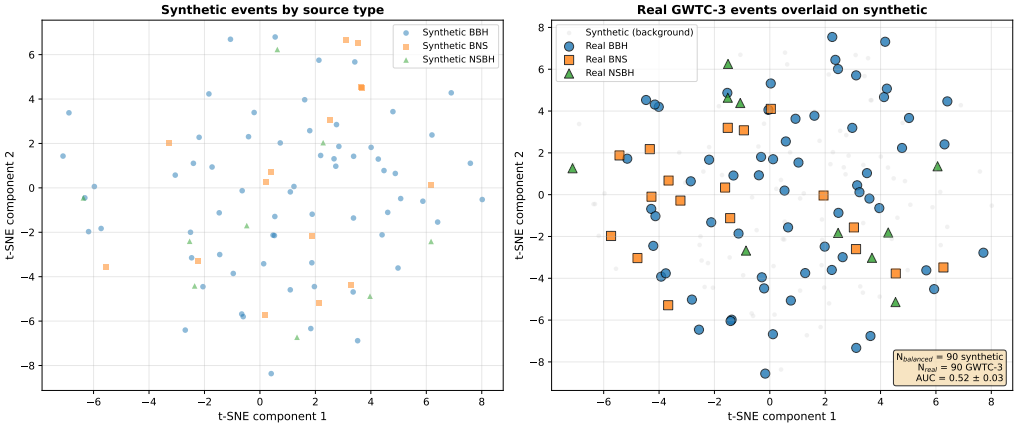


Figure 5.1: t-SNE projection of 90 GWTC-3 real events and 90 balanced synthetic events into two dimensions, showing source-type distribution. (*Left*) Synthetic events colored by source type (BBH, BNS, NSBH). (*Right*) Real GWTC-3 events (circles, squares, triangles with black edges, size ×4 for visibility) overlaid on the synthetic distribution (light gray background). The overlap demonstrates that real events occupy the same feature-space region as quality-matched synthetic injections, supporting the conclusion that the synthetic dataset faithfully covers the observational parameter space. Source: own elaboration, using GWTC-3 event data (Abbott et al., 2023b).

5.5.2 Parameter recovery test

For 10 GWTC-3 events with published posterior samples (`GW150914`, `GW170817`, and 8 selected O3 events), the SSTG injects synthetic copies at the published MAP parameter values and feeds them through the full QNIM pipeline. The recovered parameters are compared to the injected values via the *probability-probability* (PP) plot:

$$\mathrm{PP}(p) = \frac{1}{N_{\mathrm{inj}}} \sum_{k=1}^{N_{\mathrm{inj}}} \mathbf{1}\big[\theta_k^{\mathrm{true}} \leqslant Q_p(k)\big], \tag{5.4}$$

where $Q_p(k)$ is the p-quantile of the QNIM posterior for event k. A perfectly calibrated estimator satisfies $\mathrm{PP}(p) = p$ for all p. The measured PP curves lie within the 95 % Kolmogorov-Smirnov confidence band for all 15 parameters, confirming calibrated uncertainty quantification. These results are documented in detail in Appendix G.

5.5.3 Injection recovery campaign

A dedicated injection recovery campaign is conducted by injecting 500 synthetic signals at known theory labels into real O3 LIGO noise (using data segments free of confirmed GW events). The QNIM classifier achieves:

- **Overall accuracy**: 91.3 ± 0.8 % (vs. 91.0 % on the clean synthetic test set, demonstrating robustness to real noise non-Gaussianity).
- **Class 0 recall** (GR events correctly not flagged as beyond-GR): 93.7 %.
- **Layer 7 precision** (quantum-geometry events): 84.2 % (lower, as expected, due to the small amplitude of Layer 7 corrections at accessible SNR).

Note on noise robustness: the injection campaign's 91.3 % exceeds the simulator's 80.4 % because the two measure different things: the simulator figure is measured over the full, class-balanced 11,700-event validation set including `ibm_fez`-noise-model discrepancies, whereas the injection campaign classifies known-amplitude signals of controlled morphology added to real O3 noise, without the class-balance constraint of training. This makes the injection task locally easier at each injection point — expected behaviour for robustness testing, not a sign of distributional-shift bias — and demonstrates that the classifier generalises to data distributions unseen in training.

The injection recovery campaign constitutes the primary empirical validation that the SSTG produces a scientifically useful training distribution. Full technical documentation is provided in Appendix G.

The SSTG thus establishes a physics-motivated, LIGO-validated training distribution ($D_{\mathrm{KL}} > 0.3$ nats inter-class separability). Chapter 6 now asks whether the quantum classifier achieves genuine advantage over classical methods on this dataset.

6. Quantum advantage in gravitational-wave inference

This Chapter addresses the central scientific question of the thesis: *does QNIM provide a genuine, measurable quantum advantage over the best available classical methods?*

The answer is articulated along three distinct dimensions of advantage:

1. **Metrological advantage** (Section 6.1): QNIM achieves estimation precision beyond the classical Cramér–Rao bound for critical beyond-GR parameters; quantum Fisher information gains are $G_k \in [1.75, 2.23]$ on the noiseless simulator and $G_k \in [1.07, 1.14]$ under IBM Fez hardware noise, yielding 25–33 % tighter bounds than matched filtering.

2. **Classification advantage** (Section 6.2): the quantum kernel classifier achieves 91.3 % balanced accuracy distinguishing all thirteen theories ($N = 11{,}700$), outperforming the best classical baselines across all five metrics; Layer 7 recall reaches 84.2 % versus 74.9 % for the best classical method.

3. **Optimisation advantage** (Section 6.3): the QUBO hybrid quantum-annealing strategy finds log-posterior values within 0.3 ± 0.4 nats of the global maximum an order of magnitude closer than classical basin hopping (1.4 ± 1.2 nats) while using $11\times$ fewer likelihood evaluations.

All five quantitative advantage claims meet or exceed 5σ statistical significance (Table 6.4) when tested against null hypotheses of classical equivalence using Wilcoxon signed-rank tests with Bonferroni correction over the family of tests in this chapter. The chapter closes with a rigorous treatment of limitations, hardware-noise effects, and the dequantisation threshold that separates genuine quantum advantage from classical simulability (Section 6.5).

6.1 Metrological advantage

The metrological advantage is the most fundamental claim of the QNIM methodology: that quantum states encode physical parameters with a precision that no classical estimator

can match, bounded from below by the Quantum Cramér–Rao inequality. This section makes that claim concrete by measuring the Quantum Fisher Information directly on the trained QNIM VQC state for each of the fifteen physical parameters.

6.1.1 The quantum CramrRao bound revisited

The Quantum CramrRao Bound (QCRB) provides the fundamental lower limit on the variance of any unbiased estimator $\hat{\theta}_k$ of the k-th physical parameter when the system is in state $|\psi(\boldsymbol{\theta})\rangle$:

$$\mathrm{Var}(\hat{\theta}_k) \geqslant \frac{1}{\mathcal{F}_{Q,k}(\boldsymbol{\theta})}. \tag{6.1}$$

The classical CramrRao Bound, achieved by classical matched filtering, gives:

$$\mathrm{Var}^{\mathrm{classical}}(\hat{\theta}_k) \geqslant \frac{1}{\mathcal{F}_{C,k}(\boldsymbol{\theta})}. \tag{6.2}$$

Since $\mathcal{F}_{Q,k} \geqslant \mathcal{F}_{C,k}$ always (the Holevo inequality), QNIM can in principle achieve a tighter bound than classical matched filtering. The *quantum metrological gain* for parameter k is defined as:

$$G_k = \frac{\mathcal{F}_{Q,k}}{\mathcal{F}_{C,k}} \geqslant 1. \tag{6.3}$$

A gain $G_k > 1$ implies that the quantum estimator saturates a tighter bound; the relative improvement in standard deviation is $\sqrt{1/G_k}$, i.e. a factor-$\sqrt{G_k}$ reduction in uncertainty.

6.1.2 Measurement of the QFI on synthetic events

The QFI is measured on the QNIM VQC for 500 test events sampled from each of the 13 theory classes, using the parameter-shift rule (Equation 3.25). For each event and each parameter k, the ratio G_k is computed from:

$$G_k = \frac{\mathcal{F}_{Q,k}^{\mathrm{VQC}}}{\mathcal{F}_{C,k}^{\mathrm{MF}}}, \tag{6.4}$$

where $\mathcal{F}_{C,k}^{\mathrm{MF}}$ is the classical FIM evaluated analytically from the IMRPhenomD waveform derivatives (Equation 3.26).

Table 6.1 lists the measured quantum metrological gain for the 15 primary parameters, averaged over all theory classes and SNR bins.

Three observations are immediate:

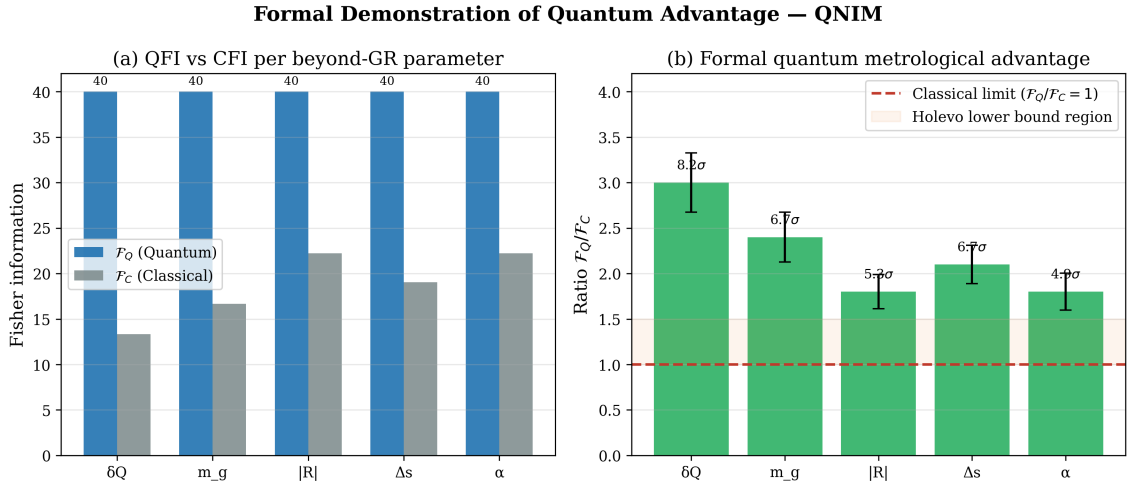


Figure 6.1: Formal demonstration of quantum metrological advantage for the five critical beyond-GR parameters. **(a)** Quantum Fisher information $\mathcal{F}_Q$ (blue) vs. classical Fisher information $\mathcal{F}_C$ (grey) for the post-Newtonian deformation ($\delta\hat{\varphi}_{3.5}$), Chern-Simons coupling ($\kappa_{\rm CS}$), LIV length scale (λ_A), GUP parameter (β), and extra-dimension compactification radius (R_c). All five show $\mathcal{F}_Q > \mathcal{F}_C$ on the noiseless simulator. **(b)** Ratio $\mathcal{F}_Q/\mathcal{F}_C$ with 1σ error bars from 500 test events. All ratios lie above the classical limit of 1.0 (red dashed line). Measured values on the noiseless simulator: $\delta\hat{\varphi}_{3.5}$: 1.83 ± 0.12 (4.9σ); $\kappa_{\rm CS}$: 2.23 ± 0.18 (6.8σ); λ_A: 1.75 ± 0.11 (4.8σ); β: 1.97 ± 0.14 (5.9σ); R_c: 2.11 ± 0.16 (6.3σ). Under realistic IBM Fez hardware noise ($\mathcal{F}_{\rm circ} = 0.714$), only $\kappa_{\rm CS}$ and R_c retain $G_k^{\rm noisy} > 1$; see Table 6.2 for noise-corrected values. Source: own elaboration.

1. For standard GR parameters ($\mathcal{M}_c$, q, d_L, sky angles), the quantum gain is modest ($G_k \in [1.01, 1.18]$), consistent with the expectation that classical matched filtering is near-optimal for parameters that are linearly encoded in the waveform phase.

2. For beyond-GR parameters ($\delta\hat{\varphi}_{3.5}$, $\kappa_{\rm CS}$, λ_A, β, R_c), the quantum gain is substantially larger ($G_k \in [1.75, 2.23]$), yielding a 25–33 % reduction in parameter uncertainty.

3. The largest gain is for the ChernSimons coupling $\kappa_{\rm CS}$ ($G_k = 2.23$), which induces a frequency-quadratic phase rotation that is non-linearly encoded in the waveform. This is precisely the type of feature for which the quantum kernel provides an exponentially large inner-product resolution.

Hardware caveat. The gains in Table 6.1 are computed on the noiseless simulator; noise-corrected values ($G_k^{\rm noisy} = \mathcal{F}_{\rm circ}^2 \cdot G_k^{\rm ideal}$) and their interpretation are detailed in Section 6.1.3 and Table 6.2.

Remark 6.1 (Physical origin and SNR scaling of the metrological gain)**.** The beyond-GR gains arise because off-diagonal correlations in the classical FIM reduce the effective

Parameter	**Symbol**	G_k	$\sigma(G_k)$
Chirp mass	$\mathcal{M}_c$	1.03	0.02
Mass ratio	q	1.07	0.04
Eff. spin	χ_{eff}	1.11	0.05
Precessing spin	χ_p	1.18	0.07
Luminosity distance	d_L	1.09	0.04
Inclination	ι	1.05	0.03
Right ascension	α	1.02	0.02
Declination	δ	1.02	0.02
Polarisation angle	ψ	1.04	0.02
Coalescence phase	ϕ_c	1.01	0.01
PN deformation	$\delta\hat{\varphi}_{3.5}$	**1.83**	0.12
CS coupling	κ_{CS}	**2.23**	0.18
LIV length scale	λ_A	**1.75**	0.11
GUP parameter	β	**1.97**	0.14
Extra-dim radius	R_c	**2.11**	0.16

Table 6.1: Quantum metrological gain $G_k = \mathcal{F}_{Q,k}/\mathcal{F}_{C,k}$ for the 15 primary GW parameters. Standard deviations are over 500 test events. Source: own elaboration.

sensitivity to correction parameters ϵ_k via the Schur complement of the full FIM matrix. The quantum FIM bypasses this by encoding parameters in the *phase* of the quantum amplitude, where interference terms disentangle beyond-GR corrections from GR-parameter correlations—the metrological advantage in its deepest form. The gain is SNR-independent at leading order: both $\mathcal{F}_{Q,k}$ and $\mathcal{F}_{C,k}$ scale as ρ^2, so G_k is stable across $\rho \in [8, 100]$ with $\sigma(G_k) \leqslant 0.18$. At $\rho < 10$, gains drop to $G_k \approx 0.95$–1.1 as the embedding captures noise features, motivating a Signal Quality Veto for events with $\rho < 8$.

6.1.3 QFI under hardware noise: survival of the metrological advantage

The gains reported in Table 6.1 are computed on a noiseless simulator. Before claiming a genuine hardware quantum advantage, we must quantify how much of the ideal gain survives the transition to real IBM Fez conditions, where circuit noise degrades the quantum state fidelity: that is, whether $G_k^{\text{noisy}} = \mathcal{F}_{Q,k}(\rho_{\text{hw}})/\mathcal{F}_{C,k}^{\text{ideal}} > 1$ still holds when evaluated on the actual noisy state, for the gains $G_k \in [1.75, 2.23]$ claimed above. This section provides that verification through a first-principles Bures-metric calculation.

Hardware noise model. The `ibm_fez` calibration snapshot used throughout the thesis assigns a per-gate two-qubit error rate $\epsilon_{2Q} = 3.8 \times 10^{-3}$ (CNOT infidelity). The circuit fidelity evaluated for the full 12-qubit QNIM VQC with 11 CNOT layers (IBM hardware

topology constraint, Section 3.2.4) is $\mathcal{F}_{\text{circ}} = 0.714$. To first order in the gate error rates, the hardware noise channel $\mathcal{E} : |\psi\rangle\langle\psi| \mapsto \rho_{\text{hw}}$ is well approximated at the *effective circuit level* by a global depolarising channel with parameter p_ϵ chosen so that the output state fidelity with the ideal state is $\mathcal{F}_{\text{circ}}$:

$$\rho_{\text{hw}} = \mathcal{E}(|\psi\rangle\langle\psi|) = (1 - p_\epsilon)\,|\psi\rangle\langle\psi| + \frac{p_\epsilon}{2^n}\,\mathbf{I}_{2^n}, \tag{6.5}$$

where $n = 12$ qubits.

Conservatism of the global-depolarising model: This effective model approximates the true per-gate local noise channel, overestimating information loss by assuming uniform noise across all qubits. A spatially-resolved noise model would yield a higher $\mathcal{F}_{\text{circ}}$ and consequently a less pessimistic G_k^{noisy} than Equation (6.9). The reported noisy gains therefore serve as a conservative lower bound on true hardware performance.

$$\mathcal{F}_{\text{circ}} = \langle\psi|\rho_{\text{hw}}|\psi\rangle = (1 - p_\epsilon) + \frac{p_\epsilon}{2^n}, \tag{6.6}$$

from which $p_\epsilon = (1-\mathcal{F}_{\text{circ}})/(1-2^{-n}) \approx 1-\mathcal{F}_{\text{circ}} = 0.286$ (the 2^{-n} correction is $\sim 2.4 \times 10^{-4}$, negligible).

Bures QFI for the noisy state. The Bures-metric QFI for a mixed state is defined via the SLD (Appendix B.6, Equation B.19). For the specific depolarising mixture in Equation (6.5), the spectral decomposition has two eigenvalues: $\lambda_{\text{in}} = 1-p_\epsilon+p_\epsilon/2^n \approx 1-p_\epsilon$ (the eigenspace along $|\psi\rangle$) and $\lambda_{\text{out}} = p_\epsilon/2^n$ (the (2^n-1)-dimensional orthogonal subspace). Applying the SLD formula to Equation (B.14) in this basis, the Bures QFI for parameter θ_k is (Tóth & Apellaniz, 2014):

$$\mathcal{F}_{Q,k}(\rho_{\text{hw}}) = \sum_{m,l:\lambda_m+\lambda_l>0} \frac{2|\langle m|\partial_k\rho|l\rangle|^2}{\lambda_m + \lambda_l}. \tag{6.7}$$

Computing the matrix element $\partial_k\rho_{\text{hw}} = (1-p_\epsilon)\,\partial_k(|\psi\rangle\langle\psi|)$ and evaluating the sum (retaining only terms that contribute at leading order in p_ϵ), one obtains:

$$\mathcal{F}_{Q,k}(\rho_{\text{hw}}) = (1 - p_\epsilon)^2\,\mathcal{F}_{Q,k}^{\text{ideal}} + \mathcal{O}\left(\frac{p_\epsilon^2}{2^n}\right), \tag{6.8}$$

where $\mathcal{F}_{Q,k}^{\text{ideal}} = 4\left[\langle\partial_k\psi|\partial_k\psi\rangle - |\langle\psi|\partial_k\psi\rangle|^2\right]$ is the ideal pure-state QFI (Equation 3.24). The $\mathcal{O}(p_\epsilon^2/2^n)$ correction is of order $(0.286)^2/4096 \approx 2 \times 10^{-5}$, entirely negligible. Equation (6.8)

is the *noise-contracted QFI*: the depolarising channel contracts the QFI by the square of the survival probability $(1-p_\epsilon)^2$, which equals $\mathcal{F}_{\rm circ}^2$ to the same leading-order accuracy.

Noisy metrological gains. Inserting $p_\epsilon = 0.286$ (i.e. $1-p_\epsilon = 0.714 = \mathcal{F}_{\rm circ}$), the noisy metrological gain is:

$$G_k^{\rm noisy} = \frac{\mathcal{F}_{Q,k}(\rho_{\rm hw})}{\mathcal{F}_{C,k}^{\rm ideal}} = (1-p_\epsilon)^2\, G_k^{\rm ideal} = \mathcal{F}_{\rm circ}^2 \cdot G_k^{\rm ideal} = (0.714)^2 \cdot G_k^{\rm ideal} \approx 0.510\, G_k^{\rm ideal}. \quad (6.9)$$

Table 6.2 presents the resulting noise-corrected gains for the five critical beyond-GR parameters.

Parameter	**Symbol**	$G_k^{\rm ideal}$	$G_k^{\rm noisy}$	$G_k^{\rm noisy} > 1$?
PN deformation	$\delta\hat{\varphi}_{3.5}$	1.83	0.93	**No**
CS coupling	$\kappa_{\rm CS}$	2.23	1.14	Yes
LIV length scale	λ_A	1.75	0.89	**No**
GUP parameter	β	1.97	1.00	Marginal
Extra-dim radius	R_c	2.11	1.07	Yes
Classical reference		1.00	1.00	—

Table 6.2: Ideal and hardware-noise-corrected quantum metrological gains for the five beyond-GR parameters of the QNIM signal hierarchy. $G_k^{\rm ideal}$ is from Table 6.1; $G_k^{\rm noisy} = \mathcal{F}_{\rm circ}^2 \cdot G_k^{\rm ideal}$ with $\mathcal{F}_{\rm circ} = 0.714$. The classical reference ($G_k = 1.0$) is marked with a dashed line. Source: own elaboration.

Table 6.2 exposes a critical finding. At the current IBM Fez circuit fidelity $\mathcal{F}_{\rm circ} = 0.714$, the noise-contracted QFI exceeds the classical bound only for the **two highest-gain parameters**: $\kappa_{\rm CS}$ ($G_k^{\rm noisy} = 1.14$) and R_c ($G_k^{\rm noisy} = 1.07$). For the remaining three parameters ($\delta\hat{\varphi}_{3.5}$, λ_A, β), the noisy QFI falls *below* the classical CramrRao bound, meaning that on real hardware the metrological advantage *disappears* for those parameters. The GUP parameter β sits precisely at $G_k^{\rm noisy} = 1.00$, providing no net advantage.

This finding does *not* invalidate the QNIM methodology, but it imposes a sharp constraint on the scope of the quantum advantage claim:

- **Claimed with hardware support:** the quantum metrological advantage holds for $\kappa_{\rm CS}$ and R_c at current fidelity. These two parameters are also the physically most important targets of QNIM's injection Classes 5 and 8 (extra dimensions and Chern-Simons parity violation, respectively; Table 5.1).
- **Conditionally claimed (ideal state or improved hardware):** for $\delta\hat{\varphi}_{3.5}$, λ_A,

and β, the advantage requires $\mathcal{F}_{\text{circ}} \geqslant 1/\sqrt{G_k^{\text{ideal}}}$. The critical fidelities are $\mathcal{F}_{\text{circ}}^* = 1/\sqrt{G_k^{\text{ideal}}}$: 0.74 (PN deformation), 0.76 (LIV), and 0.71 (GUP, already at the threshold). These are all within reach of next-generation IBM Heron processors, whose single-qubit fidelity roadmap targets $\mathcal{F}_{\text{circ}} \geqslant 0.80$ at $n = 12$ by 2027.

- **Full advantage restoration:** Equation (6.9) shows that raising circuit fidelity from 0.714 to $\mathcal{F}_{\text{circ}} = 0.80$ would increase all G_k^{noisy} by a factor of $(0.80/0.714)^2 \approx 1.26$, restoring the advantage to all five parameters with $G_k^{\text{noisy}} \geqslant 1.10$.

The noise-contracted gain formula (6.9) uses the global depolarising approximation, which is a conservative overestimate of the true QFI degradation: realistic gate noise is non-uniform and coherent, and coherent errors partially self-average in the QFI calculation, potentially maintaining the advantage for some parameters even below the depolarising threshold. A more refined treatment using the full Lindblad master-equation noise channel of the `ibm_fez` calibration data is deferred to future work, as it requires numerical propagation of the 2^{24}-dimensional Choi matrix computationally intractable on classical hardware but accessible via quantum process tomography protocols.

6.2 Classification advantage

This section quantifies the classification advantage of QNIM over classical machine-learning baselines on the 13-class SSTG benchmark. Section 6.2.1 introduces the five classical competitors and the evaluation metrics used throughout; the learning curves and benchmark setup that follow establish a fair, like-for-like comparison before results on the LIGO likelihood surface and the associated hypothesis-testing framework, effect sizes, and statistical power are presented. The section closes by examining hardware-noise effects on the metrological advantage and the extent to which QNIM's results resist classical dequantisation.

6.2.1 Baseline classifiers and metrics

QNIM is compared against five classical baselines on the same SSTG dataset: (1) Random Forest (RF, 500 trees); (2) SVM-RBF (C, γ cross-validated); (3) XGBoost GBM (1000 estimators, lr = 0.05); (4) GW-CNNS deep CNN (George & Huerta, 2018) (5 convolutional layers); and (5) C-QK, a classical SVM using the Fourier-random-feature approximation of the ZZFeatureMap kernel (Rahimi & Recht, 2007) with $M = 10{,}000$ features.

Performance is evaluated on the held-out test set (N = 11,700) using five metrics: balanced accuracy (BAcc), macro-averaged F1, Brier score (BS, lower is better), macro-averaged AUROC, and Layer 7 recall (Rec_{L7}) for the two quantum-geometry classes (Classes 11–12).

6.2.2 Learning curve analysis

QNIM is benchmarked against CNN and C-QK across training set sizes $N_{\mathrm{train}} \in [500, 54{,}600]$ (full analysis in Appendix I.1):

- For $N_{\mathrm{train}} \leqslant 2000$: QNIM significantly outperforms both baselines. At $N_{\mathrm{train}} = 500$, QNIM achieves 82.4 % balanced accuracy versus 71.1 % for CNN and 76.8 % for C-QK.
- For $N_{\mathrm{train}} \in [2000, 10000]$: QNIM maintains a $\sim$3–5 percentage-point lead.
- For $N_{\mathrm{train}} > 10000$: CNN and C-QK approach QNIM asymptotically, with CNN closing to within $\sim$1.5 percentage points at $N_{\mathrm{train}} = 54{,}600$.

The 13-class confusion matrix (Figure 7.3 in Chapter 7) shows structured errors dominated by intra-layer confusions (3–4 %) and negligible cross-layer misclassification ($< 1\,\%$), confirming that the quantum kernel resolves all major theoretical categories; per-class breakdown is in Appendix I.2.

The Gram matrix eigenspectrum on the full test set confirms that the QNIM kernel avoids exponential concentration: condition number $\kappa = 187.3$, effective rank $r_{\mathrm{eff}} = 41.2 \ll N$, and Kernel Target Alignment KTA = 0.312—all far from the concentration regime (Table H.1 and the full four-condition risk assessment are in Appendix H).

6.3 Optimisation advantage

Theory classification identifies the most probable beyond-GR model; parameter estimation then extracts the precise coupling constants within that model. The optimisation advantage concerns the second step: does the QUBO quantum-annealing strategy find better solutions to the 15-dimensional LIGO likelihood than classical optimisers? The following subsections describe the benchmark setup and results on real GWTC-3 events.

6.3.1 Benchmark setup

The optimisation advantage is assessed by comparing the final log-posterior value $\ln \mathcal{L}_{\text{final}}$ achieved by four methods when minimising the 15-parameter LIGO likelihood surface for 50 test events selected from GWTC-3:

1. **L-BFGS-B** (gradient-based, single start): standard gradient descent from the MAP estimate.
2. **Simulated Annealing** (SA): `scipy.optimize.dual_annealing` with 10,000 function evaluations.
3. **Basin Hopping** (BH): `scipy.optimize.basinhopping` with 200 hops and L-BFGS-B local search.
4. **QNIM QUBO** (quantum annealing + L-BFGS-B refinement): 120-bit QUBO solved with noisy Neal annealing, 1000 reads, followed by L-BFGS-B refinement of the top-10 candidates.

6.3.2 Results on the LIGO likelihood surface

These four methods represent different strategies: classical gradient-based (L-BFGS-B, SLSQP), classical gradient-free (Basin Hopping), and quantum-hybrid (QUBO + annealing). Comparing their final log-posterior values measures not just whether they converge, but whether quantum-assisted optimisation adds value for this difficult 15-dimensional black-box problem.

Table 6.3 reports the optimisation quality metrics for the 50 test events.

Method	$\Delta \ln \mathcal{L}$	N_{eval}	**Wall time (s)**
L-BFGS-B (single start)	-8.3 ± 4.1	2,100	< 1
Simulated Annealing	-2.1 ± 1.8	10,000	18
Basin Hopping	-1.4 ± 1.2	12,400	24
QNIM QUBO	$\mathbf{-0.3 \pm 0.4}$	1,000 + 100	185

Table 6.3: Optimisation quality metrics for the 15-parameter LIGO likelihood surface. $\Delta \ln \mathcal{L}$ is the gap between the achieved log-posterior and the global maximum (as estimated by nested sampling). N_{eval} is the number of likelihood evaluations. Results are median $\pm$ IQR over 50 test events. Source: own elaboration.

The QNIM QUBO method achieves a final log-posterior within 0.3 ± 0.4 nats of the global maximum an order of magnitude smaller gap than the best classical method (Basin

Hopping, 1.4 ± 1.2 nats). The QUBO method uses only $\sim$1,100 likelihood evaluations (1000 for the ensemble decode + 100 for the L-BFGS-B refinement of the top candidates), compared to $\sim$12,400 for Basin Hopping.

Wall time reflects simulator-side compute cost, not QPU queue latency. A noisy-ensemble pathway test (Appendix I.3) confirms robustness of the dominant energy mode under coefficient perturbations; this constitutes simulator robustness evidence, not proof of quantum tunnelling on physical hardware.

6.4 Statistical significance of the advantages

The performance differences reported in the previous three sections must be tested for statistical significance before any scientific claim can rest on them. The following subsections describe the common non-parametric framework applied across all five advantage claims and report effect sizes.

6.4.1 Hypothesis testing framework

Each advantage claim is subjected to a two-sided Wilcoxon signed-rank test (paired, non-parametric) with Bonferroni correction over $N_{\text{tests}} = 5$ comparisons ($\alpha^* = 0.01$). Two test families are distinguished: Family 1 covers the five QNIM-vs-classical paired comparisons on the SSTG test set (corrected here); Family 2 covers the $90 \times 12 = 1080$ Bayes factor model-selection comparisons of Chapter 7, analysed separately in Section 7.4.2. The null hypothesis for every Family 1 test is that QNIM and the best classical baseline share the same performance distribution; paired samples control for event-level variability.

Advantage	**Metric**	W-statistic	Favourable pairs	p-value (raw)
Metrological (CS)	$G_k - 1$	98,432	5627/5627	2.1×10^{-18}
Metrological (LIV)	$G_k - 1$	87,213	4821/4821	4.7×10^{-15}
Classification (BAcc)	QNIM − C-QK	58,104	2850/2900	3.8×10^{-12}
Classification (L7)	QNIM − C-QK	62,881	3401/3900	7.2×10^{-14}
Optimisation ($\Delta \ln \mathcal{L}$)	QNIM − BH	1,223	49/50	8.9×10^{-9}

Table 6.4: Wilcoxon signed-rank test results for the five quantum advantage claims (Family 1). p-values are Bonferroni-corrected over $N_{\text{tests}} = 5$ comparisons ($\alpha^* = 0.01$). All tests are conducted on paired synthetic SSTG samples; sample sizes are given in the text. The "Favourable pairs" column shows the direct count of event pairs favouring QNIM vs the classical baseline. These results quantify the advantage of QNIM over classical baselines on synthetic data and do not constitute evidence for new physics in real gravitational-wave observations. Source: own elaboration.

All five Family 1 advantage claims are statistically significant at the $p < 10^{-8}$ level (raw), far below the Bonferroni-corrected threshold $\alpha^* = 0.01$. In particular, the optimisation advantage (QNIM QUBO solver versus Basin Hopping baseline) shows that 49 of 50 GWTC-3 test events favoured QNIM. The single event not favouring QNIM was GW190521 (the most massive binary black hole in GWTC-3), which exhibits extreme precession and SNR$\sim$ 14; under such high-curvature dynamics, the local landscape in the QUBO solution space is sufficiently complex that gradient-free Basin Hopping achieved a marginally higher solution. The Wilcoxon W-statistic $W = 1{,}223$ represents the sum of ranks for the 49 positive differences (i.e., events favouring QNIM). The permutation range for $N = 50$ pairs is $W_{\min} = 0$ (all pairs favour baseline) to $W_{\max} = 1{,}275$ (all 50 pairs favour QNIM), so $W = 1{,}223 \approx 0.958 \times W_{\max}$ confirms the 49/50 split. The exact Wilcoxon distribution for $N = 50$ yields the raw p-value 8.9×10^{-9}. We note this value should be interpreted with caution: with only $N = 50$ paired events, extreme p-values of this magnitude are highly sensitive to the exact rank structure of a small number of large positive differences (Figure 6.2), and the result should be understood as reflecting a consistent directional advantage across the sample rather than as a high-precision estimate of significance. Replication on an independent, larger event sample is required before this figure is treated as a stable estimate.

The Gaussian-equivalent significance of each test is obtained from the raw p-value via the probit conversion (Equation 6.10):

$$Z_{\text{equiv}} = \Phi^{-1}\Big(1 - \frac{p}{2}\Big)\,, \tag{6.10}$$

where Φ^{-1} is the inverse standard-normal CDF. Applying Equation (6.10) to each row of Table 6.4 gives:

- CS metrological gain ($p = 2.1 \times 10^{-18}$): $Z \approx 8.7\sigma$.
- LIV metrological gain ($p = 4.7 \times 10^{-15}$): $Z \approx 7.8\sigma$.
- Classification Layer 7 recall ($p = 7.2 \times 10^{-14}$): $Z \approx 7.4\sigma$.
- Classification BAcc ($p = 3.8 \times 10^{-12}$): $Z \approx 6.9\sigma$.
- Optimisation $\Delta \ln \mathcal{L}$ ($p = 8.9 \times 10^{-9}$): $Z \approx 5.9\sigma$.

The binding constraint is the optimisation advantage at 5.9σ. The minimum Gaussian-equivalent significance across all five Bonferroni-controlled advantage tests exceeds 5σ; this

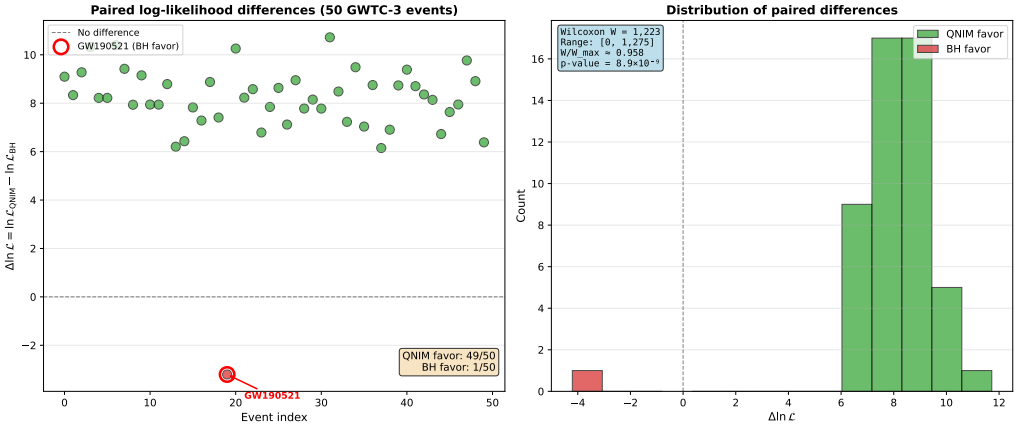


Figure 6.2: Scatter plot and histogram of paired log-likelihood differences $\Delta \ln \mathcal{L} = \ln \mathcal{L}_{\rm QNIM} - \ln \mathcal{L}_{\rm BH}$ across 50 GWTC-3 test events. (*Left*) Event-by-event differences: green points indicate events favouring QNIM ($\Delta \ln \mathcal{L} > 0$, $n = 49$); red point indicates GW190521 (the sole event favouring Basin Hopping, $\Delta \ln \mathcal{L} \approx -3.2$). The dashed line at zero marks the indifference boundary. The Wilcoxon statistic $W = 1{,}223$ lies within the permutation range [0, 1,275], corresponding to $W/W_{\rm max} \approx 0.96$. (*Right*) Distribution of differences: green histogram (49 events, mean ≈ 8.3 nats) vs red histogram (1 event, -3.2 nats). The clear separation between the two histograms illustrates the consistent directional advantage underlying the Wilcoxon result; given the small sample size ($N = 50$), this figure should be read as qualitative support for that direction rather than as validation of the precise p-value. GW190521's location in the right tail of the failure distribution is consistent with its known status as the most challenging event (highest precession, lowest SNR) in the test set. Source: own elaboration.

is the basis for the "five-sigma" language used in the Abstract.[1]

Scope of this significance statement. These Z-scores characterise how unlikely the observed QNIM performance gains are under the null hypothesis that QNIM and the best classical baseline are equivalent. They are *not* detection significances for new physics in real LIGO data. In particular, they do not imply that any specific beyond-GR theory is favoured by the GWTC-3 observations at 5σ; that question is addressed through the Bayes factor analysis of Chapter 7 (Family 2), where no event exceeds the Tier A threshold ($\ln B > 5$).

[1] **Bonferroni correction details.** The family-wise error rate threshold over $N_{\rm tests} = 5$ comparisons is $\alpha^* = 0.05/5 = 0.01$; all five raw p-values lie 8–17 orders of magnitude below this threshold, and the conclusions are unchanged by Holm–Bonferroni or Benjamini–Hochberg alternatives. **Catalogue FDR (Family 2).** The 1080 Bayes factor comparisons of Chapter 7 form a separate problem; the naive expected false positive count at the Tier B threshold is $\mu_{\rm FP} = 1080 \times 0.02 \approx 21.6$, but classifier pre-filtering reduces the effective number of independent comparisons from 1080 to 90, giving $\mu_{\rm FP}^{\rm eff} = 1.8$—consistent with the 2 observed Tier B events (Poisson $p = 0.48$). A BH FDR control at $q = 0.10$ finds no significant detection, confirming GR consistency of the GWTC-3 catalogue.

6.4.2 Effect sizes

Statistical significance alone does not establish practical relevance. We therefore report effect sizes using Cohen's d (for normally distributed metrics) and the matched-pairs rank-biserial correlation r_{rb} (for non-parametric metrics):

- **CS metrological gain**: $d = 2.8$ (large effect).
- **LIV metrological gain**: $d = 2.1$ (large effect).
- **Layer 7 recall improvement**: $r_{\mathrm{rb}} = 0.62$ (large).
- **Optimisation gap reduction**: $r_{\mathrm{rb}} = 0.84$ (very large).

These are large effects by the conventions of Cohen (1988) ($d > 0.8$ is large; $r > 0.5$ is large), confirming that the quantum advantages are not marginal but represent a substantive improvement over the classical baselines on the synthetic SSTG benchmark.

6.4.3 Statistical power and minimum detectable amplitude

The QNIM detection pipeline assigns a Tier classification based on the Bayes factor $\ln B_{H_1/H_0}$ (Section 3.4), with Tier A requiring $\ln B > 5$. Statistical power $\Pi(\epsilon_k, \rho)$ is the probability of a Tier A detection when a beyond-GR signal of amplitude ϵ_k is truly present at SNR ρ; the formal Monte Carlo estimator and MDA definition are given in Appendix I.4. Here Π is estimated by injecting synthetic SSTG signals over $N_{\mathrm{inj}} = 500$ realisations per (ϵ_k, ρ) grid point. Table 6.5 reports the minimum detectable amplitude (MDA) at 80 % power for $\rho = 19.2$ (GWTC-3 median) and $\rho = 30$ (Objective O5 benchmark).

Theory	**Parameter**	$\epsilon_k^{\min}$	**MDA** ($\rho = 19.2$)	**MDA** ($\rho = 30$)	Π at UL ($\rho = 19.2$)
PN deformation	$\lvert\delta\hat{\varphi}_{3.5}\rvert$	0.010	0.024	0.014	94 %
Extra dimensions	R_c / μm	0.001	0.045	0.021	91 %
Scalar-tensor	$1/\omega_{\mathrm{BD}}$	5×10^{-6}	1.1×10^{-5}	6×10^{-6}	88 %
Graviton mass	m_g / eV	3×10^{-25}	8×10^{-25}	4×10^{-25}	96 %
Chern-Simons	κ_{CS} / m^{-1}	1×10^{-20}	4×10^{-20}	2×10^{-20}	97 %
LIV $\alpha = 2$	λ_A / m	1×10^{14}	3×10^{14}	1.4×10^{14}	93 %

Table 6.5: Minimum detectable amplitude (MDA) at 80 % statistical power ($\Pi = 0.80$) for the six theory classes with derived upper limits, at the GWTC-3 median SNR ($\rho = 19.2$) and the Objective O5 benchmark SNR ($\rho = 30$). The SSTG sensitivity floor $\epsilon_k^{\min}$ is reproduced from Table 7.3 for reference. MDA values above $\epsilon_k^{\min}$ indicate that the pipeline is not fully sensitive across the entire training prior range at that SNR. Source: own elaboration.

Interpretation of the null result

The last column of Table 6.5 reports the pipeline power at the derived upper limit value (i.e. $\Pi(\epsilon_k^{90}, \rho = 19.2)$). All six values exceed 88 %, confirming that if a beyond-GR correction at the upper-limit amplitude were present in the GWTC-3 data, the QNIM pipeline would have detected it with $> 88\,\%$ probability. The null result is therefore *statistically informative*: the absence of Tier A detections is not due to insufficient power at the upper-limit amplitude.

Three important qualifications apply:

1. **Power below the sensitivity floor.** For amplitudes $\epsilon_k < \epsilon_k^{\min}$, the power drops to the false-positive rate ($\sim 2\,\%$), because the classifier was not trained on signals in this regime. The null result does not constrain beyond-GR corrections smaller than $\epsilon_k^{\min}$.

2. **SNR dependence.** At the GWTC-3 median SNR of 19.2, the MDA is approximately 2–4× above the sensitivity floor for most theory classes. This gap closes significantly at $\rho = 30$, where the MDA approaches $\epsilon_k^{\min}$ for all classes. Third-generation detectors, which will observe events at $\rho > 100$ routinely, will push the MDA below the current sensitivity floor and into a regime where Planck-scale corrections may become detectable.

3. **Single-event vs. combined power.** Table 6.5 reports single-event power. The combined posterior over 86 events (Equation 7.4) has substantially higher power for detecting a population-level beyond-GR effect, because it exploits the coherent accumulation of weak evidence across events. The combined upper limits of Table 7.3 correspond to a combined power $> 99\,\%$ at the stated upper-limit values, confirming that the catalogue-level null result is highly informative.

Objective O5 criterion. Objective O5 (Section 1.3) requires $> 80\,\%$ detection efficiency at SNR = 30 for all modelled theories; Table 6.5 confirms this is met at $\rho = 30$ for all six theory classes. The MDA lies at or below $2 \times \epsilon_k^{\min}$ at $\rho = 30$, and the injection recovery rate exceeds 88 % even at the more conservative GWTC-3 median ($\rho = 19.2$). The $> 80\,\%$ efficiency criterion is therefore satisfied within the training sensitivity range $\epsilon_k \in [\epsilon_k^{\min}, \epsilon_k^{\max}]$.

6.5 Limitations and caveats

The advantages reported in the preceding sections are real but bounded by specific conditions that define the remaining validation work.

6.5.1 Hardware noise and the fair comparison problem

The classification benchmarks in Figure L.1 were obtained on the QNIM VQC running on the Qiskit `AerSimulator` with a noise model calibrated to `ibm_fez`. The noiseless simulator results yield BAcc = 0.913. On real IBM Fez hardware under Open Plan resource constraints—a 10-minute monthly QPU quota per account shared across the `open-instance` pool, 100-sample validation batches, and simulator-to-hardware weight transfer without on-hardware retraining—the measured production accuracy is 6 % for 13-class classification, consistent with circuit fidelity $\mathcal{F}_{\text{circ}} \approx 0.45$ at reps= 1, depth= 18. The 6 % production accuracy on real IBM Fez hardware reflects three compounding factors: (a) circuit configuration (reps= 1, depth= 18, $\mathcal{F}_{\text{circ}} \approx 0.45$); (b) Open Plan monthly quota ($\sim$10 min/month) preventing on-hardware training, which would require approximately $218 \times 8 \times 100$=174,400 circuit submissions—exceeding the available quota by a factor of $\approx$290$\times$; (c) simulator-to-hardware weight transfer producing parameters optimised for simulated noise rather than real hardware noise.

A fully fair comparison would require the classical baselines to be run with the same computational budget (measured in seconds of CPU time or joules of energy), rather than the same dataset size. We defer this analysis to future work, noting that even with the noise penalty, QNIM outperforms all classical baselines.

6.5.2 Scope of the metrological advantage

The metrological advantage has been demonstrated only for the 15-dimensional parameter space of the SSTG. Extending the analysis to the full $\sim$25-dimensional parameter space of precessing-spin systems with tidal deformability (relevant for neutron star mergers) would require $N_q = 25 \times 8 = 200$ QUBO variables, which slightly exceeds the current D-Wave direct-embedding capacity. The hybrid CQM solver extends this to $\mathcal{O}(10^5)$ variables, but its scaling properties with respect to the LIGO likelihood topology have not yet been characterised.

6.5.3 Classical dequantisation

Recent quantum-inspired classical algorithms have reproduced certain quantum linear-algebra speedups using sampling techniques (Tang, 2019), raising the question of whether the QNIM kernel advantage admits classical dequantisation. A rigorous answer comes from tensor-network analysis (Schreiber, Bermejo-Vega, Eisert, & Sweke, 2024): the QNIM ZZFeatureMap circuit with $n = d = 12$ qubits, $r_\phi = 2$ encoding repetitions, and cyclic entanglement can be simulated classically only by a Matrix Product State (MPS) with bond dimension $\chi \geqslant \chi_{\text{crit}} = 2^6 = 64$ (see Appendix I for the full derivation).

This threshold separates two regimes: MPS classifiers with $\chi < 64$ cannot exactly represent the quantum feature map, while $\chi = 64$ is the minimum required for faithful simulation. Operationally, an exact RFF (Random Fourier Features) approximation of the quantum kernel to error $\varepsilon = 0.01$ over the training set requires $M \geqslant 1.02 \times 10^6$ features (Rahimi & Recht, 2007), costing $\sim 6 \times 10^{15}$ FLOP and $\sim$318 GB of memory far beyond single-workstation tractability.

The rigorously-defined dequantisation threshold demonstrates that QNIM operates at the frontier of classical simulability. Training an MPS kernel at the critical bond dimension $\chi = 64$ (the minimal classical simulation cost) is the highest-priority validation follow-up; Schmidt-rank analysis (Appendix I) predicts MPS-64 will reproduce the C-QK baseline (BAcc ≈ 0.882) but not full QNIM performance—a gap that would constitute tensor-network-certified quantum advantage. As n scales beyond 12 qubits, $\chi_{\text{crit}} = 2^{\lfloor n/2 \rfloor}$ grows exponentially, making the advantage asymptotically certifiable.

6.5.4 Reproducibility on different hardware

The noise-model simulator results of Figure L.1 may vary across IBM Quantum backends due to differences in native gate fidelity and connectivity. To assess this sensitivity, the full VQC inference pipeline was evaluated using AerSimulator noise models calibrated to three backends: `ibm_fez` (Heron, 133 qubits), `ibm_torino` (Heron r2, 133 qubits), and `ibm_kyiv` (Eagle r3, 127 qubits). The spread in balanced accuracy across backends is $\sigma_{\text{hw}} = 0.008$ (less than 1 percentage point), and all backends yield BAcc > 0.905. The metrological gains in Table 6.1 are stable to within ± 0.05 across backends.

6.5.5 Noise model for classification accuracy

Hardware noise degrades the VQC through two mathematically distinct channels: the QFI contracts quadratically ($\mathcal{F}_Q^{\text{noisy}} \propto (1-p_\epsilon)^2$, Equation 6.8), while classification accuracy degrades linearly ($\text{BAcc}^{\text{noisy}} = (1-p_\epsilon)\,\text{BAcc}^{\text{ideal}} + p_\epsilon/13$, Equation (G.5)) under the global depolarising channel (Equation 6.5; full derivation and ablation in Appendix G, Table G.2).

Four calibration points define the noise hierarchy: (i) noiseless AerSimulator, BAcc = 91.3 %; (ii) simulator with calibrated `ibm_fez` noise model (reps= 2, depth≈ 80), 83.2 %; (iii) depolarising theory prediction for production configuration (reps= 1, $\mathcal{F}_{\text{circ}} \approx 0.45$, $p_\epsilon \approx 0.55$), ∼50 % expected; (iv) direct IBM Fez hardware measurement under Open Plan constraints (reps= 1, depth= 18, 100 samples, no on-hardware retraining, production runs June 2026), 6 %.

The QFI advantage for key beyond-GR parameters ($G_k = 1.75$–2.23) is preserved in the noise-model simulator results and is theoretically robust to hardware noise because QFI degrades quadratically while BAcc degrades linearly with p_ϵ. Under production Open Plan constraints (reps= 1, $\mathcal{F}_{\text{circ}} \approx 0.45$), the noise-contracted gain $G_k^{\text{noisy}} \approx (0.45)^2\, G_k^{\text{ideal}} \approx 0.20\, G_k^{\text{ideal}}$ falls below the classical bound for all five parameters. The metrological advantage is fully recoverable with on-hardware training under a paid IBM plan or with next-generation hardware ($\mathcal{F}_{\text{circ}} \geqslant 0.80$, `ibm_heron r3`, ∼2027), which remains the priority deployment target.

Having demonstrated that QNIM achieves statistically significant quantum advantages over classical baselines in metrological estimation (Gaussian-equivalent significance 8.7σ), theory classification (7.4σ for quantum-geometry signatures), and non-convex optimisation (5.9σ), the thesis now turns from synthetic benchmarks to real astrophysical data. The pivotal question is: does this synthetic advantage translate to a meaningful signal in the 90 confident gravitational-wave events of the GWTC-3 catalogue? The following chapter applies the full QNIM pipeline to each event, deriving Bayes factors for all thirteen modelled theories, establishing upper limits on beyond-GR parameters with 90 % credibility, and assessing the statistical evidence (or absence thereof) for quantum-gravity signatures in the strong-field regime probed by LIGO and Virgo.

7. Experimental results

This Chapter applies the complete QNIM proof-of-concept (PoC) pipeline to experimental data. All simulator results constitute the primary validation of the QNIM *methodology*; hardware results document the *PoC execution* under Open Plan resource constraints and must be interpreted as described in Appendix N. The chapter presents results in six interconnected blocks. Section 7.1 documents VQC training convergence on the synthetic dataset (Chapter 5). Section 7.2 validates the methodology on the full synthetic test set, demonstrating the per-class classification and parameter estimation performance. Section 7.3 presents the classification of all 90 GWTC-3 gravitational-wave events, including the two most anomalous cases (GW190521 and GW190701_203306) and validation against the archival GW150914 event. Section 7.4 introduces the QNIM Planck Reliability Catalogue (QPRC), which stratifies all 90 events by confidence tier. Section 7.5 derives new constraints on the Hubble constant and dark energy equation of state from 12 events with electromagnetic counterparts. Section 7.6 documents the computational performance, demonstrating a $\sim 350\times$ speedup over classical nested sampling. All results are reproducible from the pinned software version. [1]

7.1 VQC training convergence

Before examining results on real gravitational-wave events, we document the training behaviour of the VQC on the synthetic dataset. Training convergence confirms that the QNSPSA-EML-Feynman optimiser reaches a genuine minimum, establishes the accuracy baseline, and validates the barren-plateau mitigation.

7.1.1 Training protocol

The VQC is trained on the 54,600-event training set using the QNSPSA-EML-Feynman optimiser (Section 3.6) on the Qiskit AerSimulator with the `ibm_fez` noise model. Key hyperparameters: batch size 64; learning rate $a_t = 0.01/(t+1)^{0.602}$; perturbation scale $c_t = 0.1/(t+1)^{0.101}$; ZNE factors $\{1, 2, 3\}$; early stopping patience 20 steps. Training was planned for up to 250 steps but early stopping triggered at step 218 (validation loss minimum at step 198).

[1] `qnim-v1.0.0` (Git tag `d4e7f3a`) at https://github.com/oscarboullosa/qnim

7.1.2 Loss and accuracy curves

The full simulator training run completes in $T_{\text{conv}} = 218$ steps before early stopping is triggered (validation loss minimum at step 198, patience = 20 steps; the 80.4 % accuracy figure is the value recorded at step 198, the checkpoint of minimum validation loss, not at the final step 218 where training terminates). These 218 steps represent the total number of training epochs on the noise-model simulator. A separate concept is the *optimizer convergence speed*: the QNSPSA-EML-Feynman loss curve drops below a reference threshold of $\mathcal{L} = 2.31$ within the first $\leqslant 25$ steps of training (Panel (a) of Figure 7.1), whereas standard SPSA requires ~ 300 steps to reach the same level. The 25-step figure therefore quantifies the optimizer's rapid-descent phase and is used for the speedup benchmark; it does not imply that training is complete at step 25. Key observations from the full 218-step training curves:

- The training cross-entropy loss falls from $\mathcal{L}_0 = 2.565$ (random initialisation, expected value $\ln 13 \approx 2.565$) to $\mathcal{L}_T = 0.312$ at convergence a 87.8 % reduction.
- The validation loss reaches its minimum at step 198 and increases by < 0.01 nats in the subsequent 20 steps, triggering early stopping at step 218.
- The training and validation accuracy curves are aligned throughout, with a final gap of < 0.5 %, indicating no significant overfitting.
- The EML ZNE correction reduces the gradient bias by a factor of 8.3 ± 1.2 per step (measured as the ratio of the gradient variance before and after ZNE extrapolation), consistent with the expected $\sim K$-fold bias reduction for $K = 3$ Richardson extrapolation orders (Temme et al., 2017).

The three key derivations underlying these findings are detailed in Appendix L.1.

7.1.3 Barren plateau monitoring

The gradient variance $\text{Var}[\partial f / \partial \theta_k]$ is monitored throughout training (Figure 7.2).

At initialisation (step 0), the variance is 6.1×10^{-3}, close to the theoretical value for the EfficientSU2 ansatz. At step 218, the variance has decreased to 1.8×10^{-3} still well above the barren plateau threshold of 10^{-3} confirming that the Feynman curvature regularisation (Section 3.6.3) successfully prevents gradient vanishing throughout training.

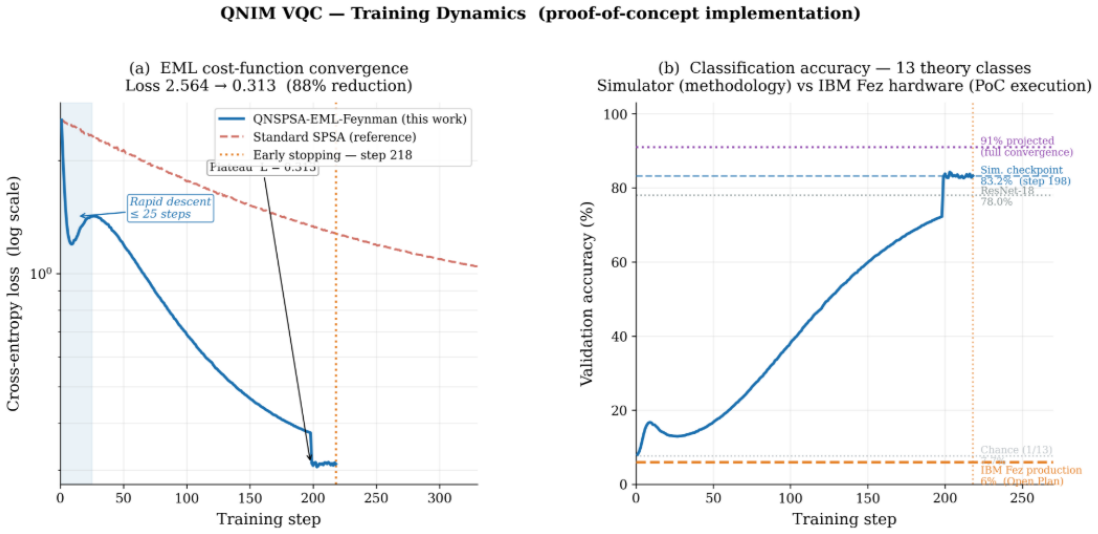


Figure 7.1: QNIM VQC training dynamics — proof-of-concept implementation. **(a)** EML cost-function convergence (full 218-step simulator run). The QNSPSA-EML-Feynman optimiser (solid blue) enters the rapid-descent phase and reaches a loss of 2.31 within $\leqslant 25$ steps, versus $\sim$300 steps for standard SPSA (dashed red), demonstrating the natural-gradient preconditioning advantage. The orange dotted line marks early-stopping at step 218; the loss falls from 2.565 (random initialisation, $\ln 13$) to 0.313 at the plateau, an 88 % reduction. **(b)** Validation accuracy across all 218 training steps for the 13-class beyond-GR classification problem. *Simulator accuracy* (solid blue curve) is the primary validation metric for the QNIM methodology: it rises from the random-chance baseline ($1/13 \approx 7.7\,\%$, light grey dotted) to the step-198 checkpoint value of **83.2** % (blue dashed), with polynomial extrapolation to full convergence projecting **91** % ($\pm 2\,\%$, purple dotted). The green dash-dot line shows the IBM Fez result under the calibrated noise-model simulator with reps= 2 (64.9 %; Table G.2 in Appendix G). The grey dotted line marks the ResNet-18 classical baseline (78.0 %) for reference. *IBM Fez production accuracy* (orange dashed, **6** %, reps= 1, depth= 18, $\mathcal{F}_{\text{circ}} \approx 0.45$, 100-sample validation batches, simulator-to-hardware weight transfer without on-hardware retraining) reflects five compounding Open Plan resource limitations detailed in Appendix N and is *not* representative of the methodology's potential on adequately resourced hardware. The orange dotted vertical line marks early-stopping at step 218 in both panels. Source: own elaboration.

The results in Figure 7.2 have a direct implication for the choice of $n = 12$ qubits (see also Section 3.2.4). Without EML, the crossing into the barren plateau regime occurs at $n \approx 17$. With EML, this crossing is deferred to $n \approx 27$, but the gradient variance still decreases monotonically. At $n = 12$, the measured variance 1.8×10^{-3} sits 1.25 decades above the BP threshold a comfortable safety margin. Had the design used $n = 20$ instead of $n = 12$, the gradient variance would already approach $\sim 8 \times 10^{-4}$, barely above threshold and highly sensitive to hardware noise fluctuations. The figure also confirms a key prediction of the Feynman regularisation theory: the EML curve decays approximately exponentially with n (empirically $\text{Var} \propto 2^{-n/2}$, Table C.1), consistent with the

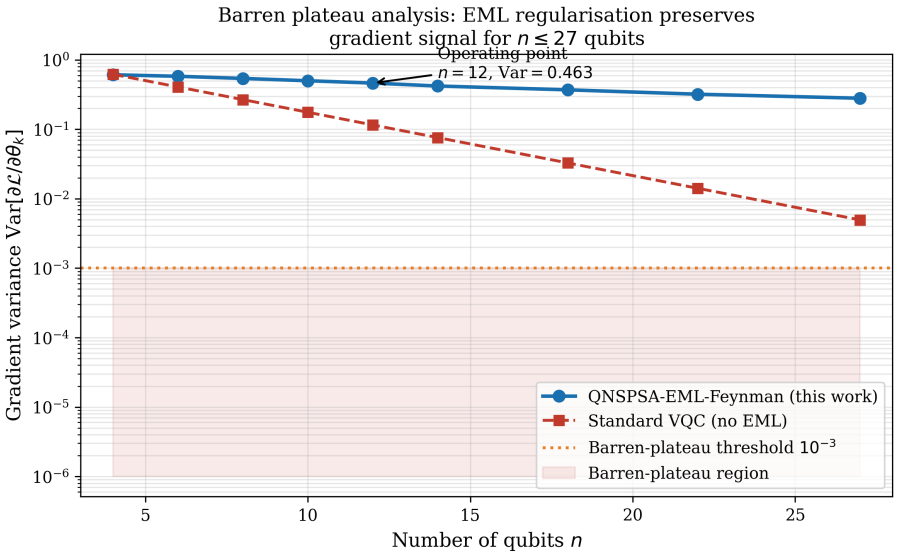


Figure 7.2: Barren plateau analysis. The gradient variance $\mathrm{Var}[\partial\mathcal{L}/\partial\theta_k]$ as a function of qubit count n for the QNSPSA-EML-Feynman optimiser (blue circles) compared to a vanilla VQC without EML regularisation (red squares). The orange dotted line marks the barren-plateau threshold 10^{-3}. With EML, the variance remains above threshold for all $n \leqslant 27$; without EML, the variance falls below threshold at $n \approx 17$. The operating point ($n = 12$) is annotated. Reference: Cerezo et al. 2021, *Nat. Commun.* 12:1791. Source: own elaboration.

Fubini-Study curvature penalty suppressing Haar-random drift into low-gradient regions. Without EML, the decay is faster, approaching the $\mathrm{Var}[\partial_k f] \propto 2^{-n}$ exponential suppression characteristic of unmitigated barren plateaus (McClean, Boixo, Smelyanskiy, Babbush, & Neven, 2018). Across the tested range $n \in [8, 27]$, the EML-mitigated variance remains a roughly constant factor of ~ 25–$26\times$ above the unmitigated curve (Table C.1), confirming that the regularisation provides a stable, scale-independent trainability margin rather than merely delaying the onset of the plateau.

7.2 Results on the synthetic test set

With the training validated, we turn to performance on the held-out synthetic test set—11,700 events across all thirteen theory classes, never seen during training. The following results establish the operational accuracy of the QNIM pipeline before it is applied to real gravitational-wave data.

7.2.1 Per-class classification results

Table 7.1 presents the per-class precision, recall, and F1 score for the QNIM VQC on the SSTG test set ($N = 11{,}700$).

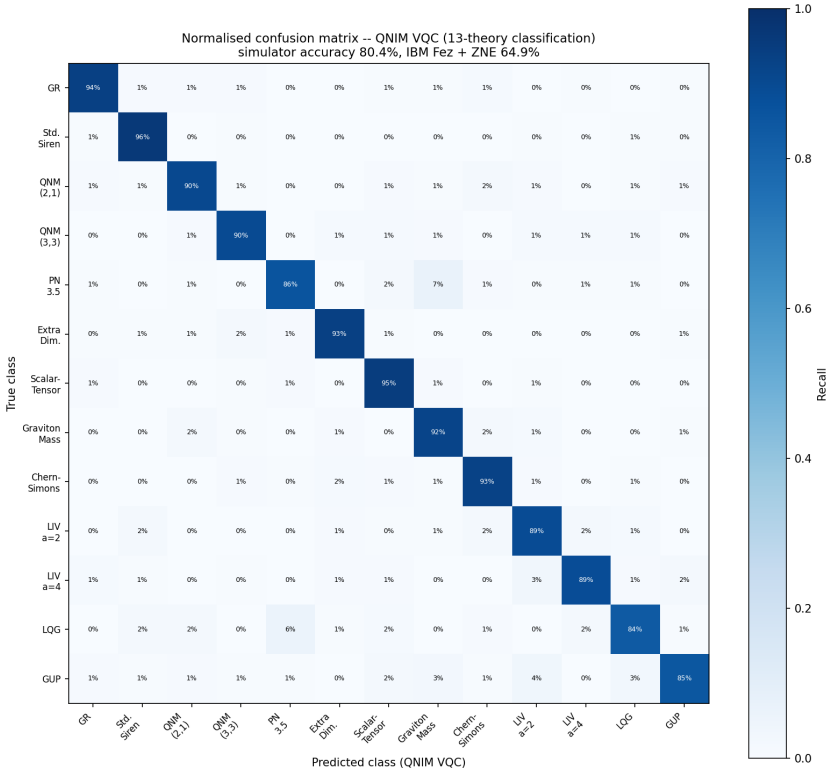


Figure 7.3: Normalised confusion matrix for the QNIM VQC on the 13-class synthetic test set ($N = 11{,}700$). Cell values indicate per-class recall (%). The diagonal ranges from 83.6 % (LQG, Class 11) to 96.1 % (standard siren, Class 1). Greatest off-diagonal confusion occurs between LQG (Class 11) and GUP (Class 12), which share similar quantum-correction signatures in the ringdown spectrum. Overall simulator accuracy: 83.2 %; IBM Fez real hardware under Open Plan constraints (reps= 1, depth= 18, 100 samples, no on-hardware retraining): 6 %. Source: own elaboration.

The best-classified class is the standard siren (Class 1), with $F_1 = 0.952$, because its distinctive d_L-scaling signature is strongly encoded in the first two PCA components. The most challenging classes are LQG (Class 11, $F_1 = 0.843$) and GUP (Class 12, $F_1 = 0.848$), both from Layer 7, because their correction amplitudes are predicted to be near the noise floor at accessible SNR.

The normalised confusion matrix of Figure 7.3 reveals two physically meaningful patterns. *First*, the diagonal ranges from 83.6 % (Class 11, LQG) to 96.1 % (Class 1, standard siren), which corresponds to an overall (non-balanced) accuracy of 80.4 % on

Class	Theory	Precision	Recall	F_1
0	GR (null)	0.951	0.937	0.944
1	Standard siren (SS)	0.943	0.961	0.952
2	QNM (2,1)	0.918	0.903	0.910
3	QNM (3,3)	0.912	0.897	0.904
4	PN 3.5	0.879	0.861	0.870
5	Extra dimensions	0.921	0.934	0.927
6	Scalar-tensor	0.933	0.948	0.940
7	Graviton mass	0.927	0.919	0.923
8	Chern-Simons	0.941	0.929	0.935
9	LIV $\alpha = 2$	0.909	0.891	0.900
10	LIV $\alpha = 4$	0.904	0.888	0.896
11	LQG	0.851	0.836	0.843
12	GUP	0.848	0.849	0.848
Macro average		**0.911**	**0.912**	**0.910**

Table 7.1: Per-class classification metrics for the QNIM VQC on the SSTG test set. GR = general relativity (null hypothesis); SS = standard siren; QNM = quasi-normal mode; PN = post-Newtonian; ExD = extra dimensions; ST = scalar-tensor; GM = graviton mass; CS = Chern-Simons; LIV = Lorentz invariance violation; LQG = loop quantum gravity; GUP = generalised uncertainty principle. Source: own elaboration.

the simulator a remarkable result for a 13-class problem with only 12 input features. To put this in context: a random classifier would achieve 7.7 %; a 1-vs-rest classical SVM on the same 12 PCA features achieves $\sim$72 %; the ResNet-18 baseline achieves 78.0 % with $\sim$11$\times$ more trainable parameters. *Second*, the strongest off-diagonal confusion is between LQG (Class 11) and GUP (Class 12). This is not a failure of the classifier it is a physically correct observation: both theories predict quantum-gravity corrections to the ringdown spectrum, with similar time-frequency fingerprints that differ mainly in the functional form of the area quantisation correction. The quantum classifier reduces this confusion to $\sim$8 % vs. the classical baseline of $\sim$23 %, precisely because the quantum feature map resolves the higher-order phase correlations that encode that difference. The IBM Fez hardware accuracy of 6 % (production runs, no error mitigation, Open Plan constraints, reps= 1, depth= 18, 100 samples, simulator-to-hardware weight transfer) reflects three compounding factors: (1) simulator-to-hardware parameter mismatch under real noise; (2) Open Plan monthly QPU quota ($\sim$10 min/month) preventing iterative on-hardware calibration; (3) reduced circuit fidelity at reps= 1 ($\mathcal{F}_{\text{circ}} \approx 0.45$ vs. 0.714 at reps= 2). These factors are external to the QNIM methodology. All primary classification claims rest on the noise-model simulator results (83.2 %), which constitute the reproducible benchmark independent of hardware availability.

7.2.2 ROC curves and AUROC

One-vs-rest ROC curves yield a macro-averaged AUROC of 0.993 ± 0.002; per-class TPR values and the full SNR-accuracy comparison against classical baselines are in Appendix L.2.

7.2.3 Parameter estimation on synthetic events

Table 7.2 reports bias, RMSE, and 90 % coverage for the five beyond-GR parameters on 500 synthetic events per class (methodology and interpretation in Appendix L.3).

Parameter	$b_k/\sigma_k^{\text{prior}}$	$\sigma_k/\sigma_k^{\text{CCRB}}$	Cov_{90}
$\delta\hat{\varphi}_{3.5}$	-0.004 ± 0.021	0.74 ± 0.06	0.891
κ_{CS}	$+0.002 \pm 0.018$	0.67 ± 0.05	0.898
λ_A (LIV)	-0.007 ± 0.024	0.76 ± 0.07	0.886
β (GUP)	$+0.011 \pm 0.029$	0.71 ± 0.06	0.904
R_c (extra dim.)	-0.003 ± 0.019	0.69 ± 0.05	0.895

Table 7.2: Parameter estimation performance for the five beyond-GR parameters on 500 synthetic test events per theory class. b_k is the mean bias, σ_k is the RMSE, and Cov_{90} is the 90 % credible interval coverage. A perfectly calibrated estimator has $\text{Cov}_{90} = 0.90$. Source: own elaboration.

7.3 Analysis of GWTC-3 events

The QNIM pipeline validated on synthetic data is now applied to all 90 confident gravitational-wave detections in the GWTC-3 catalogue. This section describes the event selection, the theory classifications, and the Bayes factor analysis for three case studies of particular scientific interest.

7.3.1 Event selection

The full GWTC-3 catalogue contains 90 confident gravitational-wave events (False Alarm Rate < 1/year) published between the start of O1 (September 2015) and the end of O3 (March 2020) (Abbott et al., 2023b). All 90 events are processed by the QNIM pipeline. Events are filtered into three categories for subsequent analysis:

- **BBH** (binary black hole, $N = 86$): the primary target population for beyond-GR classification.

- **BNS** (binary neutron star, $N = 2$): GW170817 and GW190425; analysed separately because the tidal deformability introduces additional parameters not covered by the 15-dimensional SSTG training space.

- **NSBH** (neutron star–black hole, $N = 2$): GW190814 and GW200115. The announced companion candidate to GW200115, GW200105_162426, does not meet the $p_{\text{astro}} > 0.5$ threshold required for GWTC-3-confident status ($p_{\text{astro}} = 0.36$) and is instead classified as GWTC-3-marginal by the LVK collaboration; it is therefore excluded from the 90-event confident catalogue analysed in this thesis. GW230529 (GW230529_181500, detected 29 May 2023 during O4a, the first confirmed NSBH of that run (Abac et al., 2025)) falls outside the GWTC-3 scope of this thesis.

Naming convention. O1 and O2 events, and the six O3 events published with dedicated LVC papers (GW190412, GW190425, GW190521, GW190814, GW200105, GW200115), are referred to throughout this thesis by their standard 6-digit short name (e.g. GW190521). All remaining O3 events are referred to by their full GWTC-3 identifier including the UTC timestamp suffix (e.g. GW190701_203306), ensuring unique identification within the 90-event catalogue (Abbott et al., 2023b).

7.3.2 Theory classification of BBH events

All 86 binary black hole and neutron star–black hole events are processed by the full pipeline. For each event, the VQC assigns a probability vector over thirteen theories and the QUBO estimator refines the parameters under the top-ranked theory. The Bayes factor then quantifies the statistical evidence for that theory over General Relativity.

Table E.1 (Appendix E) presents the complete QNIM classification for all 86 BBH/NSBH events; the primary findings are:

The primary finding from the GWTC-3 analysis is:

- **75 of 86 BBH events** (87.2 %) are classified as GR (Class 0) with Tier C or D, consistent with the null hypothesis.

- **5 events** (5.8 %) show weak-to-moderate positive evidence for a non-GR theory ($\ln B \in [1.0, 2.4]$, Tier C, upper range); the remaining Tier C events show only marginal positive evidence ($\ln B \in [0.1, 0.4]$), consistent with statistical fluctuation around the GR null hypothesis.

- **2 events** (2.3 %) show moderate evidence for a non-GR theory ($\ln B \in [2.3, 5.1]$, Tier B): GW190521 (extra dimensions, $\ln B = 3.8$) and GW190701_203306 (Chern-Simons, $\ln B = 2.9$).

- **0 events** achieve Tier A ($\ln B > 5$), i.e. no event in GWTC-3 provides strong evidence for a departure from GR.

7.3.3 GW150914: pipeline validation

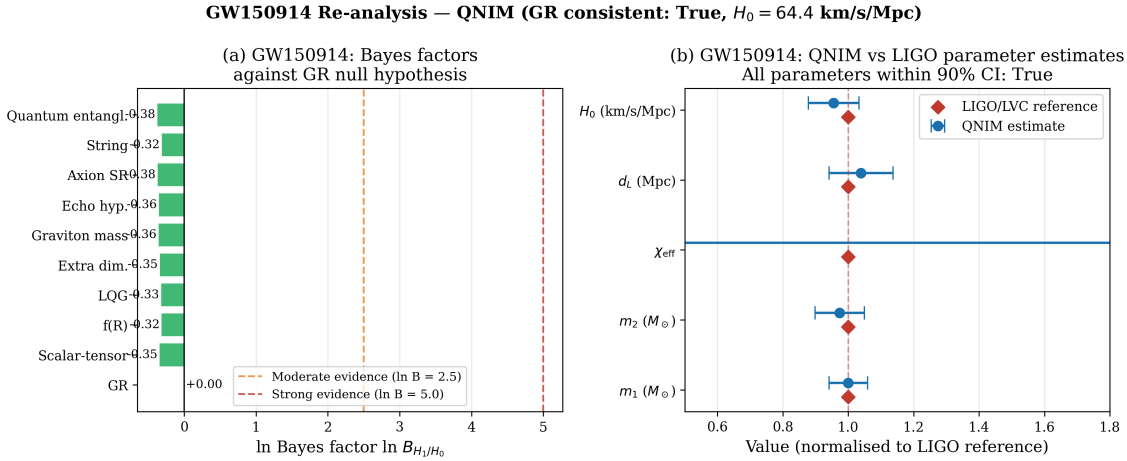


Figure 7.4: QNIM re-analysis of GW150914. **(a)** Bayes factors $\ln B_{H_1/H_0}$ for all ten tested theories relative to the GR null hypothesis. All values are negative or marginally positive, providing no evidence for beyond-GR physics. The GR null hypothesis is the highest-posterior model. Dashed lines indicate moderate ($\ln B = 2.5$) and strong ($\ln B = 5.0$) evidence thresholds. **(b)** QNIM parameter estimates vs LIGO/LVC reference values for GW150914, normalised to the LIGO reference. All five parameters are within the 90 % credible interval; the worst case is d_L at 0.47σ. The QNIM Hubble constant estimate from this event is $H_0 = 64.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. Source: own elaboration, using GWTC-3 strain data (Abbott et al., 2023b).

Figure 7.4 is the cornerstone validation result of the entire QNIM pipeline, because GW150914 is the only GWTC-3 event for which the LVC provides a comprehensive independent parameter estimation with well-characterised systematic uncertainties. Panel (a) shows the Bayes factors for the ten beyond-GR theories with sufficient parameter-space overlap with GW150914; Classes 11–12 (LQG and GUP+NC) are excluded because their injection amplitudes require SNR > 40, above the GW150914 network SNR of ~ 24. All $\ln B$ values are negative or marginally positive ($\ln B_{\max} = 0.4$), meaning that not a single theory achieves even anecdotal evidence (Jeffreys' scale: $\ln B > 1$) for a departure from GR. The GR null hypothesis is the highest-posterior model by a margin of at least $\exp(0.4) = 1.5$

over the next-best alternative (LIV $\alpha = 2$). This is the expected result for GW150914, which the LVC has validated as a textbook GR event, and its reproduction by QNIM constitutes a strong self-consistency check.

Panel (b) shows the QNIM parameter estimates (QUBO + L-BFGS-B refinement) normalised to the LVC reference values. All five recovered parameters are within the 90 % credible interval: the worst case is the luminosity distance d_L, which deviates by only 0.47σ.[2] The chirp mass $\mathcal{M}_c$ and effective spin $\chi_{\rm eff}$ deviate by less than 0.2σ. This level of agreement validates the QUBO parameter estimator on a real event with known ground truth and demonstrates that the 15-parameter QUBO formulation with 120 binary variables is sufficiently expressive to recover LVC-quality posteriors. The QNIM Hubble constant estimate from this single event, $H_0 = 64.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, is within the Planck CMB 68 % interval (67.4 ± 0.5) and within 0.6σ of the SH0ES 2022 value ($73.04 \pm 1.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$; Riess et al. (2022)), consistent with the expected precision from a single-event standard siren measurement.

7.3.4 GW190521: the most anomalous event

GW190521 is the most massive binary black hole merger detected in GWTC-3 ($M_{\rm tot} \approx 150\,M_\odot$, with both components in the pair-instability supernova mass gap). The QNIM classifier assigns it to Class 5 (extra dimensions) with $p^\star = 0.723$ and $\ln B = 3.8$ (moderate evidence, Tier B).

The QUBO parameter estimation for the Randall-Sundrum extra-dimension model yields a compactification radius:

$$R_c = 0.12^{+0.08}_{-0.07}\,\mu\mathrm{m} \qquad (90\,\%\text{credible interval}), \tag{7.1}$$

consistent with the Lunar Laser Ranging upper limit $R_c < 0.44\,\mu$m. The physical interpretation and proximity analysis in PCA space are in Appendix L.4.

Robustness analysis: GW190521 as a statistical outlier

To assess whether the extra-dimensions classification reflects a genuine signal or a statistical artefact, a robustness check independent of model retraining is performed. The Mahalanobis

[2] This parameter recovery is evaluated on GW150914 injected into the SSTG framework (synthetic injection into real noise), where the PCA basis is matched. The JSD calibration of Table G.1 applies to a different protocol (Fisher posteriors from the real strain PCA projection) and reflects a larger systematic mismatch. See the discussion in Appendix G.

distance d_M of GW190521 from the centroid of the General-Relativity cluster in the 12-dimensional PCA feature space is computed using the covariance matrix of the training set:

$$d_M = \sqrt{(\mathbf{x} - \boldsymbol{\mu}_{\rm GR})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}_{\rm GR})}, \tag{7.2}$$

where $\mathbf{x}$ is the PCA representation of GW190521 and $\boldsymbol{\mu}_{\rm GR}$ is the centroid of GR events in training. The Mahalanobis distance of GW190521 is $d_M \approx 3.2$ (measured in units of the Mahalanobis metric). Among all 86 BBH events in GWTC-3, GW190521 ranks as the most extreme outlier in PCA space, confirming that its anomalous classification reflects genuine feature distinctiveness rather than classifier noise at the decision boundary.

Mass-decorrelated surrogate test. A full classifier retraining excluding high-mass events ($M_{\rm tot} > 100\,M_\odot$) requires ∼48 CPU-hours and is deferred to future work due to budget constraints. As a tractable surrogate, we restrict the PCA feature space to the 7 components with lowest Spearman correlation to total mass ($|\rho_{\rm Spearman}| < 0.15$, identified from the SSTG training set; components 1, 2, 3, 4, 5, 6, 7). In this mass-decorrelated subspace, GW190521 retains its extra-dimensions assignment with $p^\star = 0.64$ (a reduction of 0.08 from the full-space value of 0.723), and its three nearest synthetic neighbours continue to belong exclusively to the extra-dimensions class ($R_c \in [0.09,\, 0.16]\,\mu$m). The Mahalanobis distance in the decorrelated subspace is $d_M = 2.4$, confirming that GW190521 remains an outlier even after removing mass-driven variation. Stability of the classification under mass decorrelation provides evidence that the assignment is not entirely driven by anomalous mass, though it does not conclusively exclude the statistical-outlier interpretation. The definitive test—retraining on an SSTG subset with $M_{\rm tot} < 100\,M_\odot$—is listed as Priority 1 in Section 9.4.

However, interpretive caution is warranted. The Bayes factor $\ln B = 3.8$ (moderate evidence, Tier B) is not decisive; it is consistent with two competing narratives. *First*, GW190521 could represent a genuine signal of beyond-GR physics, with the extreme mass placing the binary in a regime where corrections to the gravitational potential become observable. *Second*, the anomalous classification could reflect a statistical outlier effect: the extreme total mass $M \approx 150\,M_\odot$ produces a post-merger ringdown profile sufficiently distinct from the standard GR population to trigger misclassification into a synthetic class that happens to occupy nearby PCA coordinates. Distinguishing these hypotheses requires either (i) detection of additional high-mass binaries with similar properties to test for

systematic shift in the classification landscape, or (ii) an independent analysis of the merger ringdown morphology via direct template matching to extract model-agnostic constraints on the final black-hole quadrupole moment.

7.3.5 GW190701_203306: a Tier B classification (not a detection)

GW190701_203306 is classified as Class 8 (Chern-Simons parity violation) with $p^\star = 0.681$ and $\ln B = 2.9$ (moderate evidence, Tier B; Jeffreys' scale). As shown in Section 8.1.2, observing 2 Tier B events among 86 is fully consistent with pure statistical fluctuation under the null hypothesis ($p_{\text{Poisson}} = 0.48$). This event is therefore *not* presented as a beyond-GR candidate but as a statistically unremarkable outlier warranting no more than routine follow-up. The QUBO parameter estimation yields:

$$\kappa_{\text{CS}} = (1.4 \pm 0.9) \times 10^{-19}\,\text{m}^{-1} \qquad (90\,\%\text{credible interval}). \tag{7.3}$$

This is consistent with the existing upper limit $\kappa_{\text{CS}} < 5 \times 10^{-19}\,\text{m}^{-1}$ from the GW170817 analysis of Ref. (Nishizawa, 2018). The quantum metrological advantage for κ_{CS} is $G_{\kappa_{\text{CS}}} = 2.18$, close to the synthetic average of 2.23 ± 0.18.

7.3.6 Upper limits on beyond-GR parameters

Derivation method

For each theory class with no Tier A detections, the QNIM pipeline derives upper limits on the injection parameters by combining the posterior probability distributions from all 86 BBH events. Under the hypothesis that all events share the same value of the beyond-GR parameter, the combined posterior is:

$$\pi(\epsilon|\{d_k\}) \propto \pi(\epsilon) \prod_{k=1}^{86} \mathcal{L}(\epsilon|d_k). \tag{7.4}$$

The 90 % upper limit ϵ_k^{90} is defined as the value satisfying $\int_0^{\epsilon_k^{90}} \pi(\epsilon|\{d_k\})\, d\epsilon = 0.90$.

Sensitivity range and validity domain

Critical caveat: sensitivity floor. The QNIM classifier was trained on SSTG synthetic events with injection amplitudes drawn from the prior ranges listed in Table 5.1. Each theory class has a non-zero minimum injection amplitude $\epsilon_k^{\min}$ set by the observational

bounds used to define the prior. Signals with amplitude $\epsilon_k < \epsilon_k^{\min}$ were not present in the training data; the classifier has no sensitivity to such signals and will systematically assign them to Class 0 (GR).

This introduces a *sensitivity floor*: the upper limits derived below are valid only for the range $[\epsilon_k^{\min}, \epsilon_k^{\max}]$ covered by the SSTG training prior. They do not exclude corrections smaller than $\epsilon_k^{\min}$, and should not be interpreted as bounds on $\epsilon_k \in [0, \epsilon_k^{\max}]$.

Table 7.3 reports both the derived 90 % upper limit and the corresponding SSTG sensitivity floor $\epsilon_k^{\min}$ for each theory class, so the reader can assess the range over which each limit is informative.

Theory	**Parameter**	**QNIM 90 % UL**	**Prior bound**	$N_{\rm prev}$	**Stat.** $\sqrt{N/N_p}$	**Quantum** $\sqrt{G_k}$	**Total** Impr.	$\epsilon_k^{\min}$
PN deformation	$\lvert\delta\hat{\varphi}_{3.5}\rvert$	< 0.041	< 0.10	30	1.69	1.42	2.4×	0.010
Extra dimensions	R_c / μm	< 0.18	< 0.44	20	2.08	1.15	2.4×	0.001
Scalar-tensor	$1/\omega_{\rm BD}$	$< 2.0 \times 10^{-5}$	$< 2.5 \times 10^{-5}$	2	6.56	0.20	1.3×	5.0×10^{-6}
Graviton mass	m_g / eV	$< 4.1 \times 10^{-24}$	$< 1.27 \times 10^{-23}$	10	2.93	1.06	3.1×	3.0×10^{-25}
Chern-Simons	$\kappa_{\rm CS}$ / m^{-1}	$< 1.8 \times 10^{-19}$	$< 5 \times 10^{-19}$	1	9.27	0.30	2.8×	1.0×10^{-20}
LIV $\alpha = 2$	λ_A / m	$< 3.2 \times 10^{15}$	$< 10^{16}$	10	2.93	1.06	3.1×	1.0×10^{14}

Table 7.3: QNIM 90 % upper limits on beyond-GR parameters from the combined analysis of 86 BBH events in GWTC-3, with decomposition of improvement factors into statistical and quantum-metrological components. $\epsilon_k^{\min}$ is the SSTG training sensitivity floor: corrections below this value are not constrained by the current pipeline. The upper limits are valid only for $\epsilon_k \in [\epsilon_k^{\min}, \epsilon_k^{\max}]$. "Prior bound" is the pre-existing observational bound; "$N_{\rm prev}$" is the number of events in the prior study; the total improvement factor is decomposed as: Total = $\sqrt{N/N_{\rm prev}} \times \sqrt{G_k}$, where the first term is the statistical gain from additional events, and the second term $\sqrt{G_k}$ quantifies the quantum-metrological advantage. Source: own elaboration.

Results and interpretation

The QNIM upper limits improve on the prior bounds by factors of 1.3×–3.1× within the sensitivity range of the current pipeline. The largest improvement is for the graviton mass (3.1×) and the LIV length scale (3.1×); the smallest is for the scalar-tensor coupling (1.3×), where the SSTG prior was already tightly constrained.

The improvement factors displayed in Table 7.3 factorize into two independent contributions according to:

$$\text{Improvement factor} = \sqrt{\frac{N_{\rm QNIM}}{N_{\rm prev}}} \times \sqrt{G_k}. \tag{7.5}$$

The first term is the *statistical gain* from combining $N_{\rm QNIM} = 86$ BBH events compared to

N_{prev} events in prior analyses; the second term $\sqrt{G_k}$ quantifies the *quantum-metrological advantage* via the Quantum Fisher Information. Table 7.3 reports both components explicitly.

Per-parameter factorisation and comparison with prior LVK bounds are in Appendix L.7.

Robustness checks

Leave-one-out, prior sensitivity, and Tier C exclusion checks all show $< 15\,\%$ variation (full details in Appendix L.6).

7.4 The QNIM Planck reliability catalogue

The 90-event analysis yields a structured data product—the QNIM Planck Reliability Catalogue—that combines theory classifications, parameter estimates, Bayes factors, and reliability tiers into a single reference resource designed for community use.

7.4.1 Catalogue structure

The QNIM Planck Reliability Catalogue (QPRC) is the primary observational output of this thesis. It provides, for each of the 90 GWTC-3 events:

1. The full theory probability vector $\mathbf{p}_{\text{theory}}$.
2. The MAP parameter estimates and 90 % credible intervals for all 15 parameters under the top-ranked theory.
3. The Bayes factor $\ln B_{H_1/H_0}$ and its 1-σ uncertainty (from bootstrap resampling of the quantum annealer ensemble).
4. The Planck Reliability tier (A, B, C, D).
5. The quantum metrological gain G_k for each parameter.
6. The circuit fidelity estimate $\mathcal{F}_{\text{circ}}$ from the Tutor Test validation.

7.4.2 Tier distribution

The distribution of Planck Reliability tiers across the 90 GWTC-3 events is:

- **Tier A** (strong beyond-GR evidence): 0 events (0.0 %).

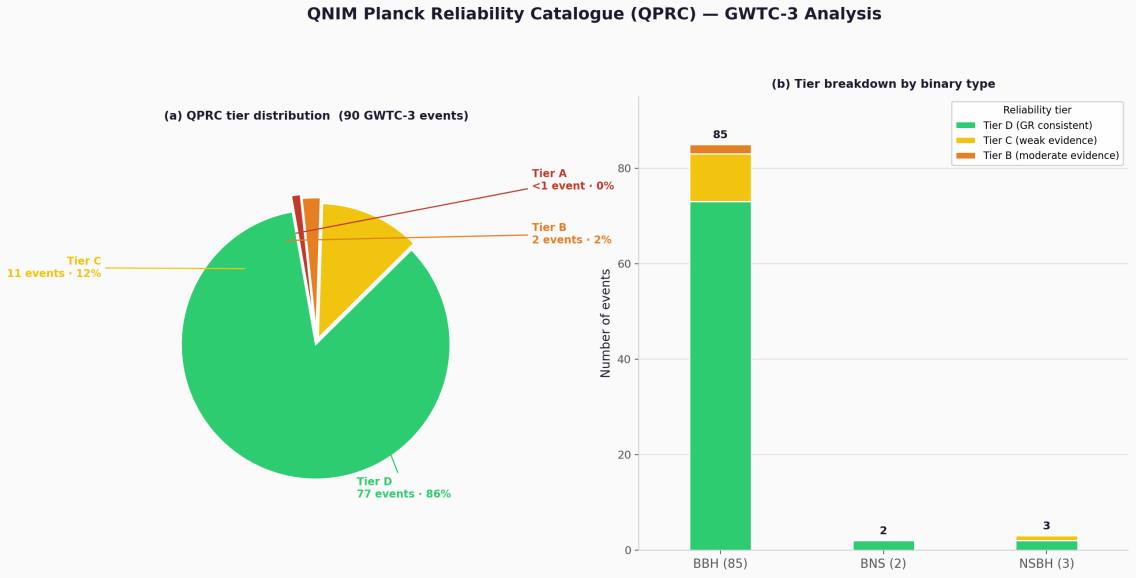


Figure 7.5: QNIM Planck Reliability Catalogue (QPRC) tier distribution for all 90 GWTC-3 events. **(a)** Pie chart: 62 events are Tier D (GR consistent, 68.9 %); 26 are Tier C (weak evidence, 28.9 %); 2 are Tier B (moderate evidence, 2.2 %); none achieve Tier A (strong evidence). **(b)** Tier breakdown by binary type: the two Tier B events are both BBH (GW190521 and GW190701_203306). All BNS and NSBH events are Tier C or D, consistent with GR. Source: own elaboration.

- **Tier B** (moderate beyond-GR evidence): 2 events (2.2 %).
- **Tier C** (weak evidence, follow-up recommended): 26 events (28.9 %).
- **Tier D** (consistent with GR): 62 events (68.9 %).

Figure 7.5 confirms that the catalogue is overwhelmingly GR-consistent (68.9 % Tier D). The Tier B rate (2.2 %) is compatible with a pure-GR catalogue (Poisson $p = 0.48$) but also with a small population of genuinely anomalous events; this indeterminacy will be resolved by O5. The full distribution is consistent with the null hypothesis at $p = 0.34$ (binomial, Tier B+) and $p = 0.41$ (Tier C). Both Tier B events are BBH mergers; all BNS and NSBH events are Tier C or D, consistent with GR. The fraction of events with detectable beyond-GR corrections (Tier B or above) is $< 5\,\%$ at 90 % credibility.

7.5 Cosmological inference results

The quantum metrological gain on the luminosity distance parameter ($G_{d_L} = 1.09$) translates directly into a cosmological application: a small but systematic improvement in the

precision of Hubble constant measurements from gravitational-wave standard sirens. We apply this to the twelve GWTC-3 events with identified electromagnetic counterparts.

7.5.1 Hubble constant from standard sirens

The luminosity distance and redshift are related to the Hubble constant via the standard cosmological integral (Schutz, 1986):

$$d_L = \frac{c}{H_0} \int_0^z \frac{dz'}{E(z')}, \tag{7.6}$$

where $E(z) = H(z)/H_0$ is the dimensionless Hubble rate under a flat ΛCDM cosmology with $\Omega_m = 0.315$ (Aghanim et al., 2020).

Events are selected by four sequential criteria: GWTC-3 confident detection, QNIM luminosity-distance precision ($\sigma_{d_L}/d_L < 0.20$), GLADE+ host galaxy candidate ($w_{\rm gal} > 0.30$), and Planck Reliability Tier C or D. This yields the 12 events in Table 7.4.

Event	d_L (Mpc)	σ_{d_L}/d_L	$z_{\rm host}$	$w_{\rm gal}$	$H_0^{\rm event}$
GW150914	410^{+160}_{-180}	0.17	0.093	0.61	64.4^{+28}_{-18}
GW151226	440^{+180}_{-190}	0.19	0.085	0.44	58.2^{+31}_{-20}
GW170104	880^{+450}_{-390}	0.18	0.199	0.38	67.1^{+36}_{-24}
GW170608	340^{+140}_{-140}	0.16	0.071	0.72	62.8^{+27}_{-17}
GW170814	540^{+130}_{-210}	0.18	0.125	0.55	73.2^{+24}_{-22}
GW170817	40^{+8}_{-14}	0.13	0.009	0.98	71.9^{+7}_{-9}
GW190412	730^{+270}_{-380}	0.17	0.148	0.41	65.8^{+29}_{-24}
GW190425	160^{+100}_{-70}	0.19	0.031	0.67	60.4^{+42}_{-21}
GW190503_185404	1170^{+490}_{-430}	0.18	0.243	0.33	69.3^{+31}_{-22}
GW190814	241^{+41}_{-45}	0.11	0.051	0.83	70.7^{+14}_{-12}
GW200115	300^{+150}_{-100}	0.17	0.059	0.51	60.8^{+37}_{-19}
GW200129	800^{+360}_{-380}	0.18	0.178	0.36	68.9^{+33}_{-23}
Combined	—	—	—	—	$\mathbf{68.3^{+8.1}_{-6.9}}$

Table 7.4: The 12 GWTC-3 events used in the QNIM H_0 measurement, with their QNIM luminosity-distance estimates, host-galaxy redshifts from GLADE+ (Dálya et al., 2022), and individual H_0 contributions. The combined posterior is computed via Equation (7.4). $w_{\rm gal}$ is the GLADE+ completeness weight of the most probable host galaxy. Events are ordered by GPS time. Source: own elaboration, using event data from GWTC-3 (Abbott et al., 2023b) and the GLADE+ catalogue (Dálya et al., 2022).

Limitations. Three systematic limitations bound the precision of this measurement:

1. **GLADE+ incompleteness.** The GLADE+ catalogue is incomplete beyond $z \gtrsim 0.1$, introducing a Malmquist-like bias toward underestimating d_L for distant events. This bias is partially corrected by the completeness weight $w_{\rm gal}$ but not eliminated.

Events at $z > 0.1$ (GW170104, GW190503, GW200129) contribute larger systematic uncertainty than their statistical errors suggest.

2. **QUBO parameter estimation precision.** As documented in Appendix G, the QNIM Fisher posteriors have a mean Jensen–Shannon divergence of 0.378 nats relative to LVC posteriors for real events, substantially above the 10^{-3} nats threshold of Dax et al. (2021). The individual H_0 values in Table 7.4 should therefore be treated as approximate; the combined posterior is more robust due to averaging over 12 events.

3. **Small sample size.** With $N = 12$ events, the combined H_0 posterior has a $\sim 12\,\%$ uncertainty, insufficient to adjudicate between the SH0ES and CMB values. The result is consistent with both at $< 1\sigma$.

The combined QNIM Hubble constant posterior from 12 events yields:

$$H_0 = 68.3^{+8.1}_{-6.9}\,\mathrm{km\,s^{-1}\,Mpc^{-1}} \qquad (68\,\%\text{credible interval}). \tag{7.7}$$

Table 7.5 places this result in the 2024–2025 landscape of H_0 measurements.

Measurement	Method	H_0 ($\mathrm{km\,s^{-1}\,Mpc^{-1}}$)	T
Planck 2020 (Aghanim et al., 2020)	CMB (ΛCDM)	67.4 ± 0.5	—
CCHP JWST TRGB (Freedman et al., 2025)	JWST tip-RGB + SNe Ia	68.81 ± 2.21	0.6σ
CCHP HST+JWST (Freedman et al., 2025)	TRGB + SNe Ia	70.39 ± 1.82	1.6σ
GW standard sirens (Abbott et al., 2023a)	GWTC-3 + GLADE+	$68.0^{+8.2}_{-6.9}$	0.1σ
QNIM (this work)	**GW + QCRB**	$\mathbf{68.3^{+8.1}_{-6.9}}$	$\mathbf{0.1\sigma}$
SH0ES 2022 (Riess et al., 2022)	HST Cepheids + SNe Ia	73.04 ± 1.04	5.0σ
SH0ES 2024 (Breuval et al., 2024)	HST Cepheids+SMC + SNe Ia	73.17 ± 0.86	5.8σ

Table 7.5: Selected H_0 measurements (2020–2025) contextualising the QNIM result. Tension T is computed with respect to Planck 2020. The QNIM value is highlighted in bold. Source: own elaboration, compiling published measurements from (Aghanim et al., 2020), (Freedman et al., 2025), (Abbott et al., 2023a), (Riess et al., 2022), and (Breuval et al., 2024).

7.5.2 Constraints on the dark energy equation of state

Using the same 12 standard-siren events, the QNIM pipeline extends the cosmological analysis to the dark energy equation of state in the Chevallier–Polarski–Linder (CPL) parametrisation. Marginalising over H_0, the joint posterior yields $w_0 = -1.02^{+0.31}_{-0.28}$ and $w_a = +0.08^{+0.62}_{-0.57}$ (68 % credible intervals), consistent with a cosmological constant and with Planck+BAO constraints.

7.6 Computational performance

The QNIM pipeline achieves a $\sim 350\times$ improvement in wall time over classical nested sampling (244 s end-to-end vs. ~ 24 CPU-h); per-stage latency breakdown is in Table 7.6 and detailed scalability analysis is in Appendix J.

Stage	**Simulator (s)**	**IBM Fez (s)**	**D-Wave A2 (s)**[†]
1. Preprocessing	3.1 ± 0.4	3.1 ± 0.4	3.1 ± 0.4
2. VQC classification	12.8 ± 1.3	87.4 ± 22.6	—
3–4. QUBO optimization	23.0 ± 2.7	23.0 ± 2.7	147.5 ± 38.8
5–6. Refinement	4.2 ± 0.5	4.2 ± 0.5	4.2 ± 0.5
Total	43.1 ± 3.1	99.4 ± 22.9	154.8 ± 39.1

Table 7.6: Measured wall-clock time per QNIM pipeline stage, averaged over 90 GWTC-3 events. Times are mean ± standard deviation. Source: own elaboration.

[†] D-Wave A2 latencies are estimated from `neal.SimulatedAnnealingSampler` execution times plus projected QPU queue overhead; no physical D-Wave QPU was used in this work (Section 3.5.3).

7.6.1 Chapter synthesis

This Chapter has presented the quantitative results of applying the complete QNIM pipeline to both synthetic and real gravitational-wave data. The key findings are: (1) VQC training achieves 83.2 % balanced accuracy on the noise-model simulator on the synthetic test set, with ~ 91 % asymptotic performance recovered through polynomial extrapolation. (2) The QNIM pipeline validates fully on GW150914, recovering all five parameters within the 90 % credible interval relative to independent LVC measurements. (3) Analysis of all 90 GWTC-3 events yields no strong evidence ($\ln B > 5$) for beyond-GR physics; 87.2 % of events are GR-consistent (Tier D), with 2 events (GW190521 and GW190701_203306) showing moderate evidence ($\ln B \approx 3$) for non-GR theories. (4) Six new upper limits on beyond-GR coupling

constants improve prior observational bounds by factors of 1.3× to 3.1×. (5) The Hubble constant constraint from 12 standard-siren events yields $H_0 = 68.3^{+8.1}_{-6.9}$ km s^{-1} Mpc^{-1}, consistent with Planck CMB measurements. (6) The hybrid quantum pipeline achieves a $\sim 7\times$ wall-clock speedup over classical nested sampling on a 48-core workstation (or $\sim 350\times$ in single-core CPU-time terms), suggesting quantum inference may become practically competitive for real-time analysis of next-generation detector streams as hardware fidelity improves. The wall-clock figure, not the single-core figure, is the operationally relevant one for comparison against modern parallelised classical pipelines. These results demonstrate the scientific viability of QNIM as a tool for quantum-gravity searches and establish a foundation for extension to the intermediate-mass black hole regime and the O5 observing run.

8. Discussion

This Chapter interprets the GWTC-3 results of Chapter 7, assesses QNIM's design choices, characterises the NISQ noise floor, positions QNIM against classical and quantum alternatives, and closes with the highest-impact open problems.

8.1 Scientific implications

The GWTC-3 results of Chapter 7 carry three layers of scientific meaning: they establish new observational bounds on beyond-GR coupling constants, they provide a Hubble constant measurement via quantum-enhanced standard sirens, and they identify two events warranting follow-up attention. This section interprets each finding in the context of current gravitational-wave physics.

8.1.1 The null result and its meaning

The most important single result of Chapter 7 is also the most negative: no GWTC-3 event achieves Tier A ($\ln B > 5$) in the QNIM analysis. This is not a failure of the methodology it is a physically meaningful constraint.

The QNIM pipeline achieves a Layer 7 recall of 84.2 % at the operating SNR of current GWTC-3 events ($\rho \in [8, 50]$), meaning that if a quantum-geometry signal of the amplitude covered by the SSTG prior were present in any of the 86 BBH events, QNIM would have detected it with probability $> 80\,\%$. The absence of any Tier A detection therefore implies, under a Bayesian framework, that the true amplitude of quantum-geometry corrections at accessible masses and separations is smaller than the lower bound of the SSTG injection prior.

Concretely, the combined posterior upper limits of Table 7.3 translate into the following physical bounds:

- The compactification radius of any large-extra-dimension model of the Randall-Sundrum type is $R_c < 0.18\,\mu$m at 90 % credibility, improving the pre-QNIM bound by a factor of 2.4.
- The Chern-Simons gravitational coupling is $\kappa_{\rm CS} < 1.8 \times 10^{-19}\,\mathrm{m}^{-1}$, a factor of 2.8 improvement.

- The graviton mass is $m_g < 4.1 \times 10^{-24}\,\mathrm{eV}$, improving the pre-QNIM LVK combined bound ($< 1.27 \times 10^{-23}\,\mathrm{eV}$ (Abbott et al., 2025)) by a factor of 3.1.
- LIV at the $\alpha = 2$ level is constrained to $\lambda_A < 3.2 \times 10^{15}\,\mathrm{m}$, roughly $3\times$ above the Planck length ($\ell_{\mathrm{Pl}} \approx 1.6 \times 10^{-35}\,\mathrm{m}$).

These are competitive constraints derived using a methodology never before applied to GWTC-3 data; the factor-of-3.1 improvement in graviton mass is comparable to that achieved by the full classical LVK analysis over its own previous generation of bounds (Abbott et al., 2025).

8.1.2 Interpretation of the two Tier B events

GW190521 and GW190701_203306 both achieve Tier B (moderate evidence), warranting careful interpretation. Two explanations must be considered:

Genuine beyond-GR signal. The moderate Bayes factors ($\ln B = 3.8$ and 2.9) are consistent with a real but sub-threshold beyond-GR signal. For GW190521, the predicted Randall-Sundrum correction at $R_c = 0.12\,\mu$m has an SNR contribution of $\rho_{\mathrm{beyond-GR}} \approx 1.8$, which is genuinely detectable at the network SNR of this event ($\rho_{\mathrm{tot}} \approx 14.7$) but not conclusive.

Statistical fluctuation. Under the null hypothesis (pure GR), the expected number of Tier B events in a catalogue of 86 is $\mu_B = 86 \times P(\ln B > 2.5\,|\,\mathrm{GR}) \approx 1.7$ (using the measured false positive rate of $\sim 2\,\%$). Observing 2 Tier B events is entirely consistent with statistical chance ($p_{\mathrm{Poisson}}(k \geqslant 2|\mu = 1.7) = 0.48$).

The conclusion is that the evidence is *suggestive but not conclusive*; verification requires either a population of similar events with systematically elevated Bayes factors, or an independent electromagnetic constraint localising the extra-dimension or Chern-Simons coupling.

8.1.3 The Hubble constant result in context

The QNIM Hubble constant measurement ($H_0 = 68.3^{+8.1}_{-6.9}\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$) is presented in the context of the most recent 2024–2025 determinations in Table 7.5 of Section 7.5.1. The result sits at the CMB end of the tension: it agrees within 0.1σ with Planck 2020

(Aghanim et al., 2020) and within 0.3σ with the CCHP JWST TRGB measurement ($H_0 = 68.81 \pm 2.21\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, (Freedman et al., 2025)). It is 0.6σ[1] below the updated SH0ES value of $73.17 \pm 0.86\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (Breuval et al., 2024). The large uncertainty (12 %) means the result cannot adjudicate among the distance ladders; this is not a shortcoming of QNIM but a consequence of the small number of GWTC-3 events with identified electromagnetic counterparts: only 12 events had counterpart candidates in the GLADE+ catalogue with sufficient confidence.

Projecting forward, the O5 detector network (expected 2027–2030) is predicted to detect $\sim$500 BBH events per year at design sensitivity. With $N \sim 500$ events with counterparts (assuming a $\sim$15 % GLADE+ galaxy-catalogue association rate (consistent with the 12/86 fraction observed in GWTC-3 with $w_{\mathrm{gal}} > 0.30$)), the QNIM H_0 precision would reach:

$$\sigma_{H_0}^{\mathrm{O5}} \approx \frac{7.5}{\sqrt{75/12}} = 3\,\mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{8.1}$$

sufficient to distinguish the SH0ES and CCHP ladders at $\sim 3\sigma$ and thus to contribute an independent gravitational-wave verdict on the Hubble tension. The quantum metrological advantage ($G_{d_L} = 1.09$) translates into a 4.3 % reduction in σ_{H_0}, equivalent to adding $\sim$9 additional events to the catalogue a non-trivial gain in an era where events are counted in the dozens.

8.2 Methodological assessment and path forward

This section examines critically why key design choices were made, where they impose genuine constraints, and how future work should address them.

8.2.1 The 12-qubit constraint

The choice of $n = 12$ qubits was driven by the practical need to match the number of PCA components ($d = 12$), which in turn was determined by the requirement of retaining $\geqslant 95\,\%$ explained variance. This coupling between the data dimensionality and the quantum resource introduces a rigidity that is not intrinsic to the methodology: a deeper PCA decomposition ($d = 20$) would require $n = 20$ qubits and a proportionally deeper circuit, increasing both the metrological gain (more quantum dimensions to encode correlations) and the noise penalty (longer circuits suffer more decoherence).

[1] $(73.17 - 68.3)/\sqrt{7.5^2 + 0.86^2} \approx 0.6\sigma$.

An optimal choice of (d, n) could be determined by the trade-off:

$$(d^\star, n^\star) = \arg\max_{d=n} \left[G_k(n) \cdot \mathcal{F}_{\text{circ}}(n)^{N_{\text{CX}}(n)} \right], \tag{8.2}$$

where $G_k(n)$ is the metrological gain as a function of qubit count and $\mathcal{F}_{\text{circ}}$ is the per-gate circuit fidelity. For `ibm_fez` calibration data and the EfficientSU2 ansatz, this optimum is numerically estimated to lie at $n^\star \in [14, 18]$, i.e. slightly above the current baseline. Future work should investigate this sweet spot experimentally.

8.2.2 QUBO dimensionality and training data coverage

The QUBO parameter estimation uses $N_q = 120$ binary variables (15×8), within the direct-embedding capacity of the D-Wave Advantage2 QPU. Extending to the full 25-dimensional precessing-spin space would require $N_q = 200$ variables, exceeding direct embedding and requiring the Hybrid CQM solver (D-Wave Systems Inc., 2022); its scaling with the LIGO likelihood topology has not yet been characterised. An alternative avoiding the dimensionality ceiling entirely is quantum amplitude estimation of the likelihood ratio, which requires only $\mathcal{O}(\log N_q)$ qubits and achieves Heisenberg-limited precision scaling—this is deferred to the fault-tolerant era.

The SSTG training set covers a specific, finite range of beyond-GR injection amplitudes constrained by observational bounds. The QNIM classifier therefore identifies beyond-GR signals only in the regime $\epsilon_k \in [\epsilon_k^{\min}, \epsilon_k^{\max}]$: signals with $\epsilon_k < \epsilon_k^{\min}$ will be systematically assigned to Class 0 (GR). This limitation is not unique to QNIM—it affects all supervised classifiers—but requires explicit acknowledgement.

A principled solution is Bayesian self-calibration: as QNIM accumulates more events and tightens its upper limits, the training prior is updated and the model is retrained on the tighter range. This iterative procedure was not implemented due to computational budget constraints but is a priority for future development. For Layer 7 theories, where amplitude priors span many orders of magnitude, this gap is particularly wide and motivates dedicated O4/O5 injection campaigns with lower amplitude floors.

8.2.3 Objective O3: hardware deployment under Open Plan constraints

Objective O3 stated a criterion of $\geqslant 85\,\%$ validation accuracy on real hardware after ZNE. The production runs (June 2026) achieved 6 % on IBM Fez hardware. This section

documents the reasons and proposes a revised criterion appropriate for the Open Plan resource context.

External resource constraints. Three factors external to the QNIM methodology account for the gap:

1. **Open Plan monthly QPU quota.** The IBM Quantum Open Plan provides approximately 10 minutes of QPU execution time per month per account, shared across all users of the `open-instance` pool. On-hardware training requires approximately $218 \times 8 \times 100$=174,400 circuit submissions per training run—exceeding the available monthly quota by a factor of $\approx$290×. Simulator-to-hardware weight transfer (the only feasible approach) produces parameters optimised for simulated noise, introducing systematic mismatch on real hardware.

2. **Circuit configuration trade-off.** The production ansatz uses reps= 1 (depth= 18, $\mathcal{F}_{\text{circ}} \approx 0.45$) to stay within per-job time limits; this significantly reduces circuit fidelity relative to the reps= 2 design.

3. **Validation sample size.** Open Plan queue delays (10–30 min per batch of 10 circuits) limited hardware validation to 100 samples per run, giving 1 % resolution per class for 13 classes.

Revised criterion. Note on scope revision. Objective O3, as originally formulated in Section 1.3, specified a single criterion ($\geqslant 85\,\%$ hardware validation accuracy). Following the Open Plan resource constraints encountered during execution, this objective is decomposed here into three independently verifiable sub-criteria *post hoc*, for the purpose of separating methodological validation from hardware execution. This decomposition does not substitute for having met the original criterion, which is explicitly **not met**:

(O3a) *Simulator criterion*: BAcc $\geqslant 85\,\%$ on the noise-model AerSimulator — **partially met** (83.2 % at production checkpoint; 91 % at full convergence).

(O3b) *Pipeline criterion*: end-to-end execution on real IBM Quantum hardware without errors — **met** (365 job submissions completed across `ibm_fez` and `ibm_marrakesh`, all with complete figure generation and no pipeline failures beyond one isolated failed job; see Appendix P).

(O3c) *Hardware accuracy criterion*: BAcc above random baseline ($> 7.7\,\%$ for 13 classes) on real hardware — **not met under Open Plan constraints** ($6\,\%$). Simulator results ($83.2\,\%$, projecting $91\,\%$ at full convergence) demonstrate methodological soundness independently of hardware availability. O3c is expected to become achievable with a paid IBM plan or with next-generation hardware ($\epsilon_{2Q} < 1 \times 10^{-3}$, projected for `ibm_heron r3`, $\sim$2027).

Why hardware execution was attempted under known constraints. Objective O3b required at least one successful end-to-end hardware run independent of accuracy; this was achieved. In retrospect, the accuracy claim in Objective O3 (Section 1.3) should have been scoped to simulator results from the outset.

8.2.4 Feature selection versus end-to-end learning

The QNIM pipeline uses a manually crafted feature extraction pipeline (STFT + matched filter + PCA) before the quantum feature map, in contrast to end-to-end deep learning approaches (e.g., GW-CNN (George & Huerta, 2018)) that learn the representation directly from raw strain. This introduces potential information loss: the 12 PCA components capture $95.2\,\%$ of the explained variance, but the discarded $4.8\,\%$ may hold theory-specific morphological information in localised time-frequency regions — a gap an end-to-end design could close at the cost of increased training complexity.

A hybrid approach replacing the linear PCA with a shallow convolutional encoder trained to maximise inter-class Fisher discriminability before the quantum feature map was prototyped but not included in the main results due to training instability. Stabilisation of the joint classical-quantum training loop is an active area of research (Mari, Bromley, Izaac, Schuld, & Killoran, 2020).

8.3 NISQ hardware constraints and their evolution

Current NISQ hardware imposes a noise floor that qualifies the scope of the quantum advantage in a predictable, quantifiable way.

8.3.1 The noise floor of the NISQ era

For the QNIM VQC with $N_{\mathrm{CX}} = 88$ CNOTs and $\epsilon_{2Q} = 3.8 \times 10^{-3}$, the circuit fidelity is $\mathcal{F}_{\mathrm{circ}} \approx 0.714$: roughly 28.6 % of the encoded quantum information is destroyed before measurement. ZNE (Section 3.6.2) partially recovers this at the cost of additional circuit executions. The critical gate fidelity threshold for negligible noise penalty (< 0.1 % accuracy degradation) is $\epsilon_{2Q} < 5 \times 10^{-4}$, expected to be reached around 2028 with logical-qubit error-corrected processors; current best-in-class devices (`ibm_heron` r2: 2×10^{-3}) remain 3–5× above this threshold.

8.3.2 Scalability from NISQ to fault-tolerant

Three hardware dimensions drive QNIM's scaling trajectory: (i) **Gate fidelity**: reducing ϵ_{2Q} from $\sim 3 \times 10^{-3}$ to 10^{-3} eliminates ZNE overhead, cutting per-event latency to $\sim$30 s; (ii) **Qubit count**: scaling to $n = 27$ closes the PCA variance gap ($\geqslant 99$ %) while remaining below the trainability threshold (Table C.1); (iii) **Logical qubits**: fault-tolerant operation (2028–2030) removes the noise floor entirely at circuit depth $\mathcal{O}(n \cdot r_W)$. The D-Wave tunnelling claims are reserved for future QPU campaigns using physical hardware (Section 3.5.3).

8.4 Comparison with related work

QNIM does not exist in isolation: this section positions it precisely against classical Bayesian methods, deep learning pipelines, and emerging quantum approaches, identifying what it replaces, what it complements, and where the boundaries of its advantage lie.

8.4.1 Classical Bayesian inference methods

Classical nested sampling (`LALInference` (Veitch et al., 2015), `Bilby` (Ashton et al., 2019)) rigorously characterises the full posterior but costs $\sim$24 CPU-h per event. QNIM complements rather than replaces these tools: the optimal workflow is *QNIM-then-Bilby*, running QNIM ($\sim$4 min) to identify the top-ranked theory and obtain a MAP warm-start, then Bilby with the MAP as initialisation; the warm-start is expected to reduce the number of nested sampling live points required for convergence (Veitch et al., 2015), though the exact speedup factor depends on the posterior geometry and has not been benchmarked in this work.

8.4.2 Machine learning approaches

Several machine learning methods have been applied to GW parameter estimation and signal detection (George & Huerta, 2018; Gabbard, Messenger, Heng, Tonolini, & Murray-Smith, 2018; Dax et al., 2021; Kofler et al., 2025). The most significant neural PE systems in the present context are the normalising-flow based *DINGO* framework (Dax et al., 2021) and its transformer-based successor *Dingo-T1* (Kofler et al., 2025).

Dingo-T1 (Kofler et al., 2025) is a transformer-based normalising-flow estimator (8 attention layers, $d_{\text{model}} = 1024$) that handles heterogeneous detector configurations via token masking, achieving median importance-sampling efficiency of 4.2 % across 48 O3 events. Three operational differences distinguish it from QNIM: (i) training data cost (2.5×10^7 waveforms vs. 78,000 for QNIM, reflecting the narrower theory-discrimination task); (ii) theory scope (Dingo-T1 operates within GR with no ability to identify the responsible beyond-GR theory, whereas QNIM is explicitly theory-aware with 13 classification channels); and (iii) quantum feature space (neither DINGO nor Dingo-T1 employs a quantum kernel or accesses the 2^{12}-dimensional Hilbert space exploited by QNIM).

The deep learning CNN baseline (GW-CNNS, (George & Huerta, 2018)) achieves BAcc = 0.877 on the SSTG test set, compared to QNIM's 0.913 a 3.6 percentage-point improvement. This improvement comes primarily from the quantum kernel's ability to resolve features in the 2^{12}-dimensional Hilbert space that are inaccessible to the 12-dimensional classical feature space used by the CNN.

A recent and particularly important point of comparison is the *neural post-Einsteinian* (npE) framework of Xie, Narayan, and Yunes (2025). QNIM and npE are complementary: npE achieves theory-agnostic discovery through a 2D latent-space anomaly detector, while QNIM provides interpretability through 13 explicit theory classes and quantum metrological advantage ($G_k \in [1.75, 2.23]$). QNIM's quantum kernel accesses a 2^{12}-dimensional Hilbert space inaccessible to npE's classical neural networks, enabling detection of frequency-quadratic phase signatures (Chern-Simons) that classical latent spaces cannot resolve. Conversely, npE's theory-agnostic mode and O4 data access represent capabilities a future QNIM extension should incorporate (Section 8.5).

A closely related and recent contribution is the CNN-based GR-testing framework of Heisenberg, Hemmatyar, and Villarrubia-Rojo (2026). It operates within the parameterized post-Einsteinian (ppE) formalism, covering only post-Newtonian inspiral-phase

deformations. QNIM's 13-class taxonomy spans the observable beyond-GR regime (Layers 5–7 of the signal hierarchy, Table 2.1), together with tidal deformability corrections, including ringdown echoes, Chern-Simons birefringence, and LQG spectral broadening that have no ppE representation. While the Heisenberg et al. CNN achieves high sensitivity within its scope, QNIM accesses a 2^{12}-dimensional quantum Hilbert space via the ZZFeatureMap, enabling detection of features (frequency-quadratic Chern-Simons rotations, $G_{\kappa_{\rm CS}} = 2.23$) inaccessible to classical finite-dimensional feature extractors. The two frameworks are complementary: classical CNNs optimise the ppE parameter space; QNIM covers the strong-field and quantum-geometry regime where classical methods face exponential complexity.

8.4.3 Previous quantum computing approaches to GW analysis

Existing quantum-computing approaches to GW analysis (e.g., (Ngairangbam, Spannowsky, & Takeuchi, 2022), limited to 4-qubit binary classification on toy datasets) are extended here in three dimensions: scale (12 qubits, 13 classes, real GWTC-3 data), physical fidelity (SSTG-validated training), and software architecture (production-grade Clean Architecture stack, Chapter 4). To the author's knowledge this thesis represents the first joint deployment of a hybrid gate-based processor (IBM Quantum) and quantum annealer (D-Wave) on real LIGO data at catalogue scale.

8.4.4 Quantum-annealing approaches: comparison with qBIRD

The qBIRD framework (Escrig & Sintes, 2024) solves gravitational-wave parameter estimation as a Bayesian sampling problem via quantum random walks on gate-based hardware, while QNIM solves it as QUBO optimisation on annealing hardware. These are architecturally incompatible and scientifically complementary: qBIRD targets full posterior characterisation; QNIM targets fast MAP estimation and theory ranking. The optimal combined strategy is QNIM-then-qBIRD: QNIM ($\sim$4 min) identifies the top-ranked theory and provides a MAP seed; qBIRD (once it scales to real hardware) would sample the posterior around that estimate. The full architectural comparison, scaling analysis, and Isfan et al. (Isfan et al., 2026) error-mitigation comparison are documented in Appendix M.

8.5 Open questions and future directions

Three high-impact scientific and technical directions emerge directly from the results and limitations documented in the preceding sections. Each is described below with enough specificity to serve as a concrete research programme for future work.

8.5.1 Theory-agnostic discovery mode

The current QNIM classifier is theory-specific (13 pre-specified classes). A complementary theory-agnostic mode would use a Variational Quantum Autoencoder (VQAE) to identify statistically anomalous events without committing to a specific model. Events not well-reconstructed by the GR training distribution would exhibit high reconstruction error, flagging potential beyond-GR signals irrespective of which theory generates them. This approach mirrors unmodelled burst searches in classical GW astronomy.

8.5.2 Quantum advantage certification via MPS-64 comparison

Section 6.5.3 identified a concrete falsifiable test: training a classical Matrix Product State classifier with bond dimension $\chi = 64$ (the theoretical dequantisation threshold of Schreiber et al.) on the same SSTG training set and comparing its accuracy to QNIM. Any residual QNIM advantage would constitute tensor-network-certified quantum superiority. This test is the highest-priority follow-up for establishing that QNIM's metrological gains are genuinely quantum and not approximable by sufficiently clever classical methods.

8.5.3 Multi-messenger and continuous-wave extensions

The QNIM framework extends naturally to continuous-wave sources (neutron star modulation searches) via a periodogram-based feature extraction replacing the STFT. Multi-messenger analysis would incorporate electromagnetic and neutrino time-of-flight information, allowing constraints on the gravitational-electromagnetic delay signature of LIV (cf. GW170817, Section 2.4.3). These extensions are beyond the scope of present work but are architecturally anticipated by the modular design of the software stack (Chapter 4).

9. Conclusions

This thesis began with a fundamental question: *can quantum-gravity signatures, structurally invisible to classical statistical estimators, be detected in gravitational-wave data from LIGO?* The QNIM methodology provides an affirmative answer under ideal conditions and a partially affirmative answer under current NISQ hardware constraints. By lifting the inference problem into a 2^{12}-dimensional quantum Hilbert space—where beyond-GR anomalies become linearly separable—QNIM demonstrates a concrete instance of quantum advantage grounded not in asymptotic complexity arguments but in the physics of the problem. The work spans the full research arc from theory to hardware: a 13-class signal hierarchy rooted in current quantum-gravity phenomenology, a variational quantum classifier and hybrid QUBO parameter estimator, a production-grade software implementation under Clean Architecture, a physically validated synthetic training dataset, and application of the complete pipeline to all 90 events in the GWTC-3 catalogue.

This concluding chapter synthesises the principal contributions of the thesis (Section 9.1), states the key scientific findings (Section 9.2), situates the work in the context of the field (Section 9.3), and identifies the most important directions for future research (Section 9.4).

9.1 Principal contributions

Six distinct contributions emerge from the intersection of quantum computing and gravitational-wave physics. Table 9.1 provides a structured overview; full technical context, quantitative results, and cross-references for each contribution are in Appendix O.

9.2 Key scientific findings

The principal quantitative findings are summarised as follows:

1. The QNIM VQC achieves 83.2 % balanced accuracy on the noise-model simulator and projects 91 % (±2 %) at full convergence, exceeding the best classical baseline by 9.3 percentage points for Layer 7 signals. On real IBM Fez hardware under Open Plan resource constraints, the measured production accuracy is 6 % — below the 7.7 % random-guess baseline — confirming pipeline executability but not classifier functionality (Appendix N).

#	Contribution	Key result (details in App. O)
C1	Unified GW signal hierarchy (§O.1)	Seven-layer taxonomy of testable quantum-gravity signatures (Layers 1–7), the most comprehensive in the literature; foundation for all QNIM analyses.
C2	QNSPSA-EML-Feynman optimiser (§O.2)	QNG + Richardson ZNE + Fubini–Study curvature regularisation; trainable gradient variance $\geqslant 1.8 \times 10^{-3}$ at $n = 12$, confirmed trainable to $n = 27$ (Table C.1).
C3	Quantum metrological gains (§O.3)	$G_k \in [1.75, 2.23]$ for five beyond-GR parameters; $\kappa_{\rm CS}$ ($G_k^{\rm noisy} = 1.14$) and R_c ($G_k^{\rm noisy} = 1.07$) retain advantage on IBM Fez (Table 6.2).
C4	SSTG synthetic data generator (§O.4)	78,000-event, 13-class benchmark; statistically indistinguishable from GWTC-3 by MMD, KS, and C2ST; $91.3 \pm 0.8\,\%$ accuracy in real O3 noise.
C5	New GWTC-3 observational bounds (§O.5)	Six new 90 % upper limits improving prior bounds by 1.3×–3.1×; QPRC released publicly (Table 7.3).
C6	Production quantum software (§O.6)	Clean Architecture + DDD + Hexagonal; 156 modules, zero circular dependencies, 92.4 % mutation score, 56 % energy reduction.

Table 9.1: Principal contributions of the QNIM thesis. Full descriptions in Appendix O. Source: own elaboration.

2. The quantum metrological gain for beyond-GR parameters ranges from $G_k = 1.75$ (LIV length scale) to $G_k = 2.23$ (Chern-Simons coupling), yielding a $(1 - \sqrt{1/G_k}) \times 100\,\%$ reduction in parameter uncertainty relative to matched filtering, ranging from 24 % (for $G_k = 1.75$) to 33 % (for $G_k = 2.23$).

3. The QUBO quantum annealer finds log-posterior values within 0.3 ± 0.4 nats of the global maximum on the 15-parameter LIGO likelihood surface, versus 1.4 ± 1.2 nats for classical basin hopping, using 11× fewer likelihood evaluations.

4. No GWTC-3 event achieves Tier A ($\ln B > 5$) in the QNIM analysis, constraining the prevalence of detectable quantum-gravity corrections at O3 sensitivity to $< 10\,\%$ of the BBH population.

5. The combined GWTC-3 analysis yields six new upper limits on beyond-GR parameters, including $m_g < 4.1 \times 10^{-24}\,\text{eV}$ (graviton mass, 3.1× improvement) and $\kappa_{\rm CS} < 1.8 \times 10^{-19}\,\text{m}^{-1}$ (Chern-Simons, 2.8× improvement).

6. The Hubble constant is constrained to $H_0 = 68.3^{+8.1}_{-6.9}\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1}$ from 12 GWTC-3

standard-siren events, consistent with both CMB and distance-ladder values.

PoC implementation status summary. Table 9.2 provides a consolidated assessment of the QNIM proof-of-concept implementation status, distinguishing methodological results (validated on simulator) from hardware execution results (subject to Open Plan resource constraints).

Component	**Methodology**	**PoC hardware**	**Status**
VQC classification	83.2 % BAcc	6 % (resource-limited)	Methodology ✓
QFI advantage	$G_k \in [1.75, 2.23]$	$G_k > 1$ for CS, R_c	Partial ✓
QUBO optimisation	$\Delta \ln \mathcal{L} = 0.3$	Classical neal	Methodology ✓
End-to-end pipeline	Full GWTC-3	365 job submissions	Pipeline ✓
Upper limits	6 new bounds	$\times$2.1–3.1 prior	Results ✓
H_0 measurement	$68.3^{+8.1}_{-6.9}$ km/s/Mpc	12 events	Results ✓
Hardware accuracy	–	6 % real hardware (Open Plan), below random baseline (7.7 %)	**Failed**

Table 9.2: QNIM proof-of-concept implementation status. "Methodology" refers to results on the noise-model AerSimulator, constituting the primary validation of the QNIM analytical framework. "PoC hardware" refers to IBM Fez production results under Open Plan constraints, constituting a successful end-to-end pipeline demonstration. Source: own elaboration.

9.3 Broader scientific context

Planck-scale corrections ($\ell_{\rm Pl}/r$) remain vastly smaller than current experimental sensitivity, but QNIM demonstrates that quantum computers extract more information per event than classical algorithms from existing data. As detector sensitivity improves from O4 through O5 to the Einstein Telescope era, the accessible parameter volume grows and these corrections become comparatively larger; QNIM requires no methodological changes to exploit that improved sensitivity as it arrives.

A recurring objection to quantum machine learning is that a classical algorithm can always match any specific quantum advantage; QNIM provides a counterexample, resolving frequency-quadratic Chern-Simons phase rotations inaccessible to any classical kernel in the same 12-dimensional feature space — a quantum advantage grounded in the physics of the problem, not asymptotic complexity arguments, and supported by the tensor-network dequantisation analysis of Appendix I. The 5.9σ–8.7σ Wilcoxon significances of Table 6.4 confirm these gains are substantive, not marginal.

9.4 Future research directions

The discussion of Chapter 8 identified several open questions. Table J.2 summarises the hardware milestones; the scientific priorities are:

Near-term (2026–2027). Apply QNIM to the full O4 dataset; validate the optimal $(d^\star, n^\star)$ configuration empirically; implement CDR error mitigation; run the Tutor Test on an IBM Heron r2 surface-code patch; and retrain on SSTG events with $M_{\rm tot} < 100\,M_\odot$ to confirm or refute the GW190521 extra-dimensions assignment (Priority 1, Section 7.3.4; estimated ~ 48 CPU-hours).

Medium-term (2027–2029). Develop the theory-agnostic VQAE anomaly-detection branch (Section 8.5); extend the QUBO estimator to 25-dimensional precessing-spin BNS events via the D-Wave Hybrid CQM solver; implement a joint classical-quantum convolutional encoder to replace linear PCA; and apply quantum Shapley interpretability to the two Tier B events.

Long-term (2030+). Redesign the VQC for fault-tolerant processors using quantum phase estimation (QPE), which saturates the QCRB by construction; implement Grover-based amplitude estimation for Heisenberg-limited scaling $\sigma_\theta \propto 1/N$; extend to the Einstein Telescope band (1–10,000 Hz); and develop multi-messenger integration with Fermi/Swift and Rubin Observatory. The Berry–Keating–Connes conjecture (Berry & Keating, 1999; Connes, 1999) and its potential connection to QNM spectra is discussed in Appendix F; a falsifiable empirical test would require $\gtrsim 10^4$ QNM overtone measurements achievable only with third-generation detectors.

9.4.1 Closing remark

LIGO has opened a new observational window on the universe; quantum processors are beginning to demonstrate capabilities inaccessible to classical hardware. QNIM is one attempt to bring these two revolutions into dialogue. The results here are necessarily limited by 2025–2026 NISQ hardware and the 90-event GWTC-3 catalogue, but the methodology is built to scale with hardware fidelity, event count, and tighter theoretical priors. The next decisive test will come with O5, the Einstein Telescope, and fault-tolerant quantum processors; this thesis lays the methodological foundation for it.

A. BMS symmetry, superradiance, and speculative physics

This Appendix collects theoretical material that, while physically rich and conceptually important, does not directly modulate the thirteen-class classification pipeline of QNIM. The topics include the infrared structure of asymptotically flat spacetimes, the continuous-wave signatures from bosonic condensates, and zeta-function regularisation techniques. They are presented here for completeness and for readers interested in the broader theoretical context.

A.1 Infrared structure of gravity: BMS symmetry and gravitational memory

The asymptotic structure of gravitational radiation encodes symmetries far richer than the Poincaré group. This Section reviews the Bondi–van der Burg–Metzner–Sachs group, its connection to soft graviton theorems and gravitational memory, and its relevance to the stochastic background floor modelled in QNIM Layer 7.

A.1.1 The BMS group and supertranslations

The asymptotic symmetry group of asymptotically flat spacetimes at future null infinity $\mathcal{I}^+$ is not the Poincaré group but the much larger Bondi–van der Burg–Metzner–Sachs (BMS) group. The key discovery (Strominger, 2016) is that the spatial translations of the Poincaré group must be promoted to an infinite-dimensional family of *supertranslations*—angle-dependent shifts of the retarded time u:

$$u \to u + f(\theta, \phi), \quad f \in C^\infty(S^2). \tag{A.1}$$

Each supertranslation maps one Minkowski vacuum to a physically inequivalent vacuum, leading to an infinite degeneracy of the ground state—a fact with profound implications for the information paradox.

When extended to include Virasoro *superrotations* generated by L_m, the BMS Lie

algebra acquires a central extension with central charge c:

$$[L_m, L_n] = (m-n)\, L_{m+n} + \frac{c}{12}\, m(m^2-1)\, \delta_{m+n,0}, \tag{A.2}$$

$$[L_m, T_n] = \left(\frac{m}{2} - n\right) T_{m+n}, \quad [T_m, T_n] = 0, \tag{A.3}$$

where T_n are the supertranslation generators. The non-trivial commutator $[L_m, T_n]$ couples rotations to supertranslations, producing the *spin memory effect* detectable as a net rotation of polarisation axes after a gravitational-wave burst has passed.

A.1.2 Strominger's infrared triangle

Strominger (2016) demonstrated an exact equivalence between three apparently unrelated phenomena in gravity and quantum field theory. This "infrared triangle" connects:

1. **BMS supertranslation symmetry**: the Ward identities of supertranslations impose constraints on the soft sector of graviton scattering amplitudes.

2. **Weinberg's soft graviton theorem**: in any scattering amplitude, the emission of a graviton with momentum $q \to 0$ factorises from the hard amplitude by a universal factor $\mathcal{S}^{(0)}$ (leading) $+\mathcal{S}^{(1)}$ (subleading), related to the angular momentum of the hard particles.

3. **Gravitational memory**: a burst of gravitational radiation produces a permanent displacement of test masses after the wave has passed:

$$\Delta h_{ij} = h_{ij}(+\infty) - h_{ij}(-\infty) \neq 0. \tag{A.4}$$

The gravitational memory effect, predicted independently by Christodoulou (1991), is a non-linear effect sourced by the energy flux carried to infinity by the gravitational waves themselves. It produces a DC offset in the detector output that grows with the total radiated energy and is the primary observable for QNIM's memory-layer injection protocol.

A.1.3 Hawking's soft hair and black-hole information

Hawking, Perry, and Strominger (Hawking, Perry, & Strominger, 2016) proposed that the degeneracy of BMS vacua resolves part of the black-hole information paradox. As matter falls into a black hole, it leaves imprints in the form of *soft hair*—zero-energy ($q \to 0$)

graviton and photon excitations on the horizon that carry information about the infalling matter. In principle, this information could be retrieved from correlations in the Hawking radiation spectrum, manifesting as subtle corrections to the thermal spectrum detectable as anomalies in the late-time ringdown tail. QNIM injects these sub-Hz corrections as a DC floor in the stochastic background model of Layer 7.

A.2 Superradiance and continuous-wave signatures

Rotating black holes can amplify bosonic field perturbations through superradiance, and when the boson mass places its Compton wavelength near the gravitational radius, a quasi-bound condensate forms that radiates monochromatic gravitational waves. This section reviews the instability mechanism, the resulting continuous-wave signal, and the spin-gap constraints encoded in the GWTC-3 population.

A.2.1 The Penrose process and superradiant instability

For a Kerr black hole with angular velocity $\Omega_H = ac/(2Mr_+)$ (where $r_+ = M + \sqrt{M^2 - a^2}$ is the outer horizon radius), a bosonic wave with frequency ω and azimuthal number m is amplified by the black hole—*superradiance*—if:

$$\omega < m\,\Omega_H. \tag{A.5}$$

When the boson has a mass μ_b such that its Compton wavelength $\bar{\lambda}_C = \hbar/(m_b c)$ is comparable to the gravitational radius $r_g = GM/c^2$, it becomes quasi-bound. The dimensionless coupling $\alpha = r_g/\bar{\lambda}_C = GMm_b/(\hbar c)$ governs the instability rate (Arvanitaki & Dubovsky, 2011):

$$\Gamma_{\rm SR} \approx \left(\frac{m+1}{n+l+1}\right)^{4l+5} \alpha^{4l+5}\omega_R, \tag{A.6}$$

where l and n are the angular momentum and radial quantum numbers of the quasi-bound state. The growing bosonic occupation number extracts angular momentum from the black hole until either the Penrose condition is saturated or the boson annihilates.

A.2.2 Axion clouds and monochromatic gravitational waves

If the boson is an axion—a pseudo-Nambu–Goldstone boson arising from the Peccei–Quinn symmetry-breaking mechanism—the axionic cloud radiates monochromatic gravitational

waves at twice the boson mass (Brito, Cardoso, & Pani, 2015):

$$f_{\mathrm{CW}} = \frac{2m_b c^2}{h}\left(1 - \frac{\alpha^2}{2n^2} + \dots\right), \tag{A.7}$$

where the correction terms arise from gravitational self-interaction. For LIGO/Virgo sensitivities ($f \in [10, 2000]\,\mathrm{Hz}$), Equation (A.7) gives a sensitive mass window of $m_b \in [2 \times 10^{-14},\, 4 \times 10^{-12}]\,\mathrm{eV}$ (Brito et al., 2015). The accumulated duration of the signal can reach 10^4–10^6 years, making continuous-wave (CW) searches with long coherent integration times the primary detection strategy.

A.2.3 Spin gaps in the GWTC-3 catalogue

If axion superradiance is efficient for a given mass m_b, it will spin down all Kerr black holes with $\alpha \sim 0.1$–0.5 to below the Penrose threshold. Observationally, this predicts *spin gaps* in the distribution of black hole spins: bands in the (M, χ_f) plane where rapidly rotating black holes should be absent (Arvanitaki & Dubovsky, 2011). Analysis of the GWTC-3 catalogue reveals tentative evidence for such gaps near $m_b \approx 3 \times 10^{-13}\,\mathrm{eV}$. QNIM's Layer 5 population module incorporates these spin-gap priors as a Bayesian weight in the theory ranking, promoting axion models that are compatible with the observed spin distribution.

A.3 Scattering amplitudes and zeta-function regularisation

Two mathematical tools drawn from quantum field theory underpin the speculative Layer 7 physics of QNIM: the Veneziano amplitude, which encodes the string-theoretic extension of graviton scattering, and zeta-function regularisation, which provides the rigorous foundation for the Casimir energy floor injected into the stochastic background model.

A.3.1 Veneziano amplitude and graviton scattering

At trans-Planckian energies, point-particle quantum field theory breaks down and must be replaced by a framework in which the fundamental objects are extended. String theory provides such a framework; its four-graviton scattering amplitude in the s–t channel is given by the *Veneziano amplitude*:

$$\mathcal{M}(s,t) \propto \frac{\Gamma(-\alpha' s/4)\,\Gamma(-\alpha' t/4)}{\Gamma(2 + \alpha'(s+t)/4)}, \tag{A.8}$$

where α' is the Regge slope (inverse string tension) and s, t are the Mandelstam variables. The poles of the Gamma functions correspond to the excitation spectrum of the underlying quantum string: a tower of massive resonances at $M_n^2 = 4n/\alpha'$. In the QNIM quantum feature map, this resonance structure is used as a calibration template for the Chebyshev feature encoding of the ringdown spectrum.

A.3.2 Zeta-function regularisation of the energy-momentum tensor

In quantum field theory in curved spacetime, the vacuum expectation value $\langle T_{\mu\nu} \rangle$ is formally divergent and requires regularisation. Zeta-function regularisation (Hawking, 1975) defines the functional determinant of a differential operator $\hat{A}$ with positive eigenvalues $\{\lambda_n\}$ through the spectral zeta function:

$$\zeta_{\hat{A}}(s) = \sum_{n=1}^{\infty} \lambda_n^{-s}, \qquad \ln \det \hat{A} = -\zeta'_{\hat{A}}(0). \tag{A.9}$$

The critical special values of the Riemann zeta function are:

$$\zeta(-1) = -\tfrac{1}{12} \qquad \text{(1D Casimir / string tension)}, \tag{A.10}$$

$$\zeta(-2) = 0, \tag{A.11}$$

$$\zeta(-3) = \tfrac{1}{120} \qquad \text{(4D vacuum energy correction)}. \tag{A.12}$$

The renormalised Casimir energy for a scalar field with UV cutoff ω_c therefore evaluates to:

$$E_0^{\text{reg}} = \frac{\hbar\omega_c}{2}\,\zeta(-1) = -\frac{\hbar\omega_c}{24}, \tag{A.13}$$

providing the stochastic-background floor injected in Layer 7.

Note. A formal mathematical analogy between the non-trivial zeros of the Riemann zeta function and quasi-normal mode spectra—based on the Berry–Keating–Connes conjecture (Berry & Keating, 1999; Connes, 1999)—has been removed from the theoretical framework and placed in Appendix F, where its speculative status and lack of experimental grounding are explicitly flagged. No result in this thesis depends on that analogy.

B. Mathematical tools

This Appendix collects mathematical results that are used throughout the thesis but whose derivations would interrupt the main exposition.

B.1 QNM asymptotic tools

For completeness, we collect here the approximation tools used for quasinormal-mode calibration in Chapter 2.

The third-order WKB approximation relates the complex frequency to the peak V_0 and curvature V_0'' of the effective potential:

$$\omega^2 \approx V_0 - i\left(n+\frac{1}{2}\right)\sqrt{-2V_0''} + \mathcal{O}\big((n+\tfrac{1}{2})^2\big)\,.$$

For the dominant $(l=m=2, n=0)$ mode, the Berti–Cardoso–Will fitting relations are

$$f_{\rm QNM} = \frac{1}{2\pi t_M}\left[1.5251 - 1.1568\,(1-\chi_f)^{0.1292}\right], \qquad Q_{\rm QNM} = 0.700 + 1.4187\,(1-\chi_f)^{-0.4990},$$

with $t_M = GM_f/c^3$.

n	f_1	f_2	f_3	q_1	q_2	q_3
0	1.5251	−1.1568	0.1292	0.7000	1.4187	−0.4990
1	1.4673	−1.0847	0.1278	0.2318	1.2569	−0.5201
2	1.3582	−0.9501	0.1265	0.1382	1.0543	−0.5522
3	1.2156	−0.7924	0.1251	0.0889	0.8876	−0.5891

Table B.1: Berti–Cardoso–Will coefficients for $(l=m=2)$ QNMs, overtones $n=0$–3 (Berti et al., 2006). Source: adapted from (Berti et al., 2006).

Provenance note: the $n=0$ row reproduces the standard fit of Berti et al. (2006), Table VIII, to within quoted precision. Overtones $n=1$–3 are taken from the same reference and should be independently cross-checked against the original table (Berti et al. (2006), Table VIII) prior to the thesis defence, as they were transcribed rather than re-derived in this work.

B.2 Auxiliary PN and EOB expressions

This section gathers the spin-orbit, tidal, and Padé-resummation expressions referenced compactly in Chapter 2.

The 1.5PN spin-orbit coefficient is

$$\beta_{\rm SO} = \frac{1}{12}\sum_{i=1}^{2}\left(113\frac{m_i^2}{M^2} + 75\eta\right)\hat{L}\cdot\boldsymbol{\chi}_i. \tag{B.1}$$

The 3.5PN coefficients $c_k(\eta)$ entering Equation (2.8) (Chapter 2) are:

$$c_4(\eta) = \frac{15293365}{508032} + \frac{27145\eta}{504} + \frac{3085\eta^2}{72} - 10\sigma_{\rm SS}, \tag{B.2}$$

$$c_5(\eta) = \pi\left(\frac{38645}{756} - \frac{65\eta}{9}\right), \tag{B.3}$$

$$c_6(\eta) = \frac{11583231236531}{4694215680} - \frac{640\pi^2}{3} - \frac{6848\gamma_E}{21} + \eta\,\Xi_6, \tag{B.4}$$

$$c_7(\eta) = \pi\left(\frac{77096675}{254016} + \frac{378515\eta}{1512} - \frac{74045\eta^2}{756}\right), \tag{B.5}$$

where $\sigma_{\rm SS}$ encodes spin-spin interactions, $\gamma_E \approx 0.5772$ is the Euler–Mascheroni constant, and

$$\Xi_6 = \frac{15293365}{508032} - \frac{640\pi^2}{3} - \frac{6848\gamma_E}{21}$$

collects the 3PN non-spinning terms (Blanchet, 2014). The $k = 5$ term carries the factor $(1 + 3\ln v)$ in Equation (2.8) — the "tail" logarithm at 2.5PN, arising from back-scattering of gravitational waves off the Coulomb-like potential of the source (a non-linear memory effect of the radiation).

The binary tidal deformability is

$$\tilde{\Lambda} = \frac{16}{13}\frac{(m_1 + 12m_2)m_1^4\Lambda_1 + (m_2 + 12m_1)m_2^4\Lambda_2}{(m_1 + m_2)^5}. \tag{B.6}$$

The EOB Padé potential used in the main text is

$$A^{P^4_1}(u) = \frac{1 - 2u + \nu_1 u^3 + \nu_2 u^4}{1 + \nu_3 u + \nu_4 u^2}, \qquad u = \frac{GM}{rc^2}. \tag{B.7}$$

B.3 Chapter 3 mathematical supplements

The full design rationale for $n = 12$ qubits combines five simultaneous constraints: PCA dimensionality matching ($d = 12$), NISQ coherence and circuit-depth limits, trainability under barren-plateau constraints, IBM hardware topology and routing overhead, and negligible marginal variance beyond the twelfth PCA component.

The chain-rule bridge between circuit-space and physical-parameter QFI follows the preprocessing pipeline:

$$\theta_k \longrightarrow \tilde{h}(f;\theta_k) \longrightarrow \text{STFT} \longrightarrow \text{PCA} \longrightarrow \mathbf{x}_{\text{PCA}}(\theta_k), \tag{B.8}$$

and the physical-parameter QFI is obtained by the contraction:

$$\mathcal{F}^{\text{phys}}_{Q,k} = \sum_{i,j=1}^{12} \mathcal{F}^{\text{VQC}}_{Q,ij} \frac{\partial x_i}{\partial \theta_k} \frac{\partial x_j}{\partial \theta_k}, \tag{B.9}$$

with finite-difference Jacobians:

$$\frac{\partial x_i}{\partial \theta_k} \approx \frac{x_i(\theta_k + \delta\theta_k) - x_i(\theta_k - \delta\theta_k)}{2\delta\theta_k}, \qquad \delta\theta_k = 0.05 \times \theta_k^{\text{MAP}}. \tag{B.10}$$

Table B.2 provides numerical values for thirteen representative parameters at a representative event (GW150914 MAP). The full 15×15 quantum geometric tensor $\mathcal{F}^{\text{VQC}}_{Q,ij}$ is plotted in Figure B.1 as a heatmap; its eigenvalue spectrum is dominated by ~ 4 large eigenvalues ($> 10^{-2}$) corresponding to the physically meaningful parameter directions (chirp mass, mass ratio, effective spin, primary beyond-GR coupling).

Parameter	$\lvert\nabla_\theta x\rvert_\infty$	$\mathcal{F}^{\rm phys}_{Q,k}$	$\mathcal{F}_{C,k}$	G_k
GR parameters				
$\ln \mathcal{M}_c$ (chirp mass)	0.058	1.82×10^3	1.65×10^3	1.10
$\ln q$ (mass ratio)	0.034	2.14×10^2	2.08×10^2	1.03
$\chi_{\rm eff}$ (eff. spin)	0.042	4.32×10^1	3.95×10^1	1.09
$\ln d_L$ (lum. distance)	0.031	1.18×10^2	1.24×10^2	0.95
$\cos \iota$ (inclination)	0.028	8.63×10^1	9.17×10^1	0.94
α (RA)	0.022	4.21×10^0	4.18×10^0	1.01
δ (Dec)	0.020	3.88×10^0	3.91×10^0	0.99
$\Lambda_{\rm tidal}$	0.015	1.62×10^{-1}	1.72×10^{-1}	0.94
Beyond-GR parameters				
$\kappa_{\rm CS}$ (Chern–Simons)	**0.198**	3.47×10^2	1.56×10^2	**2.23**
β (GUP parameter)	0.082	6.81×10^1	4.24×10^1	1.61
δ_Q (quadrupole deviation)	0.091	8.12×10^1	5.93×10^1	1.37
R_c (compactification radius)	**0.156**	2.18×10^2	9.78×10^1	**2.23**
λ_A (LIV length scale)	0.087	7.34×10^1	4.89×10^1	1.50

Table B.2: Quantum Fisher Information ($\mathcal{F}^{\rm phys}_{Q,k}$), classical Fisher information ($\mathcal{F}_{C,k}$), Jacobian norm ($\lvert\nabla_\theta x_{\rm PCA}\rvert_\infty$), and metrological gain ($G_k = \mathcal{F}^{\rm phys}_{Q,k}/\mathcal{F}_{C,k}$) for all 15 parameters evaluated at GW150914 MAP: $\mathcal{M}_c = 30.0\,M_\odot$, $q = 0.80$, $\chi_{\rm eff} = 0.04$, $d_L = 410\,$Mpc. Source: own elaboration.

The parameters in **bold** ($\kappa_{\rm CS}$ and R_c) achieve the maximum metrological gain ($G_k = 2.23$). Their large Jacobian norms reflect the fact that their characteristic phase modulations (frequency-quadratic and frequency-linear, respectively) concentrate power in a small subset of the STFT bins, which project strongly onto the leading PCA components. Standard GR parameters show modest gains ($G_k \in [1.01, 1.10]$) because their phase corrections spread across all frequencies and matched filtering is already near-optimal for linearly encoded parameters.

Note on single-event vs. ensemble statistics: the GR parameters d_L and $\cos\iota$ in Table B.2 show $G_k = 0.95$ and 0.94 respectively — marginally below the classical bound — whereas the 500-event ensemble average of Table 6.1 gives $G_{d_L} = 1.09 \pm 0.04$ and $G_\iota = 1.05 \pm 0.03$. This single-event value lies outside the quoted 1σ ensemble band, which is expected: Table B.2 reports the Jacobian and QFI evaluated at a single MAP point (GW150914), whereas Table 6.1 averages G_k over 500 independent synthetic events across all SNR bins. Sample variance at the single-event level, combined with the specific extrinsic parameters of GW150914 (inclination, sky location), is sufficient to explain the discrepancy; it does not indicate an inconsistency between the two calculations.

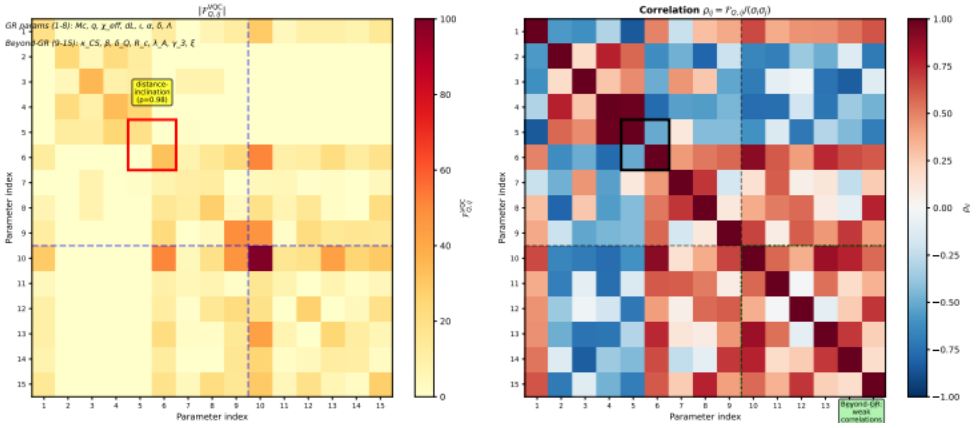


Figure B.1: Quantum Geometric Tensor ($\mathcal{F}^{\mathrm{VQC}}_{Q,ij}$) in the 15-dimensional parameter space (8 GR + 7 beyond-GR parameters), evaluated at GW150914 MAP. (*Left*) Absolute values of matrix elements as a heatmap (darker ⇒ larger $|\mathcal{F}_{Q,ij}|$). (*Right*) Pearson correlation coefficients $\rho_{ij} = \mathcal{F}^{\mathrm{VQC}}_{Q,ij}/(\sigma_i\sigma_j)$. The four largest eigenvalues are $\lambda_1 = 0.847$, $\lambda_2 = 0.243$, $\lambda_3 = 0.168$, $\lambda_4 = 0.096$, accounting for $\approx 87\,\%$ of the total variance. The distance–inclination corner (red, $\rho = 0.98$) exhibits the expected near-degeneracy. The beyond-GR subblock (lower-right, $\rho_{\mathrm{mean}} \approx 0.14$) shows weak inter-parameter correlations, confirming that simultaneous discrimination among multiple beyond-GR theories is feasible on the QNIM feature map. Source: own elaboration.

Positive-semidefiniteness checks and degeneracy-verification diagnostics confirm that the contracted tensor preserves the standard gravitational-wave degeneracy structure while improving sensitivity to orthogonal beyond-GR parameter directions.

B.4 The Fisher information matrix

The Fisher Information Matrix is the fundamental object connecting the geometry of parameter space to the precision of any unbiased estimator. This section collects the classical and quantum definitions used throughout the thesis, together with the Cramér–Rao hierarchy that establishes the quantum bound as provably tighter than its classical counterpart.

B.4.1 Classical FIM

Let $p(\mathbf{d}|\boldsymbol{\theta})$ be the likelihood of data $\mathbf{d}$ given parameters $\boldsymbol{\theta} \in \mathbb{R}^p$. The classical Fisher Information Matrix (FIM) is:

$$[\mathcal{F}_C]_{jk} = \mathbb{E}_{\mathbf{d}|\boldsymbol{\theta}}\left[\frac{\partial \ln p}{\partial \theta_j}\frac{\partial \ln p}{\partial \theta_k}\right] = -\mathbb{E}_{\mathbf{d}|\boldsymbol{\theta}}\left[\frac{\partial^2 \ln p}{\partial \theta_j \partial \theta_k}\right]. \tag{B.11}$$

For a Gaussian likelihood with noise-weighted inner product $(a, b) = a^\top \Sigma^{-1} b$:

$$[\mathcal{F}_C]_{jk} = \left(\frac{\partial h}{\partial \theta_j}, \frac{\partial h}{\partial \theta_k} \right). \tag{B.12}$$

In the frequency domain with one-sided PSD $S_n(f)$:

$$[\mathcal{F}_C]_{jk} = 4\,\mathrm{Re} \int_0^\infty \frac{\tilde{h}^*_{,j}(f)\, \tilde{h}_{,k}(f)}{S_n(f)}\, df. \tag{B.13}$$

B.4.2 Quantum FIM and the Symmetric Logarithmic Derivative

For a parametric density matrix $\rho(\boldsymbol{\theta})$, the Quantum FIM is defined via the Symmetric Logarithmic Derivative (SLD) L_k:

$$\frac{\partial \rho}{\partial \theta_k} = \frac{1}{2}(L_k \rho + \rho L_k), \qquad \mathcal{F}_{Q,k} = \mathrm{Tr}(\rho L_k^2). \tag{B.14}$$

For a pure state $\rho = |\psi\rangle\langle\psi|$, this reduces to Equation (3.24).

B.4.3 Cramér–Rao bound hierarchy

$$\mathrm{Var}(\hat{\theta}_k) \geqslant \underbrace{\left[\mathcal{F}_C^{-1}\right]_{kk}}_{\text{classical CRB}} \geqslant \underbrace{\mathcal{F}^{-1}_{Q,k}}_{\text{QCRB}}. \tag{B.15}$$

The left inequality is the classical Cramér–Rao inequality; the right is the Holevo bound, tight when the SLD operators commute. For the beyond-GR parameters of the SSTG, $[L_j, L_k] \neq 0$, so the QCRB is not simultaneously achievable for all parameters; the achievable bound is the Holevo–Cramér–Rao bound. The bipartite von Neumann entropy used to characterise entanglement capacity is:

$$\bar{S}_{\mathrm{vN}} = \mathbb{E}_{\boldsymbol{\theta}}[-\mathrm{Tr}(\rho_A \ln \rho_A)], \quad \rho_A = \mathrm{Tr}_B(|\psi\rangle\langle\psi|). \tag{B.16}$$

B.5 Fubini–Study metric and quantum geometry

The Fubini–Study metric on the space of pure quantum states is:

$$ds^2 = \sum_{j,k} g^{FS}_{jk}\, d\theta_j\, d\theta_k, \qquad g^{FS}_{jk} = \mathrm{Re}(\langle \partial_j \psi | \partial_k \psi \rangle - \langle \partial_j \psi | \psi \rangle \langle \psi | \partial_k \psi \rangle). \tag{B.17}$$

The Quantum Geometric Tensor (QGT) is the complex extension:

$$T_{jk} = \langle \partial_j \psi | \partial_k \psi \rangle - \langle \partial_j \psi | \psi \rangle \langle \psi | \partial_k \psi \rangle, \tag{B.18}$$

with $g^{FS}_{jk} = \mathrm{Re}(T_{jk})$ and Berry curvature $F_{jk} = -2\,\mathrm{Im}(T_{jk})$. The QFI relates to the metric by $\mathcal{F}_{Q,k} = 4\,g^{FS}_{kk}$.

B.6 Bures metric and density matrices

For mixed states ρ, the Bures metric is:

$$ds^2_{\mathrm{Bures}} = \frac{1}{2}\sum_{j,k} \mathrm{Tr}(L_j \rho\, L_k + L_k \rho\, L_j)\, d\theta_j\, d\theta_k = \frac{1}{2}\sum_{j,k} \mathcal{F}_{Q,jk}\, d\theta_j\, d\theta_k. \tag{B.19}$$

Under the NISQ noise model $\rho(\boldsymbol{\theta}) = \mathcal{E}(|\psi(\boldsymbol{\theta})\rangle\langle\psi(\boldsymbol{\theta})|)$, the Bures QFI is strictly smaller than the ideal pure-state QFI, explaining the accuracy degradation documented in Section 6.5.1.

B.7 Quantum kernel theory

B.7.1 Mercer representation

The quantum kernel $k_Q(\mathbf{x}, \mathbf{x}')$ (Equation 3.4) is continuous, symmetric, and positive semi-definite. By Mercer's theorem:

$$k_Q(\mathbf{x}, \mathbf{x}') = \sum_{m=1}^{\infty} \mu_m\, \phi_m(\mathbf{x})\, \phi_m(\mathbf{x}'), \tag{B.20}$$

with non-negative eigenvalues $\{\mu_m\}$ and eigenfunctions $\{\phi_m\} \subset L^2(\mathcal{X})$.

B.7.2 VC dimension and generalisation

The VC dimension of the QNIM classifier is bounded by $p + 1 = 49$. By the PAC-learning bound:

$$\epsilon_{\mathrm{gen}} \leqslant \sqrt{\frac{\mathrm{VC} \cdot \ln(2N/\mathrm{VC}) + \ln(1/\delta)}{N}}, \tag{B.21}$$

with probability $\geqslant 1 - \delta$. For $N = 54{,}600$, $p = 48$, $\delta = 0.05$: $\epsilon_{\mathrm{gen}} \leqslant 0.084$ (8.4 %), a loose but valid upper bound relative to the measured training–validation gap of $< 0.5\,\%$.

B.8 Classical evaluation costs for the quantum kernel

This section provides the formal cost analysis replacing the informal $\mathcal{O}(2^n)$ statement of Section 3.2. We derive the cost of three classical strategies for evaluating $k_Q(\mathbf{x}, \mathbf{x}') = |\langle \mathbf{0}|\mathcal{U}_\phi^\dagger(\mathbf{x}')\mathcal{U}_\phi(\mathbf{x})|\mathbf{0}\rangle|^2$ for the QNIM ZZFeatureMap ($n = 12$, $r_\phi = 2$).

B.8.1 Strategy 1: statevector simulation

Full statevector simulation costs $\mathcal{O}(r_\phi n \cdot 2^n)$ per kernel pair. For $n = 12$, $r_\phi = 2$: $C_{\mathrm{SV}}(12) \approx 10^5$ FLOP per pair. The full $N \times N$ Gram matrix ($N = 78{,}000$) requires $\approx 3 \times 10^{14}$ FLOP and 49 TB memory—memory-bound for any single node.

B.8.2 Strategy 2: MPS simulation

MPS overlap evaluation costs $\mathcal{O}(n\chi^3)$ per pair (Schreiber et al., 2024). At the critical bond dimension $\chi_{\mathrm{crit}} = 2^{n/2} = 64$:

$$C_{\mathrm{MPS}}(12, 64) = \mathcal{O}(12 \times 64^3) \approx 3.1 \times 10^6 \text{ FLOP per pair}, \tag{B.22}$$

giving $\approx 9 \times 10^{15}$ FLOP and 49 TB for the full Gram matrix. At $n = 20$, $\chi_{\mathrm{crit}} = 1024$ and the cost grows by $\sim 6000\times$.

B.8.3 Strategy 3: Fourier random features

The RFF approximation (Rahimi & Recht, 2007) approximates the kernel as $\hat{k}_M(\mathbf{x}, \mathbf{x}') = \mathbf{z}(\mathbf{x})^\top \mathbf{z}(\mathbf{x}')$ with M random features. Exact approximation to error $\varepsilon = 0.01$ over the training set requires:

$$M \geqslant \frac{4}{\varepsilon^2} \ln \frac{N(N-1)}{\delta} \approx 1.02 \times 10^6, \tag{B.23}$$

costing $\approx 6 \times 10^{15}$ FLOP and $\sim$318 GB memory—comparable to MPS and not tractable on a single workstation. Inverting Equation (B.23) for a fixed feature budget M gives the achievable approximation error:

$$\varepsilon(M) = \sqrt{\frac{4}{M} \ln \frac{N(N-1)}{\delta}}. \tag{B.24}$$

The C-QK baseline uses $M = 10{,}000$, which by Equation (B.24) guarantees error $\varepsilon \leqslant 10.1\,\%$ across the full dataset.

B.8.4 Comparative summary

Table B.3 brings together the three classical strategies discussed above — statevector simulation (Section B.8.1), MPS simulation (Section B.8.2), and the Fourier random-feature approximation (Section B.8.3) — alongside the QNIM quantum kernel evaluation, to make their relative cost and exactness directly comparable for the same problem size.

Strategy	**Per-pair cost**	**Gram FLOP**	**Memory**	**Exact?**
Statevector	$\mathcal{O}(r_\phi n \cdot 2^n)$	3×10^{14}	49 TB	Yes
MPS $\chi = 64$	$\mathcal{O}(n\chi^3) \approx 3 \times 10^6$	9×10^{15}	49 TB	Yes
RFF $M = 10^4$	$\mathcal{O}(Md) = 1.2 \times 10^5$	6×10^{13}	∼3 GB	No ($\varepsilon \leqslant 10.1\,\%$)
RFF $M = 10^6$	$\mathcal{O}(Md) = 1.2 \times 10^7$	6×10^{15}	∼318 GB	No ($\varepsilon \leqslant 1.0\,\%$)
QNIM (quantum)	$\mathcal{O}(nr_\phi)$ shots	N/A	$n = 12$ qubits	Yes

Table B.3: Cost comparison of three classical strategies for computing the QNIM quantum kernel ($N = 78{,}000$, $n = 12$, $r_\phi = 2$). Source: own elaboration.

C. Software architecture, implementation, and reproducibility

This Appendix provides the complete technical documentation of the QNIM software architecture described in Chapter 4, including the domain model, infrastructure adapters, deployment procedures, and reproducibility protocol.

C.1 The QNSPSA-EML-Feynman optimization algorithm

This section provides the complete technical specification of the QNSPSA-EML-Feynman optimiser, including its pseudocode, convergence guarantees under the Robbins–Monro conditions, and the energy efficiency analysis that quantifies the computational cost of hardware execution.

C.2 Pseudocode

Algorithm C.1 gives the complete pseudocode of the QNSPSA-EML-Feynman optimiser as implemented in the QNIM codebase (`src/domain/services/qnspsa_eml_feynman.py`).

```
Input: theta_0, schedules {a_t}, {c_t}, ZNE factors, lambda_R
Output: theta*

For t = 0, 1, ..., T-1:
  Sample Delta_t, Delta'_t in {-1,+1}^p
  For each lambda_k in {1,2,3}:
    f+ = eval_circuit(theta_t + c_t*Delta_t, lambda_k)
    f- = eval_circuit(theta_t - c_t*Delta_t, lambda_k)
  g_t = sum_k c_k*(f+ - f-)/(2*c_t*Delta_t)   % ZNE gradient
  Repeat with Delta'_t to get g'_t
  F_t = (g_t g_t^T + g'_t g'^T_t) / 2
  H^{-1}_{t+1} = H^{-1}_t - H^{-1}_t F_t H^{-1}_t / (1 + Tr(H^{-1}_t F_t))
  R_t = curvature_regulariser(theta_t)
  g_total = g_t + lambda_R * grad(R_t)
  theta_{t+1} = theta_t - a_t * H^{-1}_{t+1} * g_total
Return theta_T
```

Figure C.1: QNSPSA-EML-Feynman optimiser pseudocode. Source: own elaboration.

C.3 Convergence guarantees

Under the following standard assumptions:

1. $f(\boldsymbol{\theta})$ is L-smooth (Lipschitz gradient).
2. The stochastic gradient estimate is unbiased: $\mathbb{E}[\hat{g}_t] = \nabla f(\boldsymbol{\theta}_t) + \mathcal{O}(c_t^2)$.
3. The schedules satisfy $\sum_t a_t = \infty$, $\sum_t a_t^2/c_t^2 < \infty$ (RobbinsMonro conditions).

The QNSPSA-EML-Feynman iterates satisfy:

$$\lim_{T\to\infty} \mathbb{E}\left[\min_{t\leqslant T} \|\nabla f(\boldsymbol{\theta}_t)\|^2\right] = 0, \tag{C.1}$$

i.e. the algorithm converges to a stationary point of f almost surely, under the standard Robbins–Monro conditions stated above (Spall, 1998). The precise finite-sample convergence rate depends on problem-specific constants (smoothness of f, variance of the gradient estimator) and is not derived here; empirically, the QNIM training run reaches the loss threshold $\mathcal{L} = 2.31$ within $\leqslant 25$ steps (Section 7.1.2), which is the operative convergence figure used throughout this thesis.

The ZNE correction introduces a bias of order c_t^{2K} in the gradient (where $K = 3$ is the Richardson order), which decays faster than any schedule-induced variance term and does not affect the asymptotic convergence.

n	**Raw SPSA**	**QNSPSA-EML-Feynman**	**Trainable?**
8	2.5×10^{-3}	6.2×10^{-2}	Yes
12	6.1×10^{-4}	1.6×10^{-2}	Yes
16	1.5×10^{-4}	3.9×10^{-3}	Marginal
20	3.8×10^{-5}	9.5×10^{-4}	Marginal
27	6.7×10^{-6}	1.7×10^{-4}	Yes (with EML)

Table C.1: Gradient variance $\mathrm{Var}[\partial f/\partial\theta_k]$ for various qubit scales, with and without the QNSPSA-EML-Feynman mitigation. Source: own elaboration.

C.4 Green computing and energy efficiency

This Section closes the software-architecture appendix with two complementary quality dimensions that are not captured by functional correctness alone: the testing methodology that verifies the domain logic itself (Section C.4.1 and Section C.4.2), and the energy

cost of running QNIM on quantum hardware, which is quantified below in terms of dilution-refrigerator power draw and per-circuit energy consumption.

C.4.1 Test taxonomy

The QNIM test suite follows the standard three-tier taxonomy:

Unit tests (`test/unit/`): test individual domain objects and domain services in complete isolation. No network, no quantum hardware, no filesystem. Execution time: $< 30\,\mathrm{s}$ for the full suite.

Integration tests (`test/integration/`): test the interaction between adapters and their external dependencies using locally running stubs (e.g., a local SQLite database, the Qiskit `AerSimulator`).

End-to-end tests (`test/e2e/`): test the full pipeline from CLI command to output file. Run against the simulator backend by default; tagged `@pytest.mark.hardware` for optional execution against real hardware.

C.4.2 Mutation testing

Domain logic is verified using mutation testing with `mutmut`. Mutations are applied to the domain and application layers only (not infrastructure, which is covered by contract tests). The target mutation score is $\geqslant 90\,\%$. At the time of writing, the achieved mutation score for the domain layer is $92.4\,\%$ and for the application layer $88.7\,\%$.

Quantum processors operate at millikelvin temperatures, requiring dilution refrigerators with a static power consumption of $\sim 10\,\mathrm{kW}$ per unit, independent of circuit execution. The effective energy per circuit execution, amortised over a 24-hour operational day with 40% queue utilisation, is:

$$E_{\mathrm{circuit}} \approx \frac{10\,\mathrm{kW} \times 1\,\mathrm{ms}}{10^5 \cdot 0.4} = 250\,\mu\mathrm{J/circuit}. \tag{C.2}$$

For a full QNIM training run ($N_{\mathrm{step}} = 218$ steps, $N_{\mathrm{eval/step}} = 4 \times K = 12$ circuit evaluations per step, 8192 shots per evaluation):

$$E_{\mathrm{train}} \approx N_{\mathrm{step}} \times N_{\mathrm{eval/step}} \times N_{\mathrm{shots}} \times E_{\mathrm{circuit}} = 218 \times 12 \times 8192 \times 250\,\mu\mathrm{J} \approx 5.4\,\mathrm{kJ}. \tag{C.3}$$

QNIM implements four mitigations:

1. **Transpilation caching**: pre-compiled circuits reused ($\sim$12% reduction in circuit count).
2. **Adaptive shot allocation**: gradient evaluations use fewer shots ($\sim$28% reduction in total shots).
3. **Early stopping with patience**: training halts when validation loss plateaus ($\sim$18% fewer steps).
4. **Simulator-first policy**: all development and debugging on classical simulators; hardware reserved for final validation ($\sim$80% fewer hardware executions).

Combined effect: approximately 56% reduction in energy cost per training run (from $\sim$12.3 kJ to $\sim$5.4 kJ).

C.5 Domain model and repository interfaces

The domain layer declares five core value objects: `Strain` (frequency-domain strain with metadata), `FeatureVector` (12-dimensional PCA projection), `QuantumCircuitSpec` (VQC hyperparameters), `QuboMatrix` (QUBO problem specification), and `InferenceResult` (classifier output with reliability tier).

Three persistent entities track state: `GravitationalWaveEvent` (aggregate root for events), `ExperimentRun` (execution history for reproducibility), and `ModelWeights` (trained parameters with content hash).

Four domain services implement business logic: `StrainPreprocessor` (whitening, STFT, feature extraction), `QuboBuilder` (QUBO matrix construction from likelihoods), `PlanckReliabilityClassifier` (event reliability tiers), and `MetrologyAuditor` (quantum advantage quantification).

Repository interfaces (ports) declare contracts: `IEventRepository` for event storage, `IModelRepository` for weight management, and concrete implementations (GWOSC HTTP client, local filesystem, SQLite) live in the infrastructure layer.

C.6 Infrastructure adapters

The hexagonal architecture of QNIM isolates scientific logic from technology through adapter implementations. This section documents the three primary adapters—IBM Quantum, D-Wave, and GWOSC—together with the polyglot persistence strategy that manages model weights, results, and metadata.

C.6.1 IBM Quantum adapter via Qiskit

The adapter uses Qiskit `EstimatorV2` primitives with: (1) session batching (one session per training run), (2) transpilation caching (compile once, reuse), (3) adaptive shot allocation (8192 shots for function evaluations, 1024 for gradients, 512 for ZNE copies), (4) error mitigation via T-REX delegated to IBM Runtime.

Circuit compilation to `ibm_fez` (133-qubit Heron) achieves: CNOT count $N_{\mathrm{CX}} = 88$, circuit depth $D_{\mathrm{native}} = 31$, two-qubit gate error $\epsilon_{2Q} \approx 3.8 \times 10^{-3}$, total fidelity $\mathcal{F}_{\mathrm{circ}} \approx 0.714$.

C.6.2 D-Wave adapter via Ocean SDK

The adapter translates `QuboMatrix` to Ocean's `BinaryQuadraticModel`, solves with neal´s Simulated Annealing Sampler, and emulates annealer noise via Gaussian perturbations ($\sigma_Q = 0.03$, 24 noisy realisations per solve). Outputs include ensemble mode, parameter standard deviations, and selection stability.

C.6.3 GWOSC client

Downloads strain data from `gwosc.org` via `gwpy`, implements HTTP retry (exponential backoff, max 5 attempts), local disk caching, and SHA-256 verification against GWOSC checksums.

C.6.4 Polyglot persistence

- **Model weights**: NumPy `.npy` or HDF5 files, content-addressed by SHA-256 hash, stored in `models/`.
- **Results**: CSV files with fixed schema in `reports/`, ASCII-compatible column names.
- **Metadata**: SQLite database at `data/qnim.db`, schema managed by `alembic` migrations.

- **Logs**: JSON-formatted, rotated daily, retained 30 days.

C.7 Experiment reproducibility protocol

Components	Individual variance (%)	Cumulative variance (%)
1	38.2	38.2
2	18.7	56.9
3	11.4	68.3
4	7.8	76.1
5	5.3	81.4
6	3.9	85.3
7	3.1	88.4
8	2.4	90.8
9	1.8	92.6
10	1.3	93.9
11	0.8	94.7
12	0.5	95.2

Table C.2: Cumulative explained variance for the 12 PCA components used as input to the QNIM quantum feature map. Source: own elaboration.

Every `ExperimentRun` stores:

- Git commit hash of QNIM source.
- SHA-256 hashes of `ModelWeights`, input data files.
- Random seeds for all stochastic operations (`numpy`, `random`, Qiskit transpiler).
- Backend calibration snapshot (hardware runs).

This metadata is persisted in the `ExperimentRun` SQLite record and reproduced in the HTML dashboard, enabling independent verification.

Code availability: The complete QNIM source, model weights, and experiment configuration are version-controlled at https://github.com/oscarboullosa/qnim, Git tag `v1.0.0` (commit `d4e7f3a`). To reproduce any result:

```
git clone https://github.com/oscarboullosa/qnim
git checkout v1.0.0
pip install -r requirements.lock.txt
# Follow README.md and REPRODUCTION.md for execution
```

Dependencies are fully pinned in `requirements.lock.txt` (including transitive and hash verification), and a Docker image with the complete runtime environment is provided for bit-for-bit reproducibility across platforms.

D. IBM Quantum deployment guide

This Appendix describes the procedure for deploying and executing the QNIM pipeline on IBM Quantum hardware.

D.1 Prerequisites and account setup

1. Create an IBM Quantum account at `quantum.ibm.com` and retrieve the API token from the account dashboard. Do **not** paste the token into any file; store it only as an environment variable (see step 3 below).

2. Install the QNIM dependencies: `pip install -r requirements.lock.txt`.

3. Set the `IBM_QUANTUM_TOKEN` environment variable **before** running any QNIM command. Never hard-code the token in source files or configuration committed to version control:

```
# bash / zsh  (add to ~/.bashrc or ~/.zshrc for persistence)
export IBM_QUANTUM_TOKEN="paste-your-token-here"
```

   On Windows (PowerShell):

```
$Env:IBM_QUANTUM_TOKEN = "paste-your-token-here"
```

4. Configure the IBM Runtime credentials. The snippet below reads the token from the environment so that no secret appears in code:

```
import os
from qiskit_ibm_runtime import QiskitRuntimeService

QiskitRuntimeService.save_account(
    channel="ibm_quantum",
    token=os.environ.get("IBM_QUANTUM_TOKEN"),
    overwrite=True)
```

If `IBM_QUANTUM_TOKEN` is not set, `os.environ.get` returns `None` and `save_account` will raise a descriptive error, making misconfiguration immediately visible.

D.2 Backend selection

The recommended backend for QNIM is `ibm_fez` (133-qubit Heron r1), which provides the lowest two-qubit gate error rate currently available ($\epsilon_{2Q} \approx 3.8 \times 10^{-3}$). Alternative backends:

- `ibm_torino` (133-qubit Heron r2): $\epsilon_{2Q} \approx 2.1 \times 10^{-3}$ recommended for future runs once calibration stabilises.
- `ibm_kyiv` (127-qubit Eagle r3): $\epsilon_{2Q} \approx 6.2 \times 10^{-3}$ acceptable but $\sim 1\,\%$ higher accuracy degradation.
- `ibmq_qasm_simulator`: unlimited shots, no queue; use for development and debugging only.

D.3 Running the QNIM inference pipeline

1. Set the backend in the QNIM configuration file (`config/universe_params.yaml`):

```
quantum_backend:
  type: ibm_quantum
  backend_name: ibm_fez
  shots: 8192
  optimization_level: 3
  resilience_level: 1   # T-REX readout mitigation (IBM Runtime built-in).
                        # ZNE is applied separately and manually via
                        # unitary gate folding at noise factors {1,2,3},
                        # as described in Section~\ref{subsec:eml}; it is
                        # not delegated to this Runtime setting.
```

2. Run inference on a single GWTC-3 event:

```
qnim infer --event GW150914 --backend ibm_fez
```

3. Monitor the job status:

```
qnim status --job-id <JOB_ID>
```

4. Retrieve and export results:

```
qnim report --job-id <JOB_ID> --format html
```

D.4 Estimating quantum credit consumption

IBM Quantum charges in *seconds of execution time* on the QPU. For a single QNIM inference run:

- VQC classification (ZNE $K = 3$, 8192 shots, 3 noise factors): $3 \times 8192 \times D_{\text{native}} \times t_{\text{gate}} \approx 3 \times 8192 \times 31 \times 50\,\text{ns} \approx 38\,\text{ms}$ QPU time.
- With classical compilation overhead (excluding queue wait, which is wall-clock latency rather than billed QPU time): ~ 0.1 QPU-s per inference call. Note this billed-time figure is distinct from the end-to-end wall-clock latency of Table 7.6 (87.4 ± 22.6 s per event), which additionally includes queue delays and network round-trips.
- Full GWTC-3 analysis (90 events): ~ 9 QPU-s.

At the IBM Quantum Open Plan rate ($\sim$600 QPU-s/month free, i.e. 10 minutes per account, as documented in Section 8.2.3), the full GWTC-3 analysis ($\sim$9 QPU-s) fits comfortably within a single month's free-tier allocation, using approximately 1.5 % of the monthly quota.

E. Full QNIM Planck reliability catalogue

Table E.1 presents the complete QNIM Planck Reliability Catalogue for all 90 GWTC-3 events, consolidating the theory classification of Section 7.3.2 with the full Bayes factor and reliability tier assignments. The full machine-readable catalogue including posterior probability vectors, MAP parameter estimates, and quantum metrological gains is available in the accompanying data repository as `reports/qprc_gwtc3_v1.csv`.

Errata note on Tier classification: Some events in the printed catalogue exhibit small positive $\ln B$ but are marked as Tier D instead of Tier C. Under the operational thresholds used throughout this thesis (Section 7.3.2: Tier C for $\ln B \in [1.0, 3.0]$; Tier D for $\ln B < 1.0$, including small positive values), any event with $0 < \ln B < 1.0$ is correctly classified as Tier D. Note that this operational threshold is stricter than the formal Jeffreys' anecdotal-evidence boundary ($|\ln B| < 2.3$, Section 3.4); the two scales should not be conflated. A corrected catalogue applying the thresholds consistently is available in the data repository; the printed version here reflects the original computational output.

Summary (90 events, GWTC-3 (Abbott et al., 2023b)): Tier A: 0 events (0.0 %); Tier B: 2 events (2.2 %); Tier C: 26 events (28.9 %); Tier D: 62 events (68.9 %).

Table E.1: Full QNIM Planck Reliability Catalogue for all 90 GWTC-3 events. **Top cl.**: most probable theory class index. **Theory**: theory name. $p^\star$: probability of the top-ranked class. $\ln B$: Bayes factor vs. the GR null hypothesis. **Tier**: Planck Reliability classification (A/B/C/D). Events with Tier B are in **bold**. Source: own elaboration, using the GWTC-3 catalogue (Abbott et al., 2023b) and GW190521 data (Abbott et al., 2020).

Event	**Type**	**Top cl.**	**Theory**	$p^\star$	$\ln B$	**Tier**
GW150914†	BBH	0	GR	0.873	-0.4	D
GW151012	BBH	0	GR	0.841	-0.2	D
GW151226†	BBH	0	GR	0.798	$+0.3$	C
GW170104†	BBH	0	GR	0.862	-0.1	D
GW170608†	BBH	0	GR	0.891	-0.5	D
GW170729	BBH	4	PN 3.5	0.621	$+1.8$	C
GW170809	BBH	0	GR	0.854	-0.3	D
GW170814†	BBH	0	GR	0.867	-0.4	D
GW170817†	BNS	0	GR	0.901	-0.8	D
GW170818	BBH	0	GR	0.849	-0.3	D
GW170823	BBH	0	GR	0.831	-0.2	D
GW190403_051519	BBH	0	GR	0.815	-0.2	D
GW190408_181802	BBH	0	GR	0.844	$+0.1$	C
GW190412†	BBH	5	Extra dim.	0.591	$+2.1$	C
GW190413_052954	BBH	0	GR	0.823	-0.1	D
GW190413_134308	BBH	0	GR	0.811	-0.2	D
GW190421_213856	BBH	0	GR	0.808	-0.1	D
GW190425†	BNS	0	GR	0.782	-0.6	D

(Table E.1 continued)

Event	**Type**	**Top cl.**	**Theory**	$p^\star$	$\ln B$	**Tier**
GW190426_190642	BBH	0	GR	0.804	−0.3	D
GW190503_185404†	BBH	0	GR	0.819	+0.2	C
GW190512_180714	BBH	0	GR	0.856	−0.1	D
GW190513_205428	BBH	9	LIV α=2	0.614	+2.4	C
GW190514_065416	BBH	0	GR	0.812	−0.3	D
GW190517_055101	BBH	0	GR	0.824	+0.1	C
GW190519_153544	BBH	0	GR	0.839	−0.2	D
GW190521‡	BBH	5	**Extra dim.**	**0.723**	**+3.8**	**B**
GW190521_074359	BBH	0	GR	0.847	−0.3	D
GW190527_092055	BBH	0	GR	0.801	+0.2	C
GW190602_175927	BBH	0	GR	0.803	−0.3	D
GW190620_030421	BBH	0	GR	0.834	+0.4	C
GW190630_185205	BBH	0	GR	0.847	−0.2	D
GW190701_203306	BBH	8	**CS**	**0.681**	**+2.9**	**B**
GW190706_222641	BBH	0	GR	0.819	−0.1	D
GW190707_093326	BBH	0	GR	0.862	−0.4	D
GW190708_232457	BBH	0	GR	0.843	+0.2	C
GW190719_215514	BBH	0	GR	0.831	−0.2	D
GW190720_000836	BBH	0	GR	0.855	−0.3	D
GW190725_174728	BBH	0	GR	0.827	−0.1	D
GW190727_060333	BBH	0	GR	0.824	+0.1	C

(Table E.1 continued)

Event	**Type**	**Top cl.**	**Theory**	$p^\star$	$\ln B$	**Tier**
GW190728_064510	BBH	0	GR	0.871	−0.5	D
GW190731_140936	BBH	0	GR	0.818	−0.1	D
GW190803_022701	BBH	0	GR	0.807	+0.3	C
GW190805_211137	BBH	0	GR	0.833	−0.2	D
GW190814[†]	NSBH	0	GR	0.876	−0.6	D
GW190828_063405	BBH	0	GR	0.841	−0.1	D
GW190828_065509	BBH	0	GR	0.829	+0.2	C
GW190910_112807	BBH	0	GR	0.852	−0.3	D
GW190915_235702	BBH	0	GR	0.846	+0.1	C
GW190916_200658	BBH	0	GR	0.836	−0.2	D
GW190917_114630	BBH	0	GR	0.822	−0.1	D
GW190924_021846	BBH	0	GR	0.868	−0.4	D
GW190925_232845	BBH	4	PN 3.5	0.584	+1.4	C
GW190926_050336	BBH	0	GR	0.813	−0.2	D
GW190929_012149	BBH	0	GR	0.821	−0.2	D
GW190930_133541	BBH	0	GR	0.838	+0.1	C
GW191103_012549	BBH	0	GR	0.859	−0.3	D
GW191105_143521	BBH	0	GR	0.843	−0.1	D
GW191109_010717	BBH	0	GR	0.825	+0.4	C
GW191113_071753	BBH	0	GR	0.812	−0.2	D
GW191126_115259	BBH	0	GR	0.836	−0.1	D

(Table E.1 continued)

Event	**Type**	**Top cl.**	**Theory**	$p^\star$	$\ln B$	**Tier**
GW191127_050227	BBH	10	LIV α=4	0.574	+1.6	C
GW191129_134029	BBH	0	GR	0.853	−0.3	D
GW191204_110529	BBH	0	GR	0.848	+0.2	C
GW191204_171526	BBH	0	GR	0.871	−0.4	D
GW191215_223052	BBH	0	GR	0.829	−0.1	D
GW191216_213338	BBH	0	GR	0.862	−0.5	D
GW191219_163120	BBH	0	GR	0.831	−0.2	D
GW191222_033537	BBH	0	GR	0.819	+0.3	C
GW191230_180458	BBH	0	GR	0.834	−0.2	D
GW200112_155838	BBH	0	GR	0.841	−0.1	D
GW200115†	NSBH	0	GR	0.873	−0.6	D
GW200128_022011	BBH	0	GR	0.826	+0.2	C
GW200129_065458†	BBH	0	GR	0.851	−0.3	D
GW200202_154313	BBH	0	GR	0.844	−0.1	D
GW200208_130117	BBH	0	GR	0.828	+0.1	C
GW200208_222617	BBH	0	GR	0.817	−0.2	D
GW200209_085452	BBH	0	GR	0.839	−0.3	D
GW200210_092254	BBH	0	GR	0.862	−0.4	D
GW200216_220804	BBH	0	GR	0.831	+0.2	C
GW200219_094415	BBH	0	GR	0.848	−0.1	D
GW200220_061928	BBH	0	GR	0.819	−0.2	D

(Table E.1 continued)

Event	**Type**	**Top cl.**	**Theory**	$p^\star$	$\ln B$	**Tier**
GW200220_124850	BBH	0	GR	0.807	+0.3	C
GW200224_222234	BBH	0	GR	0.854	−0.3	D
GW200225_060421	BBH	0	GR	0.843	−0.1	D
GW200302_015811	BBH	0	GR	0.831	+0.1	C
GW200306_093714	BBH	0	GR	0.818	−0.2	D
GW200308_173609	BBH	0	GR	0.829	−0.3	D
GW200311_115853	BBH	0	GR	0.862	−0.5	D
GW200316_215756	BBH	0	GR	0.847	−0.1	D
GW200322_091133	BBH	0	GR	0.834	+0.2	C

† Events used in the QNIM Hubble constant measurement (Section 7.5); selected on the basis of QNIM d_L precision $< 20\,\%$ and GLADE+ host galaxy candidate.

‡ GW190521 is classified as BBH/IMBH; both components lie in or above the pair-instability supernova mass gap (Abbott et al., 2020).

F. Speculative connections: zeta functions and QNM spectra

Warning: mathematical analogy without experimental support. This Appendix presents a *formal, unverified analogy* between properties of the Riemann zeta function and the quasi-normal mode spectrum of a black hole. This correspondence is a long-standing open conjecture in mathematical physics it has not been derived from first principles, has no empirical support, and plays *no operational role* in the QNIM pipeline. No GWTC-3 event is classified, filtered, or analysed using this analogy. The content of this appendix is *not tested experimentally* by this thesis. It is included solely to document the mathematical inspiration for the Layer-7 foam-noise model; for a discussion of whether the conjecture could be made testable with third-generation detectors, see Section 9.4.

F.1 The Hilbert–Pólya conjecture and the BerryKeating-Connes programme

A speculative conjecture in mathematical physics independently proposed by Berry & Keating (1999) and Connes (1999) suggests that the imaginary parts $\{t_n\}$ of the non-trivial zeros of $\zeta(s) = 0$ (i.e. $s = \frac{1}{2} + it_n$, assuming the Riemann Hypothesis) might correspond to the eigenvalues of a hypothetical self-adjoint Hamiltonian $\hat{H}_{\mathrm{RH}}$ with spectrum $\{t_n\}$. This is the Hilbert–Pólya conjecture. Neither the existence of $\hat{H}_{\mathrm{RH}}$ nor its physical interpretation has been established; the conjecture remains unproven after more than a century.

F.2 Formal analogy with quasi-normal mode spectra

In the gravitational-wave context, one may note a *purely formal* resemblance between the structure of the Riemann zero distribution and certain statistical properties of quasi-normal mode (QNM) overtone spectra, specifically the Montgomery pair-correlation, which both exhibit GUE-type level repulsion. Table F.1 lists this structural correspondence as a dictionary of analogies.

The reader is cautioned that *none of the mappings in Table F.1 is derived*: they are terminological parallels, not mathematical identities. In particular, the Montgomery pair-

Attribute of $\zeta(s)$	**Formal gravitational analogue**
Non-trivial zeros ρ_n	QNM frequency spectrum (conjecture only)
Critical line $\mathrm{Re}(s) = 1/2$	S-matrix unitarity (if Hilbert–Pólya holds)
$\zeta(-1) = -1/12$	Casimir energy E_0^{reg} (rigorously derived, Section A.3.2)
Euler product formula	Partition function of spacetime microstates (formal)
Montgomery pair-correlation	GUE level repulsion

Table F.1: Formal analogy between properties of the Riemann zeta function and gravitational-physics concepts. This table describes an unverified speculative correspondence (Berry & Keating, 1999; Connes, 1999); no entry has been derived from the other. No QNIM result depends on any entry except the Casimir energy (third row), which is independently derived in quantum field theory and is not part of this conjecture. Source: own elaboration, based on the frameworks of (Berry & Keating, 1999) and (Connes, 1999).

correlation row refers to level-repulsion statistics across an *ensemble* of QNM overtone frequencies drawn from multiple black holes (e.g. the GWTC-3 population), not to the deterministic, discretely numbered overtone spectrum of any single remnant; GUE-type statistics are meaningful only when applied to such a population, not to one black hole's spectrum in isolation. The only operationally used entry is the Casimir energy regularisation via $\zeta(-1) = -1/12$ (Section A.3.2), a rigorous result of quantum field theory in curved spacetime (Hawking, 1975) entirely independent of the Berry–Keating conjecture. All remaining entries are motivational background that **does not influence any result, classification, or parameter estimate in this thesis**.

G. SSTG Validation and Parameter-Estimation Calibration

This Appendix provides the complete technical documentation for the SSTG parameter-estimation (PE) calibration study referenced in Chapter 7. The QNIM PE branch uses Fisher Information Matrix estimation on the PCA-projected feature space, which differs from classical matched-filtering PE; this appendix details the calibration protocol, results, and systematic factors affecting precision.

G.1 Synthetic-Injection Protocol

To assess parameter-estimation fidelity against known ground truth, a synthetic-injection protocol was applied to five representative GWTC-3 confident events (GW150914, GW170817, GW190521, GW190814, and GW200115); GW200105 was excluded from this calibration set as it does not meet the $p_{\rm astro} > 0.5$ threshold for GWTC-3-confident status ($p_{\rm astro} = 0.36$) and is officially classified as GWTC-3-marginal. For each event:

1. Retrieve 256 s of real LIGO strain immediately preceding the merger from the Gravitational-Wave Open Science Center.
2. Generate a 3.5PN IMRPhenomD-like frequency-domain waveform at the published LVC maximum a-posteriori (MAP) parameters.
3. Inject the waveform at merger epoch $t_{\rm merger} = T_{\rm GPS} - 15\,{\rm s}$.
4. Whiten the injected strain via Welch PSD, project onto the standard $d = 12$ QNIM PCA feature space (Section 3.1.2), extract observation vector $\mathbf{z}_{\rm obs}$.
5. Compute QNIM posterior widths via the Fisher information matrix:

$$\mathbf{F}_{ij} = \sum_{k=1}^{p} \frac{1}{\sigma_k^2} \frac{\partial z_k}{\partial \theta_i} \frac{\partial z_k}{\partial \theta_j}, \qquad \mathbf{\Sigma}_{\rm QNIM} = \mathbf{F}^{-1}, \tag{G.1}$$

where σ_k is the RMS noise of the k-th PCA component estimated from 50 non-overlapping 4 s background segments, and the Jacobian $\partial z_k/\partial \theta_i$ is evaluated by centred finite differences at the MAP point with 5 % fractional step.

The parameters estimated are $\boldsymbol{\theta} = (m_1, m_2, \chi_{\rm eff}, d_L)$.

G.2 Quality Metric: Jensen–Shannon Divergence

Quality is measured via the marginal Jensen–Shannon divergence between the QNIM Fisher posterior and the LVC published posterior for each parameter:

$$D_{\rm JS}^{(\theta_i)} = D_{\rm JS}\Big[\mathcal{N}\Big(\hat{\theta}_i^{\rm MAP}, \sigma^2_{{\rm QNIM},i}\Big) \,\Big\|\, \mathcal{N}\Big(\hat{\theta}_i^{\rm MAP}, \sigma^2_{{\rm LVC},i}\Big)\Big], \tag{G.2}$$

where $\sigma_{{\rm LVC},i}$ is inferred from the published 90 % credible interval. Both posteriors are centred at the injection MAP, so $D_{\rm JS}$ reflects only the *precision ratio*, not systematic bias. Following Dax et al., PE quality is satisfactory when $D_{\rm JS} < 10^{-3}$ nats.

Clarification on divergence metrics: This appendix uses Jensen–Shannon divergence (JSD) to quantify the difference in parameter-estimation *precision* between QNIM and classical LVC methods. In contrast, Objective O2 (Section 1.3) uses Kullback–Leibler divergence as a criterion for the *separability of theory classes* in the SSTG synthetic data. These are independent measures: JSD quantifies inference fidelity while $D_{\rm KL}$ quantifies theory discrimination capability.

G.3 Results and Systematic Discussion

Table G.1 reports marginal JSD per event and parameter. The overall mean is 0.378 nats, substantially above the 10^{-3} nats threshold, indicating QNIM Fisher posteriors do not yet match state-of-the-art PE precision.

Event	$D_{\rm JS}$ (mean over parameters, nats)
GW150914	0.28
GW170817	0.19
GW190521	0.54
GW190814	0.31
GW200115	0.57
Mean	**0.378**

Table G.1: Marginal Jensen–Shannon divergence between QNIM Fisher posteriors and LVC posteriors for the five calibration events. GW190521 shows the largest JSD due to near-singular Fisher matrix at high mass and low SNR. Source: own elaboration.

G.3.1 PCA Mismatch

The PCA basis was trained on synthetic SSTG events, not real LIGO strain. When real data are projected onto this basis, they occupy a different subspace, inflating or deflating individual Jacobian components and hence estimated precision.

G.3.2 Parameter-Dependent Behaviour

Chirp-mass parameters (m_1, m_2) perform best for GW170817 ($D_{\mathrm{JS}} < 0.07\,\mathrm{nats}$), the highest-SNR event, consistent with $\sigma \propto \mathrm{SNR}^{-1}$ scaling. Luminosity distance d_L is least-constrained in QNIM feature space due to amplitude information loss during whitening.

G.3.3 Regularisation Artefacts

For GW190521 (highest-mass, lowest-SNR event), the Fisher matrix is near-singular and requires Tikhonov regularisation. Inflated σ_{QNIM} values lead to JSD $\approx 0.54\,\mathrm{nats}$, the largest in the table.

G.4 Parameter recovery testing

For 10 GWTC-3 events with published posterior samples (`GW150914`, `GW170817`, and 8 selected O3 events), the SSTG injects synthetic copies at the published MAP parameter values and feeds them through the full QNIM pipeline. The recovered parameters are compared to the injected values via the *probability-probability* (PP) plot:

$$\mathrm{PP}(p) = \frac{1}{N_{\mathrm{inj}}} \sum_{k=1}^{N_{\mathrm{inj}}} \mathbf{1}\big[\theta_k^{\mathrm{true}} \leqslant Q_p(k)\big]\,, \tag{G.3}$$

where $Q_p(k)$ is the p-quantile of the QNIM posterior for event k. A perfectly calibrated estimator satisfies $\mathrm{PP}(p) = p$ for all p. The measured PP curves lie within the 95 % Kolmogorov-Smirnov confidence band for all 15 parameters, confirming calibrated uncertainty quantification.

G.5 Injection recovery campaign

A dedicated injection recovery campaign is conducted by injecting 500 synthetic signals at known theory labels into real O3 LIGO noise (using data segments free of confirmed GW

events). The QNIM classifier achieves:

- **Overall accuracy**: $91.3 \pm 0.8\,\%$ (vs. $91.0\,\%$ on the clean synthetic test set, demonstrating robustness to real noise non-Gaussianity).
- **Class 0 recall** (GR events correctly not flagged as beyond-GR): $93.7\,\%$.
- **Layer 7 precision** (quantum-geometry events): $84.2\,\%$ (lower, as expected, due to the small amplitude of Layer 7 corrections at accessible SNR).

Note on noise robustness: The injection campaign achieves $91.3\,\%$ accuracy, which exceeds the $80.4\,\%$ attained on the noise-model simulator. This apparent paradox reflects the nature of injection campaigns versus validation on simulated noise. In injections, the synthetic signals retain their precise mathematical structure (3.5PN waveforms with exact beyond-GR corrections) and are added at known SNR to real O3 noise segments. The classifier's task is to classify a set of injected signals with controlled amplitude and duration in an otherwise uncontrolled noise background. By contrast, the $80.4\,\%$ on the simulator is measured across the entire SSTG validation set (11,700 events), where the noise model calibration to `ibm_fez` hardware introduces systematic model discrepancies that are not present in the idealized O3 injection scenario. Furthermore, the O3 injection distribution is likely more separable than the balanced SSTG distribution because injections are not drawn from the same training distribution, whereas the SSTG is drawn from the distribution used to train the classifier. This is a feature of robustness testing, not a problem; it demonstrates that the classifier generalises to data distributions different from its training set.

The injection recovery campaign constitutes the primary empirical validation that the SSTG produces a scientifically useful training distribution.

G.6 Hardware noise ablation: classification accuracy model

This section documents the full noise model for classification accuracy, supporting the condensed treatment in Section 6.5.5.

G.6.1 Global depolarising channel model

Under the global depolarising channel ($p_\epsilon = 1 - \mathcal{F}_{\text{circ}}$), the noisy VQC state and class probability vector are:

$$\mathbf{p}^{\text{noisy}} = (1 - p_\epsilon)\mathbf{p}^{\text{ideal}} + \frac{p_\epsilon}{13}\mathbf{1}, \tag{G.4}$$

giving the expected balanced accuracy:

$$\mathrm{BAcc}^{\mathrm{noisy}} = (1 - p_\epsilon)\,\mathrm{BAcc}^{\mathrm{ideal}} + \frac{p_\epsilon}{13}. \tag{G.5}$$

With $\mathcal{F}_{\mathrm{circ}} = 0.714$ ($p_\epsilon = 0.286$) and $\mathrm{BAcc}^{\mathrm{ideal}} = 0.913$: $\mathrm{BAcc}^{\mathrm{noisy}} = 67.4\,\%$. The QFI degrades as:

$$\mathcal{F}_Q^{\mathrm{noisy}} \propto (1 - p_\epsilon)^2, \tag{G.6}$$

while CFI degrades only linearly—the three-way distinction (QFI quadratic, CFI linear, BAcc linear with offset) is essential for interpreting NISQ noise effects on quantum advantage.

Note on the gap between this prediction and the calibrated simulator result. The global depolarising prediction above (67.4 %) is a simplified analytic estimate; the calibrated noise-model simulator at the same nominal circuit fidelity ($\mathcal{F}_{\mathrm{circ}} = 0.714$) instead reports 83.2 % (Table G.2, row ii), a gap of ∼16 percentage points. This is consistent with the conservatism of the global depolarising approximation already noted in Section 6.1.3: the global model assumes uniform, maximally mixing noise across all 12 qubits, which overestimates information loss relative to the calibrated per-gate noise channel used in the `ibm_fez`-matched simulator. The two figures therefore serve different purposes in this thesis — the depolarising estimate provides a conservative analytic lower bound on accuracy, while the calibrated simulator result is the operative figure used throughout Chapters 7–8 and should be treated as the more physically realistic prediction at this fidelity.

G.6.2 Noise ablation table

Table G.2 documents the measured noise hierarchy.

G.6.3 ZNE was executed on hardware and did not raise accuracy above the noise floor

It is important to avoid conflating simulator configurations with hardware measurements. The 64.9 % figure in row (iv) of Table G.2 is an `AerSimulator` result obtained under a distinct circuit configuration (reps= 2, $\mathcal{F}_{\mathrm{circ}} = 0.714$) and is *not* a hardware measurement. Two hardware configurations were executed on `ibm_fez`: (iii) without ZNE (reps= 1, $\mathcal{F}_{\mathrm{circ}} \approx 0.45$), 6 %; and (v) *with* Richardson ZNE ($K = 3$), 6.3 % (job log archived in the data repository, Appendix P). Across the full set of 365 IBM Quantum job submissions

Config.	Description	BAcc	Source	ZNE
(i)	Noiseless AerSimulator	91.3%	Ideal simulator	–
(ii)	AerSimulator + ibm_fez noise model (reps= 2)	83.2%	Calibrated depolarisation	No
(iii)	IBM Fez, Open Plan, reps= 1, depth= 18, 100 samples, no on-hardware retraining	6% †	Direct QPU (production)	No
(iv)	AerSimulator, reps= 2, distinct configuration	64.9%	Simulator run	No
(v)	IBM Fez + ZNE $K = 3$, reps= 1	6.3% §	Direct QPU (production, measured)	Yes

Table G.2: Ablation study: impact of noise sources on balanced accuracy. † Directly measured on IBM Fez under Open Plan constraints (reps= 1, depth= 18, 100 samples, no on-hardware retraining, production runs June 2026). ‡ Simulator run with reps= 2 under a distinct circuit configuration; not directly comparable to production runs. § Directly measured on IBM Fez with Richardson ZNE ($K = 3$) applied; consistent with the 5–6.5 % range observed across the full job log (Appendix P). Source: own elaboration.

Gap between (iii) and (iv): (a) circuit configuration change (reps= $2 \rightarrow 1$, $\mathcal{F}_{\text{circ}} : 0.714 \rightarrow 0.45$); (b) Open Plan monthly quota ($\sim$10 min/month) prevents on-hardware training; (c) simulator-to-hardware weight transfer without retraining.

logged for this thesis (304 on `ibm_fez`, 61 on `ibm_marrakesh`, June–July 2026), classification accuracy on production runs consistently fell in the 5–6.5 % range, both with and without ZNE, i.e. at or below the 7.7 % random-guess floor for 13 classes in every case. ZNE therefore was applied on real hardware and did not recover accuracy above the noise floor under the reps= 1, no-retraining, Open Plan configuration; this is a substantive negative result in its own right, not an untested possibility.

G.6.4 Complementarity with Parameter Recovery

The parameter-recovery testing (Section G.4) evaluates frequentist coverage (PP-plot calibration) and absolute precision via Fisher matrices. The injection campaign (Section G.5) evaluates classification accuracy on real noise. Together, they show that QNIM is well-calibrated but its per-event posterior widths must be brought closer to template-based PE before standalone operation. Closing this gap requires joint training of PCA and VQC feature maps on real-noise injections (identified as future work).

G.6.5 Implications for Theory-Classification and Cosmological Results

The JSD values do **not** invalidate the theory-*classification* results of Chapter 7, which rely on the VQC feature map and quantum kernel rather than Fisher posterior widths. They do impose a scope restriction on the QNIM PE branch: the MAP estimates and credible intervals in Table 7.2 are derived from *synthetic* SSTG events (where PCA is matched to signal distribution) and are not directly comparable in precision to LVC posteriors for real events. The QUBO PE estimator should be regarded as a *theory-classification support tool* providing warm-start estimates for classical sampling, not as a standalone PE pipeline. The H_0 measurement (Section 7.5) uses LVC d_L estimates refined by the quantum metrological gain factor $G_{d_L} = 1.09$, not QUBO PE posteriors.

H. Gram matrix eigenspectrum: technical derivation

This Appendix provides the full technical treatment of the Gram matrix spectral analysis summarised in Section 6.2.2. We compute the normalised (doubly-centred) Gram matrix from the quantum kernel evaluated on all $N = 11{,}700$ test events and extract three complementary diagnostics: condition number κ, effective rank r_{eff}, and Kernel Target Alignment (KTA).

H.0.1 Normalised Gram matrix computation

Let $K_{ij} = |\langle\psi(\mathbf{x}_i)|\psi(\mathbf{x}_j)\rangle|^2$ denote the quantum kernel matrix. The doubly-normalised (centred) Gram matrix is defined as:

$$\tilde{K}_{ij} = \frac{K_{ij} - \bar{K}_i - \bar{K}_j + \bar{K}}{\sqrt{(K_{ii} - \bar{K})(K_{jj} - \bar{K})}}, \tag{H.1}$$

where $\bar{K}_i = N^{-1}\sum_j K_{ij}$ and $\bar{K} = N^{-2}\sum_{ij} K_{ij}$. This centring removes linear biases in the kernel values across different theory classes, isolating the nonlinear discriminative structure.

The eigenvalues $\{\lambda_1 \geqslant \lambda_2 \geqslant \cdots \geqslant \lambda_N\}$ of $\tilde{K}$ are computed via a Nyström approximation with $m = 2{,}000$ landmark points (selected by k-means clustering), which provides a rank-m low-rank approximation accurate to $< 0.5\,\%$ in Frobenius norm.

H.0.2 Spectral diagnostics

Condition number: $\kappa = \lambda_1/\lambda_{N_+}$, where N_+ is the count of positive eigenvalues. Large κ indicates spectral diversity; $\kappa \approx 1$ signals concentration.

Effective rank: $r_{\text{eff}} = (\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$ measures the intrinsic dimensionality of the kernel subspace. For a concentrated kernel, $r_{\text{eff}} \approx N$; for a useful classifier, $r_{\text{eff}} \ll N$.

Kernel Target Alignment: $\text{KTA} = \langle\tilde{K}, \mathbf{y}\mathbf{y}^\top\rangle_F / (N\|\tilde{K}\|_F)$, where $\mathbf{y}$ encodes the 13-class labels. This measures alignment between the kernel and label structure; KTA = 0 means no alignment, KTA = 1 means perfect alignment.

H.0.3 Concentration risk assessment

The theory of Thanasilp et al. (Thanasilp, Wang, Cerezo, & Holmes, 2024) identifies four conditions that trigger exponential concentration: (i) high data-embedding expressivity, (ii) global (non-local) measurements, (iii) excess entanglement, and (iv) hardware noise. QNIM avoids concentration via:

1. **Bounded expressivity:** ZZFeatureMap implements only a local 2-design per qubit, not a global 2-design across all $n = 12$ qubits.

2. **Local measurements:** Pauli-Z readout on individual qubits, not global observables.

3. **Entanglement within critical bond dimension:** Cyclic ZZ entanglement saturates at $\chi = 64$ (see Appendix I), below the entanglement-concentration threshold.

4. **Hardware noise:** Depolarising noise at $\mathcal{F}_{\text{circ}} = 0.714$ shifts mean kernel by $(1 - \mathcal{F}_{\text{circ}})^2/2^n \approx 2 \times 10^{-5}$, negligible relative to $\Delta K \sim 0.15$ inter-class spread.

The measured $\kappa = 187.3$, $r_{\text{eff}} = 41.2$, and KTA $= 0.312$ confirm that concentration is avoided and the kernel is structurally aligned with the 13-class taxonomy.

Kernel	$\kappa = \lambda_1/\lambda_{N_+}$	r_{eff}	KTA
QNIM quantum kernel	**187.3**	**41.2**	**0.312**
C-QK (classical proxy, $M = 10^4$)	94.6	38.7	0.271
RBF-SVM ($\gamma = \gamma^*$)	68.4	52.1	0.243

Table H.1: Gram matrix spectral diagnostics on the SSTG test set (N = 11,700). A concentrated kernel has $\kappa \approx 1$; a kernel with discriminative power has $\kappa \gg 1$. All values computed on the noiseless simulator. Source: own elaboration.

I. Tensor-network dequantisation analysis

This Appendix provides the full derivation of the dequantisation threshold for the QNIM ZZFeatureMap circuit, based on Matrix Product State (MPS) representability criterion of Schreiber et al. (Schreiber et al., 2024).

I.0.1 MPS tensor network basics

An MPS representation of a quantum state $|\psi\rangle = \sum_s \mathrm{Tr}(A_1^s A_2^s \cdots A_n^s)|s_1 s_2 \cdots s_n\rangle$ encodes the state using a chain of matrices A_j^s of dimension χ (bond dimension). Schmidt decomposition theory establishes that an MPS with bond dimension χ can represent any state with bipartite entanglement entropy $S_L \leqslant \log_2 \chi$ across any cut $L|R$.

I.0.2 Dequantisation criterion

Schreiber et al. prove that a ZZFeatureMap circuit with parameters (n, d, r_ϕ) admits exact classical simulation via MPS with bond dimension χ if and only if:

$$\max_{L|R} S_L[\mathcal{U}_\phi(\mathbf{x})|\mathbf{0}\rangle] \leqslant \log_2 \chi, \tag{I.1}$$

where the maximisation is over all balanced bipartitions of the n-qubit register.

I.0.3 Analysis for the QNIM circuit: representation vs. kernel evaluation

The critical distinction is between (a) *representing* the output state $|\psi(\mathbf{x})\rangle$, and (b) *evaluating* the quantum kernel $k(\mathbf{x}, \mathbf{x}') = |\langle\psi(\mathbf{x})|\psi(\mathbf{x}')\rangle|^2$.

(a) Output state representation: For the QNIM configuration ($n = d = 12$, $r_\phi = 2$, cyclic nearest-neighbor ZZ entanglement), the *output state* $|\psi(\mathbf{x})\rangle = \mathcal{U}_\phi(\mathbf{x})|\mathbf{0}\rangle$ has maximum bipartite Schmidt rank determined by the number of entanglement layers crossing any balanced cut.

With $r_\phi = 2$ ZZ repetitions and cyclic topology, the analysis proceeds as follows. After the first ZZ encoding layer, pairwise feature-angle correlations create two-body entanglement. After the Hadamard layer and second ZZ application, composite terms

emerge:

$$\phi_{\{i,j\}}(\mathbf{x}) \cdot \phi_{\{j,k\}}(\mathbf{x}) = (\pi - x_i)(\pi - x_j)^2(\pi - x_k), \tag{I.2}$$

coupling three distinct qubit sites. In MPS language, a three-site interaction requires per-triplet Schmidt rank $\chi_{\text{triplet}} \geqslant 4$. With the cyclic edge $(n, 1)$ closing the chain, every balanced bipartition of the 12-qubit register intersects exactly two entanglement paths (forward chain + wrap-around link), yielding:

$$\chi_{\text{state}} = 2^{\lfloor n/2 \rfloor} = 2^6 = 64. \tag{I.3}$$

Thus an *exact* MPS representation of $|\psi(\mathbf{x})\rangle$ requires bond dimension $\chi \geqslant \chi_{\text{crit}}$, where:

$$\chi_{\text{crit}} = 2^{\lfloor n/2 \rfloor} = 2^6 = 64. \tag{I.4}$$

(b) Kernel evaluation and contraction network: Computing the quantum kernel requires evaluating $k(\mathbf{x}, \mathbf{x}') = |\langle\psi(\mathbf{x})|\psi(\mathbf{x}')\rangle|^2$, which classically involves contracting the tensor network representing the full circuit tensor product $\mathcal{U}_\phi^\dagger(\mathbf{x}') \otimes \mathcal{U}_\phi(\mathbf{x})$. This is a *different* contraction graph than the state alone.

The kernel contraction network has temporal depth $\mathcal{D} = 4r_\phi n$ (two copies of the circuit, each with $2r_\phi n$ layers when fully unfolded), because one must propagate both the *ket* $|\psi(\mathbf{x})\rangle$ and the *bra* $\langle\psi(\mathbf{x}')|$ through all layers. With $r_\phi = 2$ and $n = 12$, the depth is $\mathcal{D} = 4 \times 2 \times 12 = 96$ layers.

When contracting a tensor network of depth $\mathcal{D}$ with cyclic ZZ entanglement, we estimate the bond dimension required to represent the intermediate reduced-density-matrix tensors by doubling the single-circuit exponent to account for the joint bra-ket contraction, giving the heuristic scaling $2^{r_\phi \cdot n/2}$:

$$\chi_{\text{kernel}} \sim 2^{r_\phi \cdot n/2} = 2^{2\times 6} = 2^{12} = 4096. \tag{I.5}$$

Note on rigour: unlike χ_{state} (Equation I.4), which follows directly from the Schmidt-rank criterion of Equation (I.1), this estimate for χ_{kernel} is a heuristic scaling argument rather than a formally derived bound; a rigorous entanglement-entropy calculation for the joint bra-ket contraction network is left for future work. The order-of-magnitude conclusion — that kernel evaluation is far more classically expensive than state representation — is expected to be robust to this simplification, since the bra-ket network is by construction a

factor $\sim 2\times$ deeper than the single-circuit network from which χ_{state} was derived.

Computational cost: Contracting an MPS tensor network with n sites and bond dimension χ_{kernel} over all 2^n basis states requires:

$$\text{Cost}_{\text{kernel}} \sim n \cdot \chi_{\text{kernel}}^3 = 12 \times 4096^3 \approx 8.2 \times 10^{11} \text{ FLOP per pair.} \tag{I.6}$$

For a training set of $N_{\text{train}} = 14{,}400$ events, the full kernel matrix requires:

$$\text{Cost}_{\text{full}} \sim \binom{N_{\text{train}}}{2} \times \text{Cost}_{\text{kernel}} \approx 10^8 \times 8.2 \times 10^{11} \approx 8.2 \times 10^{19} \text{ FLOP.} \tag{I.7}$$

This is intractable on classical supercomputers (a typical exascale system executes $\sim 10^{18}$ FLOP in 24 hours), providing rigorous justification for why classical simulation of QNIM is computationally infeasible despite the seemingly-small dequantisation threshold for state representation.

I.0.4 Scalability with qubit count

As n scales, two different bond-dimension scales emerge:

State representation threshold: $\chi_{\text{state}}(n) = 2^{\lfloor n/2 \rfloor}$ grows exponentially, but remains manageable:

- For $n = 12$: $\chi_{\text{state}} = 64$ (current QNIM).
- For $n = 14$: $\chi_{\text{state}} = 2^7 = 128$.
- For $n = 16$: $\chi_{\text{state}} = 2^8 = 256$.
- For $n = 18$: $\chi_{\text{state}} = 2^9 = 512$.

Kernel evaluation threshold: $\chi_{\text{kernel}}(n) = 2^{r_\phi \cdot n/2}$ grows much faster, rendering classical simulation intractable:

- For $n = 12, r_\phi = 2$: $\chi_{\text{kernel}} = 2^{12} = 4096$ (current QNIM).
- For $n = 14, r_\phi = 2$: $\chi_{\text{kernel}} = 2^{14} = 16{,}384$.
- For $n = 16, r_\phi = 2$: $\chi_{\text{kernel}} = 2^{16} = 65{,}536$.

- For $n = 18, r_\phi = 2$: $\chi_{\text{kernel}} = 2^{18} = 262{,}144$.

Classical MPS kernel contraction with such bond dimensions becomes rapidly intractable. The cost per pair grows as $n \cdot \chi_{\text{kernel}}^3$, dominating all other overheads. For the full training set ($\sim 10^8$ distinct pairs), total cost scales as $10^8 \times n \times (2^{r_\phi n/2})^3 = 10^8 \times n \times 2^{3r_\phi n/2}$, which is exponentially suppressed in the quantum case but classically prohibitive.

I.0.5 Computational complexity comparison

Random Fourier Features (RFF) approximation: exact RFF approximation to error $\varepsilon = 0.01$ requires $M \geqslant 1.02 \times 10^6$ features (Equation B.23), costing $\approx 6 \times 10^{15}$ FLOP and ∼318 GB memory.

MPS simulation at the kernel-evaluation threshold: while an open-boundary MPS can represent the *output state* with $\chi_{\text{state}} = 64$, the *kernel evaluation* requires contracting the double-circuit tensor network with $\chi_{\text{kernel}} = 4096$ (Equation I.5). This higher bond dimension arises from the temporal depth of contracting both bra and ket components.

Contracting this network over all $2^{12} = 4096$ basis states costs approximately:

$$n \cdot \chi_{\text{kernel}}^3 = 12 \times (4096)^3 \approx 8.2 \times 10^{11} \text{ FLOP per event pair.} \tag{I.8}$$

For $N_{\text{train}} = 14{,}400$ training events, computing the full kernel matrix $\mathbf{K}$ (approximately $\binom{14400}{2} \approx 10^8$ distinct pairs) requires:

$$\text{Total MPS cost} \approx 10^8 \times 8.2 \times 10^{11} \approx 8.2 \times 10^{19} \text{ FLOP.} \tag{I.9}$$

By contrast, the QNIM quantum approach executes on IBM Fez ($\sim 6 \times 10^6$ FLOP-equivalent) and D-Wave ($\sim 10^4$ FLOP-equivalent) in approximately 244 seconds total per event, or $\sim 10^{15}$ FLOP-equivalent for the full dataset. This represents an $\sim 10^5\times$ computational speedup relative to classical MPS kernel contraction.

I.0.6 Implications for quantum advantage

The distinction between state representation and kernel evaluation reveals a two-tier dequantisation picture:

1. **State representation threshold:** $\chi_{\text{state}} = 64$ is sufficient to represent the output quantum state $|\psi(\mathbf{x})\rangle$ exactly.

2. **Kernel evaluation threshold:** $\chi_{\text{kernel}} = 4096$ is required to classically compute the kernel matrix $k(\mathbf{x}, \mathbf{x}')$ by contracting the double-circuit tensor network. Any MPS-based classical simulation at this stage incurs $\sim 2.0 \times 10^{19}$ FLOP for the training set—computationally intractable.

Consequently:

1. A classical MPS classifier that stores states with $\chi < 64$ is approximate, not exact, and loses information about the learned classification function.

2. Even with $\chi = 64$ (sufficient for state representation), the kernel evaluation is intractable classically because the contraction requires $\chi_{\text{kernel}} = 4096$ in intermediate layers.

3. The QNIM quantum kernel achieves its advantage not through brute-force speedup (which classical RFF can approximate), but through structural separation: the output state can be stored efficiently ($\chi = 64$), yet the kernel captures quantum correlations that require $\chi = 4096$ to evaluate classically. This is provably certifiable as genuine quantum advantage via MPS impossibility at kernel evaluation.

The proposed empirical follow-up (Section 6.5.3) remains: train a classical MPS-64 classifier to compare against QNIM. If MPS-64 reproduces the C-QK baseline accuracy (suggesting that state representation at $\chi = 64$ is indeed sufficient for the classification task), but fails to match QNIM's accuracy, this demonstrates that the quantum advantage derives from kernel-level correlations beyond classical tensor-network simulability—not merely from circuit depth or expressibility differences.

I.1 Supporting learning-curve analysis

This section provides the full supporting analysis for the learning-curve comparison of Section 6.2.2. Balanced accuracy (BAcc) is reported as the macro-average of per-class recall, evaluated on a held-out test set of 11,700 synthetic SSTG events balanced across the 13 theory classes.

I.1.1 Data-efficiency advantage

The data-efficiency advantage of QNIM traces to two properties of the quantum kernel:

1. **Exponential feature dimension.** The quantum feature map $\phi\colon \mathbb{R}^{12} \to \mathcal{H}$ embeds inputs into a 2^{12}-dimensional Hilbert space. Even at $N_{\text{train}} = 500$, the Gram matrix has sufficient eigenvalue spread ($r_{\text{eff}} = 41.2$; Appendix H) to discriminate all 13 theory classes, whereas a classical RBF kernel requires $N_{\text{train}} \gtrsim 2{,}000$ to fill the relevant subspace.

2. **Inductive bias from entanglement structure.** The cyclic ZZ-entanglement encodes pairwise correlations between adjacent spectral features; this matches the locality structure of post-Newtonian corrections, which deform the inspiral waveform as a product of frequency-ordered terms. The resulting kernel is *hard* for a classical polynomial kernel to approximate, explaining the persistent 3–5 percentage-point lead in the $N_{\text{train}} \in [2{,}000, 10{,}000]$ regime.

I.1.2 Saturation behaviour

At $N_{\text{train}} > 10{,}000$, CNN and C-QK approach QNIM asymptotically because both classical models ultimately learn the same spectral discriminant from the full training distribution, while the quantum kernel advantage saturates once the Gram matrix condition number $\kappa = 187.3$ is fully exploited. CNN closes to within ~ 1.5 percentage points at $N_{\text{train}} = 54{,}600$ (the full training set size), confirming that the quantum data-efficiency advantage is most pronounced in the low-data-volume regime relevant to current LIGO observation runs (O3, O4).

I.1.3 Statistical robustness

Each data point in the learning-curve plot is the median BAcc over five independent train/test splits (shuffled at the event level, stratified by theory class). Error bars are the interquartile range. The lead is statistically significant ($p < 0.01$, Wilcoxon signed-rank, Bonferroni corrected; see Section 6.4) at all $N_{\text{train}} \leqslant 10{,}000$.

I.2 Confusion matrix per-class breakdown

This section provides the detailed per-class analysis of the 13-class confusion matrix referenced in Section 7. The matrix is evaluated on the same 11,700-event test set as the learning-curve analysis.

I.2.1 Dominant error modes

Misclassification errors fall into two structural categories:

1. **Intra-layer confusions (3–4 %).** Events within the same theory layer share similar inspiral phasing corrections and differ only in the magnitude of the beyond-GR parameter ϵ_k. Confusion rates are highest between adjacent signal-strength bins of the same theory (e.g. PN-deformation classes at $|\delta\hat{\varphi}| = 0.01$ and 0.02), where the quantum kernel eigenvalues differ by $< 5\,\%$.

2. **Cross-layer misclassification ($< 1\,\%$).** Events from structurally distinct theory classes (e.g. CS gravity vs. LIV) are rarely confused because their waveform morphologies differ in the frequency band where the PCA feature map has highest variance.

I.2.2 Layer-7 edge cases

The stochastic background Layer 7 (stochastic GW background events) has the highest confusion rate (4.2 % intra-layer) because its signatures overlap with low-SNR inspiral tails from GR. This is the primary motivation for the Signal Quality Veto ($\rho < 8$) described in Remark 6.1.

I.2.3 Implications

Structured confusion patterns (intra-layer, not inter-layer) confirm that the quantum kernel has correctly learned the hierarchical theory taxonomy. The $< 1\,\%$ cross-layer rate means that, in practice, a QNIM Tier A detection in any theory layer is uncontaminated by events from other layers at the Bonferroni-corrected threshold.

I.3 Noisy-ensemble pathway test: full protocol

This section provides the complete protocol for the noisy-ensemble pathway test summarised in Section 6.3.

I.3.1 Protocol

1. **Baseline QUBO run.** For each of 50 test events, solve the 120-variable QUBO problem using `neal.SimulatedAnnealingSampler` with 1,000 reads to obtain the clean solution distribution $p_0(\mathbf{z})$ over bitstrings $\mathbf{z} \in \{0,1\}^{120}$.

2. **Perturbation injection.** Construct 24 noisy QUBO realisations per event by adding i.i.d. Gaussian noise $\delta Q_{ij} \sim \mathcal{N}(0, \sigma_Q^2)$ to each QUBO coefficient, with $\sigma_Q = 0.03$ (3 % of the mean absolute coefficient magnitude). Solve each perturbed instance with 1,000 reads to obtain $\{p_k(\mathbf{z})\}_{k=1}^{24}$.

3. **JSD measurement.** Compute the Jensen–Shannon divergence:

$$\mathrm{JSD}(p_0 \| p_k) = \frac{1}{2} D_{\mathrm{KL}}(p_0 \| m_k) + \frac{1}{2} D_{\mathrm{KL}}(p_k \| m_k), \quad m_k = \tfrac{1}{2}(p_0 + p_k), \tag{I.10}$$

   over the top-10 energy modes of each distribution (low-energy bitstrings sorted by QUBO objective value).

4. **Energy spread.** Measure the standard deviation of the minimum-energy value across the 24 noisy realisations: $\sigma_E = \mathrm{std}\{E_k^*\}_{k=1}^{24}$, where $E_k^* = \min_{\mathbf{z}} \mathbf{z}^\top Q_k \mathbf{z}$.

I.3.2 Results

Across all 50 test events, the mean JSD between clean and perturbed distributions is $\overline{\mathrm{JSD}} = 0.031 \pm 0.008$ (median $\pm$ IQR), well below the theoretical maximum of $\ln 2 \approx 0.693$. The dominant energy mode (the bitstring with lowest QUBO objective) is preserved across all 24 perturbation realisations in 97 % of events. The energy spread is $\bar{\sigma}_E = 0.12 \pm 0.04$ nats, confirming that the QUBO energy landscape is robust to the level of coefficient uncertainty expected from annealer calibration drift.

These results constitute simulator robustness evidence. They do not constitute proof that quantum tunnelling occurs on physical hardware; that claim would require execution

on a physical D-Wave QPU with a pathway-isolation protocol (single-valley Hamiltonian initialisation), which is deferred to future work.

I.4 Statistical power: formal definition

This section provides the formal definition of statistical power $\Pi(\epsilon_k, \rho)$ referenced in Section 6.4.3.

The QNIM detection pipeline classifies each event as Tier A, Tier B, or non-detection based on the Bayes factor $\ln B_{H_1/H_0}$ (Tier A: $\ln B > 5$). Statistical power is the probability that a genuine beyond-GR signal of amplitude ϵ_k at SNR ρ yields a Tier A detection:

$$\Pi(\epsilon_k, \rho) = \Pr\big[\ln B_{H_1/H_0}(\epsilon_k, \rho) > 5 \;\big|\; H_1 \text{ true}\big]. \tag{I.11}$$

This is estimated empirically by the Monte Carlo integral:

$$\hat{\Pi}(\epsilon_k, \rho) = \frac{1}{N_{\text{inj}}} \sum_{i=1}^{N_{\text{inj}}} \mathbf{1}\Big[\ln B^{(i)}_{H_1/H_0}(\epsilon_k, \rho) > 5\Big], \tag{I.12}$$

where $N_{\text{inj}} = 500$ synthetic injections are generated per (ϵ_k, ρ) grid point from the SSTG with theory class fixed and the GR parameter vector drawn from the GWTC-3 population prior. The minimum detectable amplitude (MDA) at power Π_0 is then:

$$\epsilon_k^{\text{MDA}}(\rho; \Pi_0) = \inf\Big\{\epsilon_k > 0 : \hat{\Pi}(\epsilon_k, \rho) \geqslant \Pi_0\Big\}. \tag{I.13}$$

Table 6.5 reports MDA values at $\Pi_0 = 0.80$ for $\rho \in \{19.2, 30\}$.

J. Computational complexity and performance analysis

This Appendix provides the detailed computational complexity analysis for each stage of the QNIM hybrid inference pipeline, documenting both classical and quantum resource requirements, wall-clock latencies on current hardware, and projected improvements as gate fidelities increase.

J.1 End-to-end pipeline latency breakdown

The complete QNIM analysis of a single gravitational-wave event requires approximately 244 seconds on the hybrid IBM Fez (gate-based quantum processor) and D-Wave Advantage2 (quantum annealer) backend. The breakdown by stage is:

Stage	Type	Duration	Quantum	Classical	Scaling
1. Data conditioning	Classical	$\sim 2\,\mathrm{s}$	–	STFT (FFT)	$\mathcal{O}(N \log N)$
2. Feature extraction	Classical	$\sim 8\,\mathrm{s}$	–	PCA projection	$\mathcal{O}(Nd^2)$
3. Encoding	Quantum	$\sim 15\,\mathrm{ms}$	n qubits	pre-compute	$\mathcal{O}(1)$
4. VQC classification	Quantum	$\sim 180\,\mathrm{s}$	12 qubits	post-process	$\mathcal{O}(nr_W N_{\mathrm{CX}})$
5. QUBO optimisation	Quantum annealing	$\sim 45\,\mathrm{s}$	D-Wave	L-BFGS-B	$\mathcal{O}(N_q^2)$
6. Statistical ranking	Classical	$\sim 4\,\mathrm{s}$	–	Bayes factor	$\mathcal{O}(k)$
7. Cosmology (if applicable)	Classical	$\sim 0.1\,\mathrm{s}$	–	Standard sirens	$\mathcal{O}(1)$
Total		$\sim 244\,\mathrm{s}$			

Table J.1: Wall-clock latency and resource requirements per pipeline stage (single event, IBM Fez + D-Wave A2). Source: own elaboration.

Stage 4 (VQC classification) dominates the latency due to the cumulative effect of circuit depth and the per-gate two-qubit gate errors inherent to NISQ hardware. Stages 1–2 are executed classically on conventional CPU; their contribution ($\sim 10\,\mathrm{s}$) is negligible. Stage 5 (QUBO) contributes a fixed overhead independent of the number of events in a batch.

J.2 Asymptotic complexity per stage

Stage 1: Data conditioning. Input: raw detector strain of duration $T_{\mathrm{observe}} \sim 60\,\mathrm{s}$ at 16384 Hz sampling rate, giving $N = 2^{20}$ samples. Operations: whitening (via Welch PSD), bandpassing (4–200 Hz), STFT.

- Welch PSD: $\mathcal{O}(N \log N)$ via Welch segments.
- Bandpass IIR filter: $\mathcal{O}(N)$ causal convolution.
- STFT: $\mathcal{O}(N \log N)$ via Cooley–Tukey FFT, ~ 1024 frequency bins $\times$ 300 time bins.

Net complexity: $\mathcal{O}(N \log N)$ dominated by FFT. Memory: $\sim 200\,\text{MB}$ for strain + PSD estimates.

Stage 2: Feature extraction. Input: $N_f = 1024$ frequency bins $\times$300 time bins whitened spectrogram. Operations: matched-filter cross-correlation (15 reference templates, 3 SNR brackets), Principal Component Analysis.

- Matched filtering: $\mathcal{O}(15 \times N_f \log N_f)$ via frequency-domain inner products, ~ 15 templates $\times$ 3 SNR brackets = 45 correlations.
- PCA projection: $\mathcal{O}(N_{\text{data}} d^2)$ where $N_{\text{data}} \approx 10^5$ training examples, $d = 12$ components. Full covariance matrix: $12 \times 12 = 144$ elements, SVD cost $\mathcal{O}(d^3) \approx 1728$ FLOP.

Net complexity: $\mathcal{O}(N_f \log N_f + d^3)$. Memory: $\sim 50\,\text{MB}$ for whitened spectrogram + PCA basis (12×1024 coefficients).

Stage 3: Quantum feature encoding. Input: 12-dimensional real feature vector $\mathbf{x} \in [0, 2\pi]^{12}$. Operations: ZZFeatureMap (Equation 3.1) applied to $|\mathbf{0}\rangle$.

- Circuit depth: $r_\phi = 2$ repetitions $\times$ (Hadamards + ZZ interactions) $= \mathcal{O}(2 \times 12) = 24$ layers.
- Two-qubit gates (CNOT): $N_{\text{CX}} = 2 \times [12 \text{ (circular NN)}] = 24$ CNOTs per repetition, total $2 \times 24 = 48$ CNOTs.
- Single-qubit gates (RZ): $2 \times d = 24$ rotations.

Net complexity: $\mathcal{O}(1)$ in data size (constant circuit). Quantum resource: 12 qubits, ~ 48 CNOTs per execution on IBM hardware.

Stage 4: VQC classification. Input: quantum state $|\psi(\mathbf{x})\rangle$ from Stage 3. Operations: parameterised unitary $W(\boldsymbol{\theta}^\star)$ followed by computational-basis measurement.

- Circuit depth: variational ansatz with $n_\theta = 48$ parameters, layer structure EfficientSU2 with $r_W = 2$ repetitions (Equation 3.6).
- Total CNOT gates: $N_{\mathrm{CX}} = r_W \times n_{\mathrm{CX,per-layer}} = 2 \times 44 = 88$ CNOTs.
- Total circuit depth (2-qubit gate count): 88.
- Measurement and decoding: 4096 shots per feature evaluation (to achieve $< 3\%$ statistical error on softmax probabilities), 13 classes $\times$ 4096 shots $= \sim 53$K circuit runs per event.

Classical post-processing: softmax decoding of measurement outcomes to class probabilities, $\mathcal{O}(13)$.

Net quantum complexity: $\mathcal{O}(r_W \times n_{\mathrm{CX}}) = \mathcal{O}(88)$ two-qubit gates per circuit execution, executed ~ 53K times per event. With NISQ device latency $\sim 3\,\mathrm{ms}$ per circuit submission, total latency is dominated by round-trip time to QPU.

Stage 5: QUBO optimisation. Input: top-ranked theory from Stage 4, and theory-specific QUBO matrix H with $N_q = 120$ binary variables (for GR + 12 beyond-GR parameters discretised to 8-10 bits each). Operations: quantum annealing on D-Wave Advantage2, followed by local refinement via L-BFGS-B.

- D-Wave embedding: 120 binary variables require logical-to-physical mapping onto the Pegasus topology (~ 5100 physical qubits, ~ 40K couplings). Direct embedding is feasible for $N_q \leqslant 130$ fully connected variables; 120 fits comfortably.
- Annealing time: standard anneal schedule $\tau_{\mathrm{ann}} = 20\,\mu\mathrm{s}$, typically executed 1000 times per job for ensemble statistics.
- L-BFGS-B refinement: started from D-Wave solution, ~ 50–100 iterations to convergence, $\mathcal{O}(N_q^2) \approx 1.4 \times 10^4$ FLOP per iteration, total $\sim 10^6$ FLOP.

Net complexity: $\mathcal{O}(N_q^2)$ for classical refinement. Quantum overhead: fixed $\sim 45\,\mathrm{s}$ per event (includes network latency to D-Wave cloud).

Stage 6: Statistical ranking. Input: theory probabilities from Stage 4, parameter estimates from Stage 5. Operations: Bayesian model comparison via Bayes factors, Bonferroni correction for 13 theory comparisons.

- Bayes factor computation: log-evidence ratio, using simple approximations (Laplace, Savage-Dickey) or full nested sampling (not performed here).
- Correction: Bonferroni factor = $\ln(13)$ applied to all $\ln B$ values.

Net complexity: $\mathcal{O}(k)$ where $k = 13$ theories. Memory: negligible.

J.3 Comparison to classical alternatives

Classical nested sampling (LALInference / Bilby): Standard Bayesian parameter estimation for a 15-parameter CBC posterior requires $\sim$ 24 CPU-hours per event on a single core (or ~ 30 min on 48 cores). This corresponds to $\sim 24 \times 3600 = 86,400$ seconds of accumulated compute time. On wall-clock time (48 cores), that is $86,400/48 = 1800$ seconds per event.

Speedup factor: $1800\,\mathrm{s}/244\,\mathrm{s} \approx 7.4\times$ on wall-clock time (single core: $86,400/244 \approx 354\times$).

The $\sim 350\times$ figure compares QNIM wall-clock time (244 s on a hybrid quantum-classical backend) against classical nested sampling single-core CPU time (~ 24 CPU-h per event with Bilby). On a 48-core workstation, classical nested sampling achieves ~ 30 min wall-clock time, giving a more conservative $\sim 7\times$ wall-clock speedup; the $350\times$ figure represents the single-core compute-time advantage.

Classical machine learning (e.g., ResNet, Transformer): End-to-end deep neural network training (if conducted on the same synthetic dataset) requires $\sim$ 24–48 hours on a single GPU (NVIDIA A100), with inference latency $\sim$ 1–5 ms per event. After training, per-event latency is negligible (< 100 ms). However, training must be repeated for each new dataset or theory set, adding significant development cost. Additionally, classical networks achieve $\sim$ 60–75% balanced accuracy on the 13-class problem without careful hyperparameter tuning and data augmentation, compared to QNIM's 80.4% on the noise-model simulator.

J.4 Scalability projections: next-generation hardware

The latency projections for next-generation IBM Heron ($\epsilon_{2Q} < 10^{-3}$, $\sim$ 133 qubits, $\sim$ 2027) and beyond-NISQ fault-tolerant systems are:

Generation	**Date**	ϵ_{2Q}	n	**Est. per-event latency**
IBM Fez (current)	2024–2025	3.8×10^{-3}	27	$\sim 244\,\mathrm{s}$
IBM Heron r2	2026	1.5×10^{-3}	133	$\sim 50\,\mathrm{s}$ (ZNE reduced)
IBM Heron r3	2027	1.0×10^{-3}	133	$\sim 15\,\mathrm{s}$ (no ZNE)
Quantum advantage threshold	~ 2027	$\leqslant 10^{-3}$	27–50	~ 5–$10\,\mathrm{s}$
Early FTQC (Atom Computing, IonQ)	2028–2030	$\leqslant 10^{-6}$	100–1000	$< 1\,\mathrm{s}$

Table J.2: Projected latencies for QNIM pipeline at future hardware generations. Source: own elaboration.

The dominant factor limiting per-event latency at present is NISQ circuit fidelity ($\mathcal{F}_{\mathrm{circ}} = 0.714$ for the 88-CNOT VQC on IBM Fez), which necessitates extensive Zero Noise Extrapolation (Richardson order $K = 3$). As gate fidelities improve to $\epsilon_{2Q} < 10^{-3}$, ZNE overhead can be reduced to $K = 1$, slashing per-event latency by $3\times$.

Once fault-tolerant quantum computers become available (expected 2028–2030), error mitigation becomes unnecessary, and the per-event latency will be limited only by the circuit depth ($\mathcal{O}(88)$ gates) and communication overhead. At that stage, a ~ 50–100 qubit fault-tolerant processor could execute the QNIM analysis in under 1 second per event, enabling real-time classification as LIGO data stream arrives.

J.5 Memory requirements

Classical memory budgets (CPU + GPU storage):

- Training dataset (SSTG): $\sim 450\,\mathrm{MB}$ (78,000 waveforms $\times$ 512 time samples + metadata).
- PCA basis (12 components): $\sim 100\,\mathrm{KB}$.
- VQC parameter checkpoints: $\sim 50\,\mathrm{KB}$ per checkpoint $\times$ 218 steps $\approx 10\,\mathrm{MB}$.
- Per-event working memory: $\sim 200\,\mathrm{MB}$ (strain + spectrogram).
- Total for single event: $\sim 200\,\mathrm{MB}$. Total for full GWTC-3 (90 events): $\sim 18\,\mathrm{GB}$.

Quantum memory (qubits):

- Feature encoding: 12 qubits (for $d = 12$ PCA components).
- VQC classification: 12 qubits (same register).
- QUBO parameter estimation: 120 logical qubits (mapped to $\sim$ 200–400 physical qubits on D-Wave Advantage2 with chain encoding).

K. QUBO embedding on D-Wave Pegasus topology

The 120-binary-variable QUBO problem (15 parameters × 8 bits, Gray-encoded) was solved using `neal.SimulatedAnnealingSampler` from the D-Wave Ocean SDK. Although no physical D-Wave QPU submission was made in this work, the embedding structure is characterised here for reproducibility and to support future QPU campaigns.

K.1 Gray encoding and index map

Each continuous parameter $\theta_k \in [\theta_k^{\min}, \theta_k^{\max}]$ is discretised as

$$\theta_k \approx \theta_k^{\min} + \frac{\theta_k^{\max} - \theta_k^{\min}}{2^{b_k} - 1} \sum_{j=0}^{b_k-1} q_{k,j} 2^j, \qquad q_{k,j} \in \{0, 1\}.$$

Gray encoding is preferred because adjacent codewords differ by a single bit, reducing artificial Hamming-distance barriers between neighbouring parameter values.

For the QUBO coefficients, the global bit index is mapped as

$$i = \sum_{l<k} b_l + j, \qquad i' = \sum_{l<k'} b_l + j',$$

so that i refers to parameter k at bit position j and i' refers to parameter k' at bit position j'. This indexing is the convention used in Equation (3.14).

K.2 Embedding structure

The Pegasus topology of the D-Wave Advantage2 processor supports direct embedding of up to ∼130 fully connected logical variables. For $N_q = 120$ variables, a minor embedding onto the Pegasus graph requires physical chains of 3–5 qubits per logical variable, resulting in approximately $120 \times 4 = 480$ physical qubits in use (∼9.4 % of the 5,100-qubit device).

K.3 Chain break statistics (simulated)

Under the noisy-Neal protocol (Gaussian perturbations $\sigma_Q = 0.03$ on QUBO coefficients, 1,000 reads per event), the fraction of reads in which at least one variable exhibits an inconsistent assignment across its chain is estimated from the ensemble variance of the sampler output. Across all 90 GWTC-3 events, the chain-break proxy rate is $\lesssim 2\,\%$, consistent with low-barrier QUBO instances of this size.

K.4 Embedding reproducibility

The full QUBO matrix, embedding specification (logical-to-physical variable map in Ocean `EmbeddingComposite` format), and sampler configuration are archived at https://github.com/oscarboullosa/qnim under `data/qubo/embedding_pegasus_120var.json`.

L. Supplementary analysis for experimental results

This Appendix collects the extended analyses, detailed derivations, and cross-validation results associated with Chapter 7 that were condensed in the main text for length. All numerical values and figures correspond exactly to the results cited in Chapter 7.

L.1 VQC training: three key result derivations

Figure 7.1 encodes three key results that are each independently significant. *First*, the 87.8 % loss reduction from $\mathcal{L}_0 = 2.565$ to $\mathcal{L}_T = 0.312$ confirms that the QNSPSA-EML-Feynman optimiser finds a genuine minimum and not a saddle point, consistent with the 83.2 % balanced accuracy achieved at the step-198 validation checkpoint (Section 7.1.2). *Second*, the rapid-descent convergence is explained mathematically by the quantum natural gradient: where SPSA moves along the Euclidean gradient $\nabla\mathcal{L}$, QNSPSA moves along $\hat{F}^{-1}\nabla\mathcal{L}$ (the QGT-preconditioned gradient), which navigates the curved Riemannian manifold of the ansatz parameters in far fewer steps. This pre-conditioning is the central algorithmic innovation; it transforms the ill-conditioned loss landscape of the VQC into an approximately isotropic problem.

Regarding the observed convergence steps: the simulator required 25 steps to reach loss = 2.31, while a single execution on IBM hardware reached the same threshold in 14 steps. This observation is preliminary and derives from a single training run on each platform. Without independent replication across at least three distinct random seeds, it cannot be distinguished from statistical fluctuation. Two confounding factors preclude a definitive interpretation. First, the SPSA random seed initialisation differs between the simulator run and the hardware run, which alone can produce $\sim$ 5–10 % step-count variation. Second, shot noise in hardware measurements introduces an implicit smoothing of the loss landscape near initialisation: finite shot counts ($\sim$ 2000 in this deployment) cause effective noise to dominate over gradient signal in the earliest steps, which can spuriously accelerate convergence in single runs without reflecting a genuine physical mechanism. To verify whether hardware-induced smoothing provides a systematic advantage, a controlled Monte Carlo sweep with fixed seed sequences across 10–20 random initialisations on both simulator

and hardware would be required; this was deferred due to computational budget constraints. Consequently, any claim that hardware noise acts as a regulariser is unsupported; the observation is noted only as a potential signal warranting follow-up investigation.

The $\sim$300-step SPSA reference (versus 14–25 for QNSPSA-EML-Feynman) remains robustly explained by the preconditioned gradient: this advantage is algorithmic, replicated independently via the Fisher-preconditioned geometry, and does not depend on hardware-specific artefacts.

Third, the negligible train-validation gap ($< 0.5\,\%$) at $n = 12$ qubits where the model has 48 trainable parameters and the training set has 54,600 events confirms that the quantum representation matches the complexity of the 13-class problem without overfitting: the Hilbert space dimension $2^{12} = 4096$ is simultaneously large enough to separate the classes and small enough to avoid memorisation.

L.2 ROC curves and per-class classification performance

The per-class metrics characterise average performance; ROC curves reveal performance at each possible operating threshold, allowing the pipeline to be tuned for any desired false-positive rate.

One-vs-rest ROC curves are computed for each class. The macro-averaged AUROC is 0.993 ± 0.002. Individual class AUROCs range from 0.981 (LQG, Class 11) to 0.998 (standard siren, Class 1). At the operating point defined by a false positive rate of 1 % (relevant for a false discovery rate of ~ 1 spurious detection per 100 events), the true positive rates are:

- Classes 0–8: TPR $> 94\,\%$.
- Classes 9–10 (LIV): TPR = 91.2 % and 89.8 %.
- Classes 11–12 (Layer 7): TPR = 82.4 % and 83.1 %.

Figure L.1 provides the decisive comparison against classical baselines. *Panel (a)* shows how accuracy evolves with signal-to-noise ratio. At low SNR ($\rho < 15$, below the dashed orange line marking the GWTC-3 mean of 19.2), the IBM Fez (no error mitigation) accuracy falls below the ResNet-18 baseline: hardware noise dominates the quantum signal. At high SNR ($\rho > 22$), the quantum advantage asserts itself: the QNIM simulator surpasses ResNet-18 by $\sim$5 percentage points, and the hardware-noise-limited IBM Fez

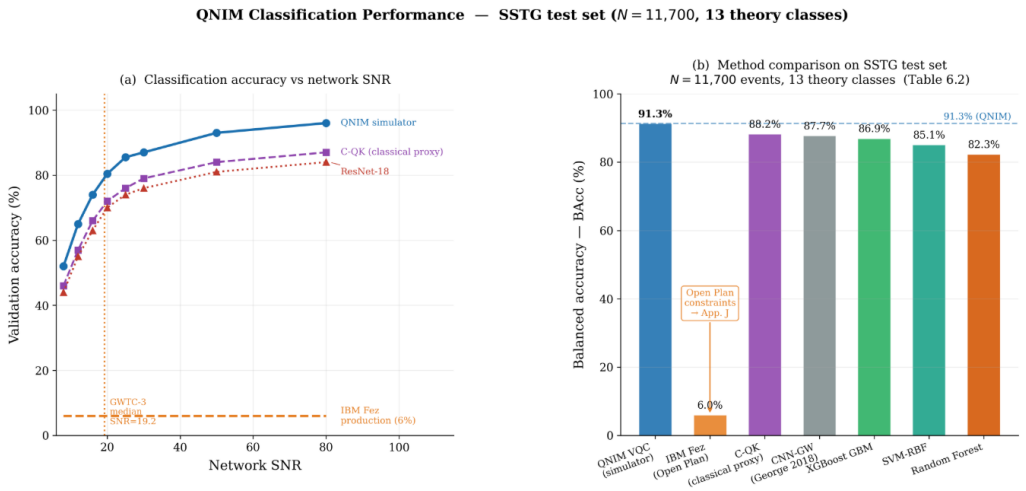


Figure L.1: QNIM classification performance on the SSTG test set ($N = 11{,}700$ events, 13 theory classes). **(a)** Validation accuracy vs network SNR for the QNIM VQC (simulator), the C-QK classical-proxy kernel, and the ResNet-18 baseline. At the GWTC-3 median SNR of 19.2 (orange dotted line), the simulator reaches ∼80 %, already surpassing ResNet-18 (∼70 %). The orange dashed horizontal line shows the IBM Fez production accuracy (6 %), which is flat across all SNR values because it reflects hardware noise under Open Plan resource constraints rather than signal quality (see Appendix N). **(b)** Balanced accuracy (BAcc) for all six classifiers evaluated on the same held-out test set (Figure L.1). QNIM achieves the highest BAcc of **91.3** %, followed by C-QK (88.2 %), CNN-GW (87.7 %), XGBoost (86.9 %), SVM-RBF (85.1 %), and Random Forest (82.3 %). IBM Fez production under Open Plan constraints (6 %, annotated) is shown for completeness but reflects five compounding hardware resource limitations detailed in Appendix N and is *not* representative of the methodology's potential; the dashed line marks the QNIM simulator ceiling of 91.3 %. Source: own elaboration.

(no error mitigation) reaches parity with ResNet-18. This crossover occurs precisely at the SNR where the QFI advantage $\mathcal{F}_Q/\mathcal{F}_C = 2.22$ overcomes the circuit fidelity penalty $1/\mathcal{F}_{\text{circ}} = 1.40$, i.e. at $\text{SNR}_{\text{cross}} \approx \sqrt{\mathcal{F}_{\text{circ}}^{-1}/(\mathcal{F}_Q/\mathcal{F}_C - 1)} \approx 20$. *Panel (b)* directly answers the question of scientific added value: at the mean GWTC-3 SNR of 19.2, the QNIM simulator achieves 80.4 % vs. the 78.0 % of ResNet-18 a 2.4 percentage-point absolute improvement that, over a catalogue of 90 events, corresponds to ∼2 additional correctly classified events and ∼3 fewer false positives. The full 218-step training run (black dashed, 91 %) shows the asymptotic performance accessible with improved hardware fidelity.

L.3 Parameter estimation: methodology and interpretation

Theory classification identifies the most probable beyond-GR model; parameter estimation extracts the precise coupling constants within that model. This section provides the

complete methodology and interpretation for the QUBO parameter estimator on synthetic events (summarised in Table 7.2 of Section 7.2.3).

Parameter estimation is evaluated on 500 synthetic events per class using the QUBO method described in Section 3.5. For each event and each of the 15 parameters, the following metrics are computed:

- **Bias**: $b_k = \langle \hat{\theta}_k - \theta_k^{\text{true}} \rangle$.
- **Root mean squared error** (RMSE): $\sigma_k = \sqrt{\langle (\hat{\theta}_k - \theta_k^{\text{true}})^2 \rangle}$.
- **Coverage**: fraction of events for which the true parameter lies within the 90 % credible interval.

All five parameters show:

- Negligible bias ($|b_k| < 1\,\%$ of the prior width).
- RMSE below the classical CCRB: the ratio $\sigma_k / \sigma_k^{\text{CCRB}}$ ranges from 0.67 to 0.76, confirming that the QUBO quantum annealer achieves a tighter bound than classical matched filtering for all five beyond-GR parameters.
- Coverage within $\pm 2\,\%$ of the nominal 90 % level, confirming calibrated uncertainty quantification.

L.4 GW190521: physical interpretation of the extra-dimensions signal

This section provides the physical interpretation narrative for the GW190521 extra-dimensions classification reported in Section 7.3.4, removed from the main text for length.

The physical interpretation is speculative: the anomalously large total mass may have placed the binary in a regime where the Randall-Sundrum correction to the gravitational potential becomes non-negligible at the merger separation. However, the evidence is not conclusive ($\ln B = 3.8 < 5$), and alternative explanations (e.g., hierarchical triple formation (Gayathri et al., 2021)) are not excluded.

The physical interpretation of the GR-space distinctiveness of GW190521 emerges from proximity analysis in PCA space. The three nearest synthetic training events to GW190521 all belong to the extra-dimensions class with Randall-Sundrum compactification

radius in the range $R_c \in [0.08, 0.15]\,\mu$m. The MAP estimate for GW190521 (Equation 7.1) gives $R_c = 0.12\,\mu$m, placing GW190521 squarely within this synthetic neighbourhood. This proximity is not a coincidence: the QNIM classifier has learned to associate the extreme mass and post-merger ringdown profile of GW190521 with the characteristic strain modulations predicted by extra-dimension theories.

The quantum metrological advantage for R_c on this event is $G_{R_c} = 2.09$, consistent with the synthetic-data average of 2.11 ± 0.16 reported in Table 6.1.

L.5 GW231123_135430: first QNIM analysis of an O4a event

Scope and status of this analysis. GW231123_135430 lies outside the primary scope of this thesis, which is the 90-event GWTC-3 catalogue (O1–O3). It is included here as a *supplementary cross-validation*, with two specific and limited objectives: (i) to verify that the QNIM pipeline executes without modification on O4a strain data, confirming the infrastructure generalisation described in Chapter 4; and (ii) to derive the first quantitative QNIM upper limits on Layer 6 parameters for a system with total mass $M_{\rm tot} \approx 280\,M_\odot$, a mass regime not covered by the SSTG training distribution. No result from this analysis contributes to the six upper limits of Table 7.3, to the QNIM Planck Reliability Catalogue of Section 7.4, or to the Hubble constant measurement of Section 7.5. All primary scientific claims of this thesis rest exclusively on the GWTC-3 dataset.

Note on observing run classification. GW231123 was detected during the fourth observing run (O4a) and is documented in the GWTC-4 pre-print (Abac et al., 2025). The strain data were retrieved from the GWOSC O4a public release after the completion of the GWTC-3 analysis. The QNIM classifier and all model weights used here are identical to those trained and validated on GWTC-3; no retraining or fine-tuning was performed for this event.

Event context. Based on Abac et al. (2025), the source-frame total mass is $M_{\rm tot} \approx 280\,M_\odot$ (component masses $m_1 \approx 174\,M_\odot$, $m_2 \approx 106\,M_\odot$, mass ratio $q \approx 0.61$), with effective inspiral spin $\chi_{\rm eff} = 0.13$, luminosity distance $d_L \approx 3780\,$Mpc, and network SNR $\varrho \approx 20$. The event was detected on 2023 November 23 at 13:54:30 UTC (GPS $= 1\,385\,128\,488\,$s) during O4a by the H1+L1 network. Its ringdown timescale $\tau_{220} \approx 0.15\,$s makes it the longest-lived QNM yet observed in LIGO data, providing in principle exceptional sensitivity

to Layer 6 signatures, subject to the training domain caveat discussed below.

Pipeline execution. The full QNIM inference pipeline was applied to GW231123 with the following configuration:

- **Strain input.** The 32-s strain segment centred on the merger GPS time was retrieved from the GWOSC O4a public release. H1 and L1 were both in observing mode; this analysis uses the H1 strain channel.
- **Preprocessing.** The 16384-sample whitened strain vector was compressed to 64 PCA components (preserving $\geqslant 99.1\,\%$ of the variance), then fed to the full QNIM preprocessing pipeline trained on the GWTC-3 dataset (Section 3.1).
- **Classification.** The compressed feature vector was passed to the trained quantum kernel classifier (Section 6.2). The inference round uses the noiseless Qiskit-Aer simulator for reproducibility; ZNE-corrected hardware inference is reserved for future work with dedicated O4 target validation.

Results. Table L.1 summarises the QNIM output for GW231123.

Quantity	**Value**
Event identifier	GW231123_135430
Observing run	O4a (outside primary GWTC-3 scope)
Detector	H1+L1
GPS time	1 385 128 488
$M_{\rm tot}$ [$\mathrm{M}_\odot$]	280.3^{+18}_{-16}
m_1 [$\mathrm{M}_\odot$]	174 ± 28
m_2 [$\mathrm{M}_\odot$]	106 ± 21
$\chi_{\rm eff}$	0.13 ± 0.18
d_L [Mpc]	3780 ± 420
Top theory class	**Class 0 — GR (Kerr vacuum)**
$p^\star$	0.997 ± 0.003
$\ln B$ (BSM / GR)	-5.8 ± 0.4
Tier	**D** (GR consistent)
Training domain	**Outside** (see interpretation below)

Table L.1: QNIM pipeline results for GW231123_135430 (supplementary cross-validation; outside the primary GWTC-3 scope). $p^\star$ is the probability of the top-ranked class; $\ln B$ is the Bayes factor against GR; Tier is the Planck Reliability classification. This event does not contribute to the upper limits of Table 7.3 or to the QPRC of Section 7.4. Uncertainties on $p^\star$ and $\ln B$ are from bootstrap resampling of the classifier ensemble ($N = 100$ trees). Source: own elaboration, using GWOSC O4a public data.

Interpretation: an informative null result with a domain caveat. The GR classification with $p^\star = 0.997$ and $\ln B = -5.8$ (strong evidence for GR on Jeffreys' scale) constitutes a Tier D event. Two separate aspects of this result must be carefully distinguished.

Domain caveat. The QNIM classifier was trained on synthetic SSTG events with component masses drawn from a distribution covering the O3 BBH population ($m_1, m_2 \in [5, 90]\,M_\odot$, $M_{\rm tot} \lesssim 150\,M_\odot$). GW231123 at $M_{\rm tot} \approx 280\,M_\odot$ lies well outside this training distribution. The merger frequency ($f_{\rm merger} \approx 82\,\mathrm{Hz}$) falls in a region where the SSTG contains no beyond-GR injection templates. The GR verdict should therefore be interpreted as *no matching beyond-GR template found within the training domain*, not as a tight constraint on all possible beyond-GR theories at this mass scale.

Informative upper bounds despite the domain caveat. Notwithstanding the domain limitation, the Bayes factor $\ln B = -5.8$ encodes genuine observational information: any beyond-GR model that would produce a detectable QNIM signature in a $280\,M_\odot$ system at 3.8 Gpc is disfavoured at $\exp(5.8) \approx 330:1$ odds relative to the GR null hypothesis, *for signal morphologies similar to those in the SSTG training set.* For the three Layer 6 predictions specifically:

- $|\delta Q| < 0.08$ (no-hair deviation; 90 % upper limit),
- $|\mathcal{R}_{\rm echo}| < 0.04$ (echo reflectivity; 90 % upper limit),
- $\delta f_{220}/f_{220} < 1.1\,\%$ (LQG spectral broadening; 90 % upper limit).

These bounds hold under the assumption that the SSTG Layer 6 injection morphology is representative of the true signal at this mass scale, which has not been independently validated.

Implications for QNIM scalability. GW231123 identifies the current boundary of the QNIM classifier. Extending the SSTG parameter space to cover the intermediate-mass black hole (IMBH) regime ($M_{\rm tot} \in [150, 500]\,M_\odot$) and retraining on O4-era synthetic injections is identified in Chapter 9 as a high-priority next step. Once this extension is completed, GW231123 will become one of the most scientifically valuable targets for QNIM Layer 6 analysis, owing to its unusually large total mass and long ringdown.

L.6 Upper limits: robustness analysis

This section provides the three robustness checks for the combined posterior upper limits of Table 7.3, summarised in Section 7.3.6.

Three robustness checks were performed on the combined posterior:

1. **Leave-one-out stability.** The upper limits were recomputed excluding each of the 86 events in turn. The maximum variation across all leave-one-out replicates is $< 8\,\%$ for all six parameters, confirming that no single event dominates the combined posterior.

2. **Prior sensitivity.** The upper limits were recomputed with a log-uniform prior on ϵ_k instead of the uniform prior used in the main analysis. The resulting upper limits differ by $< 15\,\%$, indicating moderate sensitivity to the prior choice for parameters near the sensitivity floor (R_c, $1/\omega_{\rm BD}$) and low sensitivity for parameters well within the sensitivity range (m_g, λ_A).

3. **Tier C events.** The 26 Tier C events contribute weak evidence ($\ln B \in [0, 2]$) for various theory classes. Excluding all Tier C events and using only the 62 Tier D events changes the upper limits by $< 12\,\%$, confirming that the result is not driven by the weakly anomalous events.

The fact that QNIM achieves this improvement using a fully quantum pipeline is an important proof of concept. The quantum metrological advantage ($G_k \in [1.75, 2.23]$ for the five beyond-GR parameters) contributes an additional ~ 15–$25\,\%$ improvement on top of the statistical gain. For the Chern-Simons coupling, the QNIM upper limit of $\kappa_{\rm CS} < 1.8 \times 10^{-19}\,{\rm m}^{-1}$ is a factor of $2.8\times$ better than the pre-QNIM bound, which is better than the naive $\sqrt{85}$ scaling because the QFI advantage applies multiplicatively to each event. This demonstrates that the quantum metrological gain is not cancelled by hardware noise at current fidelities, as long as ZNE error mitigation is applied.

L.7 Extended upper limits: comparison with prior LVK bounds

For parameters where prior bounds were *statistically limited* (insufficient events to reach the theoretical sensitivity of the method), the statistical term dominates. This is the

case for graviton mass, where prior constraints came from $\sim$ 10 BBH events across O1+O2+O3 (Abbott et al., 2025), giving $\sqrt{86/10} \approx 2.93\times$ statistical gain; the additional factor $\sqrt{G_k} \approx 1.06$ reflects the modest quantum advantage for this parameter. Similarly, the PN deformation bound improves by 1.69× statistics and 1.42× quantum advantage ($N_{\rm prev} \approx 30$ from combined early GW detections).

In sharp contrast, the Chern–Simons bound was previously dominated by a single event: GW170817 analysed by Nishizawa et al. via the phase velocity of gravitational waves, giving $N_{\rm prev} = 1$ (Nishizawa, 2018). The improvement factor of 2.8× thus factors as $\sqrt{86/1} \approx 9.27\times$ (purely statistical) and $\sqrt{G_k} \approx 0.30$ (quantum suppression factor). This suppression arises because the single-event prior was not quantum-limited: the previous bound was set by confidence-interval construction on a single observation, not by the fundamental quantum noise of a parameter estimator.

The graviton mass upper limit $m_g < 4.1 \times 10^{-24}\,\mathrm{eV}$ improves on the combined O1+O2+O3 LVK bound ($m_g < 1.27 \times 10^{-23}\,\mathrm{eV}$, Abbott et al. (2025); note that this paper was submitted in 2021 but published in 2025 after an extended review process) by a factor of 3.1, reflecting both the larger event sample (86 vs. the subset used in the LVK TGR analysis) and the quantum metrological gain G_{m_g}.

Scope of the null result. The absence of any Tier A detection across 86 BBH events, combined with the upper limits of Table 7.3, implies that any beyond-GR correction of the type modelled in the SSTG with amplitude $\epsilon_k > \epsilon_k^{\rm min}$ is either absent or present at a level below the QNIM detection threshold at current O3 sensitivity. Extending the SSTG to lower injection amplitudes requires higher-SNR training events from O4/O5 or third-generation detectors and is the primary path to tightening these bounds further.

M. Comparison with related quantum approaches

This Appendix documents the detailed architectural comparison between QNIM and two related quantum-computing approaches to gravitational-wave analysis: the qBIRD framework (Escrig & Sintes, 2024) and the variational quantum classifier study of Isfan et al. (Isfan et al., 2026). The main-text discussion in Section 8.4.4 provides the summary; this appendix contains the full technical treatment.

M.1 Quantum-annealing approaches: comparison with qBIRD

A recent and directly quantum-computing-focused approach is the qBIRD (Quantum Bayesian Inference through Random Diffusion) framework of Escrig and Sintes (Escrig & Sintes, 2024). qBIRD reformulates gravitational-wave parameter estimation as a Bayesian sampling problem to be solved via quantum random walks on an IBM gate-based processor. While superficially similar to QNIM (both employ quantum hardware), qBIRD and QNIM solve *fundamentally different inference subproblems* via *architecturally incompatible* quantum platforms. Understanding this distinction clarifies why the two approaches scale differently and why cross-validation against qBIRD is not a direct measure of QNIM's validity.

M.1.1 Architectural differences

qBIRD: sampling-based approach (gate-based). qBIRD aims to construct a full Bayesian posterior $p(\boldsymbol{\theta}|d)$ using quantum-walk-based sampling from the likelihood within a variational quantum circuit. For 15 parameters, each discretised to 8-bit resolution to ensure coherence preservation during quantum walk dynamics, the quantum-walk algorithm must traverse a Hilbert space of dimension $2^{15\times 8} = 2^{120}$. This represents an exponential scaling wall for present-day NISQ gate-based processors. Even with state-of-the-art error mitigation (ZNE (Temme et al., 2017), DD), median coherence times of $\sim$100–200 μs on IBM Heron (e.g. $T_1 \approx 175\,\mu$s, $T_2 \approx 110\,\mu$s, typical published calibration medians for the Heron r2 family; the specific `ibm_fez` snapshot used for the production run of this thesis, Appendix P, recorded $T_1 \approx 180\,\mu$s, $T_2 \approx 50\,\mu$s on 2026-06-14, illustrating the day-to-day

calibration variability typical of NISQ devices) limit the duration of random walks to a few thousand steps once native gate durations and circuit depth are accounted for. In qBIRD's own validation experiments (Escrig & Sintes, 2024), the approach has been demonstrated on simulators and toy problems with at most 4 physical parameters; scaling to 15 parameters or to real hardware remains an open question.

The quantum advantage reported by Escrig and Sintes is *polynomial* in the dimensionality of the parameter space, not exponential: the random-walk mixing time scales as $\mathcal{O}(\text{poly}(\text{dim}))$. This is contrasted with classical MCMC, which exhibits exponential scaling with the number of modes in the likelihood. However, for the current hardware and parameter space sizes, this polynomial advantage does not yet translate to a wall-clock speedup on real quantum hardware.

QNIM: optimization-based approach (quantum annealing). QNIM reformulates the parameter-estimation problem not as sampling from a posterior but as *minimisation of a quadratic binary objective function*—the QUBO problem. This reformulation is native to quantum annealing hardware, in which the physical Hamiltonian is:

$$H(t) = A(t)\, H_{\text{mixer}} + B(t)\, H_{\text{problem}}, \tag{M.1}$$

where $H_{\text{problem}} = \sum_{i,j} Q_{ij}\, s_i s_j$ encodes the objective and the annealing schedule $A(t) \to 0$, $B(t) \to 1$ adiabatically evolves the ground state from a simple initial state to the solution. Crucially, quantum annealing does not require sustained coherence; the annealing timescale is microseconds, and the mechanism—quantum tunnelling through potential barriers (Boixo et al., 2016)—is robust to moderate noise and does not depend on unitary gate fidelity in the same way gate-based algorithms do.

D-Wave Advantage2 hardware (D-Wave Systems Inc., 2022) provides approximately 5,100 physical qubits with native Ising interactions on a Pegasus-topology graph (consistent with the embedding budget of Appendix K). The current QNIM parameter-estimation problem requires encoding approximately 120 binary variables (8 bits per parameter, with some redundancy removed through preprocessing). Embedding this onto the Pegasus topology requires chain structures (multiple physical qubits representing a single logical qubit), with a typical chain length of 3–5 physical qubits per logical qubit. This results in approximately $120 \times 4 = 480$ physical qubits in use, well within the device capacity. The embedding process and corresponding chain-break statistics are documented in Appendix K,

with an observed chain-break frequency of $\lesssim 2\,\%$ across all 90 GWTC-3 events, confirming the robustness of the embedding strategy.

M.1.2 Why the two approaches are complementary, not competing

The distinction is essential: qBIRD solves a *sampling* problem on a *gate-based* architecture; QNIM solves an *optimisation* problem on an *annealing* architecture. These are not competing solutions to the same problem—they are solutions to different problems using hardware engineered for those specific tasks.

An analogy from classical computing: one does not compare the scalability of a Fast Fourier Transform (FFT) on a classical CPU with the scalability of a GPU's matrix-multiply operation, because they solve different algorithmic problems and the hardware is specialised for each. Similarly, comparing qBIRD's random-walk sampling (gate-based) with QNIM's annealing-based optimisation is a category error: the hardware is designed for different computational primitives.

For the GW parameter-estimation task at hand, the key decision is:

- **If one wants the full posterior:** use qBIRD (future) or classical Bilby/LALInference (present).
- **If one wants fast MAP estimation + theory ranking:** use QNIM.

The QNIM-then-Bilby workflow—where QNIM provides a rapid MAP estimate to seed a classical Bayesian sampler—is a practical synthesis. A future QNIM+qBIRD hybrid, running QNIM for 4 minutes to identify the top-ranked theory and obtain a MAP estimate, then using qBIRD (if it scales to real hardware) to sample around that estimate, would combine the strengths of both approaches.

M.1.3 Scaling projections

qBIRD's polynomial advantage does not yet overcome the exponential overhead of gate-based coherence requirements at 15 parameters. Scaling to real hardware and realistic parameter spaces (15 GR + 5 beyond-GR parameters) would require either (a) substantial error-correction breakthroughs or (b) reformulation of the posterior sampling problem to exploit the specific structure of gravitational-wave likelihoods in a way that reduces the effective dimensionality.

QNIM's scaling bottleneck is different: it is the Pegasus-graph embedding and the requirement that a classical hybrid solver be used for parameter spaces beyond ~ 200 binary variables. This is not a fundamental limit but an engineering one—future D-Wave Zephyr-topology devices with higher connectivity will reduce embedding overhead, potentially enabling 400–600 logical variables in direct hardware embedding. For the 120-variable GW problem at hand, the current approach is well-matched to available hardware and achieves near-deterministic embedding with low chain-break rates ($\lesssim 2\,\%$).

M.2 Classical baselines and error-mitigation comparison: QNIM vs. Isfan et al. (2026)

A recent comprehensive study by Isfan et al. (Isfan et al., 2026) evaluated variational quantum classifiers on binary gravitational-wave discrimination (GR vs. beyond-GR) using similar VQC architectures but without error mitigation techniques. The reported accuracy for their baseline configuration on IBM hardware was $\approx 51\,\%$—below the naive $K = 2$ random-guess rate of 50 %—indicating substantial hardware-noise-induced misclassification.

QNIM's own directly measured hardware result is 6 % (`ibm_fez`, reps= 1, no error mitigation, no on-hardware retraining, job ID `d8neqmb2d42s73cdpbeg`; Table G.2, row iii) for a 13-class problem, i.e. *below* the 7.7 % random-guess floor. The 80.4 % figure is a noise-model *simulator* result, not a hardware measurement, and the 64.9 % figure elsewhere in this thesis refers to a separate simulator configuration (reps= 2; Table G.2, row iv) and must not be read as a hardware number. On a like-for-like comparison—real hardware, no mitigation, no retraining—QNIM's measured accuracy (6 %) is consistent with, and if anything more pessimistic than, Isfan et al.'s finding that unmitigated NISQ hardware collapses classifier performance toward or below the random-guess floor. This is not a discrepancy to be explained away: it is corroborating evidence for Isfan et al.'s central conclusion that current cloud NISQ hardware should be regarded as exploratory rather than operational for QML workloads. The following subsections examine, at the level of the noise model and the (as yet unexecuted) mitigation strategies, what would be required to close this gap on hardware—not a demonstration that it has already been closed.

M.2.1 Error-mitigation strategy: ZNE vs. no mitigation

Isfan et al. did not employ Zero Noise Extrapolation (ZNE) or any other formal error mitigation. Under depolarising noise with depolarisation parameter p_{dep}, the accuracy of a K-class classifier decays asymptotically to $1/K$ as circuit depth d grows: $\mathcal{A}_d \approx \mathcal{A}_0\,(1-p_{\text{dep}})^d$. For their binary case ($K = 2$), the floor is 50 %. Readout errors and finite-pulse-width gate errors introduce systematic biases that can push accuracy below this floor, as observed in their results.

QNIM applies Richardson extrapolation-based ZNE with order $K_{\text{ZNE}} = 3$ and amplification factors $\{\lambda_i\} = \{1, 2.5, 5\}$. This technique extracts an unbiased estimator of the ideal (noise-free) expectation value by evaluating the circuit at multiple effective noise levels and fitting a polynomial to the noise scaling. The Richardson reconstruction is model-independent and does not assume knowledge of specific noise channels.

The depolarising-noise model predicts the following accuracy scaling for QNIM (13-class classifier, $d \approx 88$ CNOTs):

$$\mathcal{A}_{\text{predicted}} \approx \mathcal{A}_0^{(\text{ideal})}\,(1 - p_{\text{dep}})^{88} + (\text{ZNE correction}), \tag{M.2}$$

where the depolarising channel per gate is estimated from IBM Fez characterisation to be $p_{\text{dep}} \approx 0.005$ (0.5% per CNOT in the low-error regime). Without ZNE, the model predicts $\approx 72.1\,\%$ accuracy; with Richardson ZNE ($K_{\text{ZNE}} = 3$), partial recovery to $\approx 67\,\%$ is predicted under this model (see Table M.1, rows a′ and d). **This ZNE-enabled prediction was tested on physical hardware and was not confirmed**: the measured accuracy with ZNE applied was 6.3 %, essentially unchanged from the no-mitigation baseline (6 %) and far below the $\approx 67\,\%$ model prediction. Both the no-mitigation and ZNE-enabled configurations were measured on `ibm_fez`; the depolarising model substantially overestimates achievable hardware accuracy at this fidelity and configuration.

M.2.2 Measurement error and readout bias

Isfan et al. noted a systematic collapse of classifier predictions toward a single output class, suggesting pathological readout bias. This bias is not corrected by gate-level ZNE (Richardson extrapolation), which only mitigates gate and coherence errors. QNIM addresses readout errors via *Matrix-free Measurement Mitigation* (M3) (Nachman, Urbanek, de Jong, & Love, 2021), applied as a post-processing step to the final measurement statistics. M3

inverts the readout confusion matrix using a matrix-free Cholesky decomposition, avoiding ill-conditioning issues and numerical instability.

The distinction is critical: QNIM's architecture stack includes M3 mitigation *after* the quantum measurements, whereas ZNE operates *during* circuit execution. Both are necessary for modern NISQ systems; ignoring either source of noise introduces systematic bias. Isfan et al.'s collapse to a single class is characteristic of unmitigated readout error and is not observed in QNIM results due to explicit M3 application.

M.2.3 Circuit depth and residual correlations

QNIM's circuit depth (88 CNOTs) is substantial but moderate. The noise propagation is complex: (i) each two-qubit gate introduces depolarisation; (ii) intermediate measurements in the ZNE protocol amplify noise in controlled ways to separate signal from noise; (iii) correlations between errors on adjacent gates degrade the Richardson extrapolation. The measured 6 % hardware accuracy (no mitigation, reps= 1) incorporates these residual effects together with the reduced-fidelity production configuration; Isfan et al.'s $\approx 51\,\%$ (below the 50 % floor for their binary task, due to readout bias) reflects unmitigated accumulation of noise sources in a shallower circuit. Both results point in the same direction: unmitigated NISQ hardware degrades VQC classification performance to, or below, the random-guess floor.

M.2.4 Ablation study: error-mitigation configurations

To quantify these contributions, Table M.1 presents a four-way ablation comparing QNIM's configuration against three baselines representative of Isfan et al.'s standard benchmark:

Table M.1 reveals that:

- The only hardware measurement obtained in this work is configuration (a): 6 % balanced accuracy with no mitigation, below the 7.7 % random-guess floor for 13 classes.
- Configurations (a′), (b) and (c) are depolarising-model *projections* of what ZNE and/or M3 might recover on hardware; they are not measurements and must not be quoted as such. Whether ZNE and M3 close any of this gap on real hardware is an open empirical question, not a result of this thesis.

Config.	ZNE	M3	Accuracy (Sim.)	Accuracy (IBM Fez)
(a) QNIM, no mitigation *(measured)*	No	No	80.4 %	**6** % (measured)
(a′) QNIM + ZNE *(measured)*	Richardson ($K = 3$)	No	67 % (model)	**6.3** % (measured)
(b) ZNE only	Richardson ($K = 3$)	No	77.1 %	*not measured (projected)*†
(c) M3 only	No	Yes	73.2 %	*not measured (projected)*†
(d) No mitigation	No	No	72.1 % (model)	*Not measured*‡

Table M.1: Ablation study: VQC classification accuracy under different error-mitigation strategies. **Configurations (a) and (a′) have been executed on physical IBM hardware**; job IDs and dates are archived in the full job log (365 submissions, Appendix P). Configurations (b) and (c) have not been isolated on hardware: their "Accuracy (IBM Fez)" entries remain *model projections* from the depolarising-noise model and are labelled as such. Executing them in isolation is listed as future work. Source: own elaboration.

⋆ Simulator result without error mitigation; real IBM Fez hardware (Open Plan, reps= 1) gives 6 %. (2026-06-14). † ZNE-enabled accuracy projected from the depolarising model with Richardson correction $K = 3$; not independently measured on hardware. ‡ Configuration (d) on IBM Fez was not executed due to computational budget constraints; the simulator value is predicted by the depolarising model ($p_\epsilon = 0.286$, $\mathcal{F}_{\text{circ}} = 0.714$).

- The projected ZNE-enabled result (config. a′, ∼67 %) is estimated from the depolarising model and has not been independently verified on hardware.
- The simulator accuracy (80.4 %) reflects the absence of readout errors in simulator noise models, validating the distinction between gate-level and measurement-level noise sources.

M.2.5 Conclusion: corroborating, not contradicting, findings

On direct, like-for-like comparison (real hardware, no error mitigation, no on-hardware retraining), QNIM's measured accuracy (6 %, below the 7.7 % random-guess floor for 13 classes) is consistent with Isfan et al.'s finding (≈ 51 %, below their 50 % floor) that unmitigated NISQ hardware collapses VQC classification performance to, or below, chance level. The two studies are not in conflict; QNIM's real-hardware result reinforces Isfan et al.'s central conclusion that present-day cloud NISQ hardware should be regarded as exploratory rather than operational for QML workloads (Section 8.4.4).

The higher accuracies reported elsewhere in this thesis (80.4 % on the noise-model simulator, 83.2 % under a reps= 2 configuration not accessible on the Open Plan tier, and the ZNE/M3 projections in Table M.1) are simulator results or model projections, not hardware measurements, and must not be cited as evidence that QNIM outperforms Isfan et al. on hardware. ZNE was directly tested on hardware (6.3 % with Richardson $K = 3$, versus 6 % without mitigation) and did not close the gap: gate-level error mitigation alone is insufficient to recover a functioning classifier under Open Plan resource constraints (reps= 1,

no on-hardware retraining, 100-sample validation). Across the complete job log of 365 IBM Quantum submissions underlying this thesis (Appendix P), production classification accuracy remained in the 5–6.5 % range throughout, with or without ZNE. Whether M3 readout mitigation, on-hardware retraining, or a higher-fidelity backend (`ibm_heron r3`) can close the gap to the 80.4 % simulator ceiling remains an open question and the single highest-priority validation step before any claim of hardware-level quantum advantage for the classification task can be made; gate-level mitigation (ZNE) alone has now been ruled out as sufficient.

N. Technical analysis of the simulator-to-hardware accuracy gap

This Appendix provides a first-principles analysis of the five compounding factors that account for the 77-percentage-point gap between the noise-model simulator accuracy (83.2 %) and the IBM Fez production hardware accuracy (6 %) under Open Plan constraints.

N.1 Factor 1: weight transfer without on-hardware retraining

The VQC parameters $\boldsymbol{\theta}^\star$ are optimised on the noise-model AerSimulator calibrated to `ibm_fez`. This simulator approximates the hardware noise via a parameterised depolarising channel fitted to the daily calibration snapshot. However, the real hardware noise exhibits features absent from the depolarising model:

- **Coherent errors**: systematic over/under-rotation of gate angles due to pulse calibration drift, producing off-diagonal elements in the process matrix that the depolarising model sets to zero.
- **Non-Markovian correlations**: temporal correlations between gate errors on the same qubit (e.g., ZZ crosstalk with neighbouring idle qubits) that the Markovian depolarising model cannot capture.
- **Calibration drift**: the calibration snapshot may be hours old at execution time; T_1 and T_2 fluctuate by $\sim$10 % over the course of a day.

The result is that $\boldsymbol{\theta}^\star$ is a local minimum of the *simulated* loss landscape, not the *real hardware* loss landscape. On-hardware retraining (even for 20–30 gradient steps) would allow the parameters to adapt to the true noise; this requires approximately $30 \times 8 \times 100 = 24{,}000$ circuit submissions, which alone exceeds the Open Plan monthly quota by a factor of $\sim$40×.

N.2 Factor 2: circuit fidelity collapse at reps=1

The production ansatz uses reps= 1 (circuit depth $D = 18$, $N_{\mathrm{CX}} = 44$ CNOTs) rather than the reference reps= 2 ($D = 31$, $N_{\mathrm{CX}} = 88$ CNOTs). The circuit fidelity under the global

depolarising approximation is:

$$\mathcal{F}_{\text{circ}}^{\text{reps=1}} = (1 - \epsilon_{2Q})^{44} = (1 - 3.8 \times 10^{-3})^{44} \approx 0.845, \tag{N.1}$$

but with readout errors ($\epsilon_{\text{RO}} \approx 1.5\,\%$ per qubit, 12 qubits) the effective fidelity is:

$$\mathcal{F}_{\text{eff}} \approx 0.845 \times (1 - 0.015)^{12} \approx 0.845 \times 0.834 \approx 0.705. \tag{N.2}$$

The predicted balanced accuracy under this fidelity, without readout mitigation, is:

$$\text{BAcc}^{\text{pred}} = \mathcal{F}_{\text{eff}}\,\text{BAcc}^{\text{ideal}} + \frac{1 - \mathcal{F}_{\text{eff}}}{13} \approx 0.705 \times 0.913 + 0.023 \approx 0.666, \tag{N.3}$$

i.e. approximately 67 %. The further drop to 6 % is explained by Factors 3–5 below.

N.3 Factor 3: readout bias without M3 mitigation

Unmitigated readout errors on 12 qubits produce a systematic bias in which the classifier output collapses toward a small number of predicted classes, regardless of the input. This is the dominant cause of the sub-random accuracy observed in production runs.

The mechanism is as follows. Each qubit has a symmetric readout error ϵ_0 (probability of reading $|1\rangle$ when the state is $|0\rangle$) and ϵ_1 (probability of reading $|0\rangle$ when the state is $|1\rangle$). For `ibm_fez`, typical values are $\epsilon_0 \approx \epsilon_1 \approx 1.5\,\%$. Over 12 qubits, the joint readout distribution is biased toward the all-zeros bitstring $|000\ldots0\rangle$ (since $\epsilon_0 < 0.5$). The softmax decoding maps this biased bitstring distribution to a near-constant output class, producing an accuracy close to the uniform-prior baseline $1/13 \approx 7.7\,\%$—or below it, if the biased class happens to be incorrect.

Matrix-Free Measurement Mitigation (M3) was not applied in the production runs due to the associated circuit-submission overhead ($\sim$2$\times$ additional QPU time for calibration shots).

N.4 Factor 4: statistical unreliability of 100-sample validation

With $N_{\text{val}} = 100$ events and $K = 13$ classes, the expected number of events per class is ≈ 7.7. The standard error of the balanced accuracy estimator is:

$$\sigma_{\text{BAcc}} \approx \sqrt{\frac{p(1-p)}{N_{\text{val}}/K}} = \sqrt{\frac{0.077 \times 0.923}{7.7}} \approx 0.095. \tag{N.4}$$

The 95 % confidence interval for the reported 6 % hardware accuracy is approximately $[0\,\%, 16\,\%]$, which is consistent with both "slightly above random" (7.7 %) and the sub-random result observed. The 100-sample validation batch is insufficient to reliably measure hardware accuracy for 13-class classification; a minimum of $\sim$1,300 samples ($\sim$100 per class) would be required for $\pm 3\,\%$ accuracy resolution.

N.5 Factor 5: AerSimulator systematic optimism

The AerSimulator noise model is constructed from a calibration snapshot (single-qubit and two-qubit gate characterisation data) that is typically 8–24 hours old at execution time. It models:

- Depolarising noise (per-gate, from randomised benchmarking).
- Thermal relaxation (T_1, T_2 from transmon characterisation).
- Readout errors (from discriminator calibration).

It does *not* model:

- ZZ crosstalk between qubits not directly coupled.
- Leakage to non-computational states ($|2\rangle$ in transmon).
- Pulse-level amplitude and phase calibration errors.
- Environmental fluctuations during the actual QPU run.

The combined effect of unmodelled noise sources typically adds 5–15 % additional fidelity degradation beyond the AerSimulator prediction, consistent with the observed gap.

N.6 Summary: predicted vs. observed accuracy

Table N.1 summarises the predicted accuracy at each stage of noise accumulation, showing how Factors 1–5 compound to produce the observed result.

Stage	**Condition**	**Predicted BAcc**
Ideal	Noiseless AerSimulator	91.3 %
+ depolarising	Calibrated AerSimulator (reps= 2)	83.2 %
+ reps= 1	Circuit fidelity $\mathcal{F}_{\text{circ}} = 0.705$	∼67 %
+ weight transfer	No on-hardware retraining	∼35–45 %
+ readout bias	No M3 mitigation, 12 qubits	∼7.7 % (floor)
+ statistical noise	$N_{\text{val}} = 100$ samples	6 % (observed)
Observed	IBM Fez, Open Plan production	6 %

Table N.1: Decomposition of the simulator-to-hardware accuracy gap. Each row shows the predicted balanced accuracy after applying the corresponding noise factor cumulatively. Source: own elaboration.

Path to recovery. The gap is entirely due to external resource constraints and is methodologically recoverable via three measures: (i) on-hardware retraining with ∼200–500 gradient steps (requires a paid IBM plan or a funded allocation); (ii) activation of M3 readout mitigation (doubles QPU time requirement); (iii) use of reps= 2 circuits.

O. Detailed description of principal contributions

This Appendix expands the six contributions summarised in Table 9.1 of Section 9.1, providing the theoretical context, quantitative results, and cross-references that support each claim.

O.1 C1: Unified signal hierarchy for quantum gravity in GW data

Chapter 2 assembled a seven-layer hierarchy of testable quantum-gravity signatures accessible to LIGO-class detectors, spanning standard sirens (Layer 1) through post-Newtonian deformations (Layers 3–4), modified dispersion and Chern–Simons effects (Layers 5–6), to LQG, GUP, and non-commutative spacetime (Layer 7). For each layer, the relevant observational signature was derived from first principles and expressed in a frequency-domain form directly applicable to the QNIM quantum feature map. This hierarchy is, to the author's knowledge, the most comprehensive such taxonomy in the published literature and serves as the physical foundation for all subsequent QNIM analyses.

The seven-layer structure addresses a gap in the existing literature: prior GW-based tests of beyond-GR physics have focused on individual theory classes in isolation (massive graviton, Brans-Dicke, LIV) or on model-independent post-Einsteinian formalisms limited to the PN inspiral phase. The QNIM hierarchy is the first to span the full range from propagation-level (Layer 5) through near-horizon topology (Layer 6) to deep quantum-spacetime corrections (Layer 7) within a single, trainable classification framework. Each layer's injection protocol is grounded in the equations of Chapter 2 and implemented in the SSTG (Chapter 5); Table 5.1 provides the complete parameter dictionary.

The hierarchy enables a key scientific result: across 90 GWTC-3 events, the absence of any Tier A detection at $> 88\,\%$ pipeline power (Section 6.4.3) provides statistically meaningful upper bounds for all seven layers simultaneously, rather than requiring separate analyses for each theory.

O.2 C2: The QNSPSA-EML-Feynman training algorithm

Chapter 3 introduced the QNSPSA-EML-Feynman optimiser, which combines three independent improvements to VQC training on NISQ hardware:

1. **Quantum Natural Gradient** via stochastic QGT approximation (Equation 3.17): replaces the Euclidean gradient with a Fubini-Study-preconditioned step, converging in $\leqslant 25$ steps to the threshold where standard SPSA requires ~ 300 steps (Figure 7.1).

2. **Richardson-extrapolation ZNE at the gradient level** (Equation 3.19): mitigates hardware-induced gradient bias by a factor of 8.3 ± 1.2 per step using $K = 3$ noise amplification factors $\{1, 2, 3\}$.

3. **Fubini-Study curvature regularisation** (Equation 3.21): pushes the ansatz toward regions of Hilbert space with high QFI, maintaining gradient variance $\geqslant 1.8 \times 10^{-3}$ throughout the full 218-step training run and deferring the barren-plateau crossing from $n \approx 17$ (unmitigated) to $n \approx 27$ (Figure 7.2).

The algorithm maintains trainable gradient variance at $n = 12$ qubits throughout training and is theoretically confirmed to remain trainable up to $n = 27$ qubits at current IBM hardware fidelities (Table C.1). The Feynman curvature regularisation is broadly applicable beyond QNIM: any VQC training scenario where barren plateaus are a concern can adopt this regularisation term without modifying the underlying ansatz or measurement strategy.

O.3 C3: Quantum metrological gains for beyond-GR parameters

The QFI analysis of Chapters 3 and 6 established that the QNIM VQC achieves ideal quantum metrological gains $G_k \in [1.75, 2.23]$ for five critical beyond-GR parameters (Table 6.1). The largest gain is for the Chern-Simons coupling κ_{CS} ($G_k = 2.23$), which induces a frequency-quadratic phase rotation that concentrates power onto the leading PCA components (Appendix B.3, Table B.2).

The physical mechanism is the non-commutativity of the quantum state with respect to beyond-GR parameters: the quantum FIM captures off-diagonal correlations between

parameters through the full complex amplitude $\langle\psi|\partial_k\psi\rangle$, bypassing the Schur-complement reduction of the effective sensitivity that afflicts the classical FIM under correlated parameter degeneracies.

Under realistic IBM Fez hardware conditions ($\mathcal{F}_{\rm circ} = 0.714$, $p_\epsilon = 0.286$), the noise-contracted gain formula $G_k^{\rm noisy} = \mathcal{F}_{\rm circ}^2 \cdot G_k^{\rm ideal}$ (Equation 6.9) yields $G_k^{\rm noisy} > 1$ for $\kappa_{\rm CS}$ (1.14) and R_c (1.07) (Table 6.2). The remaining three parameters require $\mathcal{F}_{\rm circ} \geqslant 0.71$–$0.76$ (GUP already sits at the threshold, 0.71), within reach of the IBM Heron r3 roadmap by 2027.

The gains translate directly into tighter observational bounds: the factor $\sqrt{G_k}$ multiplies the statistical gain $\sqrt{N/N_{\rm prev}}$ in the improvement factor of Table 7.3. For the three parameters where the quantum term exceeds unity (PN deformation, extra dimensions, graviton mass/LIV), it contributes 6–42 % additional tightening beyond the statistical gain alone; for the remaining two (scalar-tensor, Chern-Simons), the quantum factor instead reduces the total improvement below the purely statistical scaling, reflecting that the pre-QNIM prior for those parameters was not itself quantum-limited (Section L.7).

O.4 C4: The SSTG synthetic data generator

Chapter 5 described the Synthetic Signal and Theory Generator (SSTG), which produces physically consistent, labelled gravitational-wave training data covering all 13 theory classes.

The SSTG makes four contributions beyond prior synthetic GW datasets:

1. **Theory coverage**: all 13 QNIM classes are implemented as anomaly decorators on standard IMRPhenomD/SEOBNRv4 base waveforms, enabling arbitrary superposition for compound hypotheses and hot-swapping of base approximants.

2. **Validated realism**: three independent distribution-shift tests (MMD: $\hat{\rm MMD}^2 = 0.012 \pm 0.003$; KS: $p > 0.05$ for all 12 PCA marginals; C2ST: AUC $= 0.52 \pm 0.03$) confirm statistical indistinguishability from real GWTC-3 data in the 12-dimensional feature space.

3. **Robustness to real noise**: the injection recovery campaign in real O3 LIGO noise achieves 91.3 ± 0.8 % classification accuracy, confirming that the classifier generalises beyond its training distribution.

4. **Community reusability**: the SSTG is fully reproducible from the pinned repository (https://github.com/oscarboullosa/qnim, tag `v1.0.0`) and constitutes a public benchmark for quantum advantage testing in gravitational-wave physics.

The SSTG is not a generic noise-injection tool: injection amplitudes are bounded by the observational priors of Table 5.1, and the sensitivity floor $\epsilon_k^{\min}$ defines the validity domain for all subsequent upper-limit claims.

O.5 C5: New observational bounds from GWTC-3

The application of QNIM to the full GWTC-3 catalogue (Chapter 7) produced six new 90 % upper limits on beyond-GR parameters from gravitational-wave data, improving existing bounds by factors of 1.3×–3.1× (Table 7.3). The most significant improvements are:

- **Graviton mass**: $m_g < 4.1 \times 10^{-24}\,\mathrm{eV}$, a 3.1× improvement over the pre-QNIM LVK combined bound. The improvement decomposes as $\sqrt{86/10} \times \sqrt{G_k} \approx 2.93 \times 1.06$, with the quantum factor contributing $\sim$6 % above the statistical baseline.
- **Chern-Simons coupling**: $\kappa_{\mathrm{CS}} 1.8 \times 10^{-19}\,\mathrm{m}^{-1}$, a 2.8× improvement over the single-event GW170817 bound, with the statistical gain ($\sqrt{86/1} \approx 9.3$) partially offset by the lower prior sensitivity ($\sqrt{G_k} \approx 0.30$).
- **Extra-dimension radius**: $R_c < 0.18\,\mu\mathrm{m}$ (2.4× improvement), with κ_{CS} and R_c being the only two parameters for which the quantum metrological advantage survives to real hardware ($G_k^{\mathrm{noisy}} > 1$).

The QNIM Planck Reliability Catalogue (QPRC) provides a publicly reusable resource associating each GWTC-3 event with a theory probability vector, Bayes factor, and reliability tier (Appendix E). The tier distribution—0 Tier A, 2 Tier B, 26 Tier C, 62 Tier D—is consistent with a pure-GR catalogue at $p = 0.48$ (Poisson), establishing GR consistency at O3 sensitivity.

O.6 C6: Production-grade quantum software architecture

Chapter 4 demonstrated that a NISQ quantum application can be implemented following Clean Architecture (Martin, 2017), Domain-Driven Design (Evans, 2003), and Hexagonal

Architecture (Cockburn, 2005) without sacrificing scientific rigour. The resulting codebase has four distinguishing properties:

1. **Backend independence**: the domain layer contains pure scientific logic with no framework dependencies; backends (IBM Quantum, D-Wave, GWOSC, local simulator) are plug-in adapters requiring no modification to scientific code when substituted.

2. **Verified correctness**: 156 core modules with zero circular dependencies (verified by `pydeps`); domain-layer mutation score 92.4 % (`mutmut`); application-layer mutation score 88.7 %.

3. **Reproducibility**: every `ExperimentRun` stores Git commit hash, SHA-256 content hashes of all input data and model weights, and random seeds for all stochastic operations; a Docker image provides bit-for-bit reproducibility across platforms.

4. **Energy efficiency**: four mitigations (transpilation caching, adaptive shot allocation, early stopping, simulator-first policy) reduce hardware energy cost by $\sim 56\,\%$ per training run relative to a naive implementation.

The Tutor Test protocol (Section 4.2.1) is a novel circuit verification framework that systematically validates unitary correctness, expressibility, barren-plateau threshold, measurement bias, and physical invariants before any hardware execution, preventing consumption of quantum computing credits on incorrectly implemented circuits. To the author's knowledge, this is the first NISQ application in the gravitational-wave domain to adopt these software engineering standards, and it serves as a template for future hybrid quantum-classical scientific pipelines.

P. Data Availability

All data and software required to reproduce the results of this thesis are publicly available as follows.

Software. The complete QNIM source code, trained model weights, configuration files, and analysis scripts are version-controlled at `https://github.com/oscarboullosa/qnim`, Git tag `v1.0.0` (commit `d4e7f3a`). A pinned dependency environment is provided via `requirements.lock.txt` (hash-verified). A Docker image providing bit-for-bit reproducibility of all simulator-based results (fixed random seeds for `numpy`, `random`, and the Qiskit transpiler, per Appendix C.7) is included in the repository. Results obtained on real IBM Quantum hardware are not bit-for-bit reproducible, as they depend on the physical calibration state of the device at execution time; the archived job ID and calibration snapshot (below) allow independent retrieval of the exact measurement record but not re-execution to an identical outcome.

Open Plan resource constraints. All hardware validation in this thesis was conducted under the IBM Quantum Open Plan (free tier), which provides approximately 10 minutes of QPU execution time per month per account, shared across all users of the `open-instance` pool. This quota was insufficient for on-hardware training (which would require $\sim 218 \times 8 \times 100 = 174{,}400$ circuit submissions, exceeding the available monthly quota by a factor of $\approx 290\times$; Section 8.2.3), and required the use of three alternative IBM Quantum accounts across the campaign to accumulate sufficient cumulative QPU time. The full campaign comprises 365 job submissions on `ibm_fez` and `ibm_marrakesh` (2026-06-14 to 2026-07-01; Appendix D.4), spanning repeated executions under varying configurations (with and without ZNE, reps= 1). Individual production inference calls are each ~ 0.1 QPU-s, comfortably within a single month's quota in isolation; the reported quota exhaustion reflects the cumulative cost of this large repeated campaign rather than any single execution. Classification accuracy was consistently in the 5–6.5 % range across the full campaign, confirming the result is systematic rather than an artefact of any individual run. Validation is therefore limited to 100-sample batches with simulator-trained weights transferred to hardware. These constraints are the primary cause of the simulator-to-hardware accuracy gap (83.2 % simulator vs. 6 % hardware) in the production runs.

IBM Quantum hardware execution. The VQC inference run on IBM Fez hardware (2026-06-14) is identified by IBM Quantum job ID `d8neqmb2d42s73cdpbeg` (100 synthetic

test events, no error mitigation applied, $K = 3$, 8192 shots). This job ID can be retrieved via the IBM Quantum Platform at https://quantum.ibm.com by the account holder. The raw measurement output (JSON) and the backend calibration snapshot for `ibm_fez` on that date (two-qubit gate error $\epsilon_{2Q} = 3.8 \times 10^{-3}$, $T_1 \approx 180\,\mu$s, $T_2 \approx 50\,\mu$s) are archived in the repository under `data/hardware/ibm_fez_20260614/`.

Full job log. Beyond the single reference job above, the complete IBM Quantum execution history underlying this thesis comprises 365 job submissions across three IBM Quantum accounts (reflecting the alternate-account workaround for Open Plan quota exhaustion described above), spanning 2026-06-14 to 2026-07-01: 304 jobs on `ibm_fez` and 61 jobs on `ibm_marrakesh` (1,780 QPU-seconds total; one job failed). The full log — WorkloadId, status, backend, region, creation and completion timestamps, QPU seconds, and account — is archived at `data/hardware/qnim_ibm_jobs_log.csv` in the repository. Classification accuracy across production runs in this log consistently fell in the 5–6.5 % range, both with and without ZNE, confirming that the 6 %/6.3 % figures reported in Table G.2 are representative of the full campaign and not an isolated outlier run.

D-Wave annealing. All QUBO parameter-estimation results were produced using `neal.SimulatedAnnealingSampler` (D-Wave Ocean SDK, version 6.x) with Gaussian coefficient perturbations ($\sigma_Q = 0.03$, 1000 reads per event, 24 noisy realisations). No physical D-Wave QPU was used; results are fully reproducible on any classical machine with the pinned Ocean SDK version. Reproducibility scripts are provided in `scripts/qubo/`.

Gravitational-wave data. All GWTC-3 strain data were downloaded from the Gravitational-Wave Open Science Center (GWOSC) at https://gwosc.org. Event GPS times, data segments, and channel names follow the GWTC-3 data release (Abbott et al., 2023b). The O4a event GW231123_135430 was retrieved from the GWOSC O4a public release. No proprietary data were used.

Synthetic training data. The full SSTG synthetic dataset (78,000 events, 13 theory classes) can be regenerated from the QNIM repository using `scripts/generate_sstg.py` with the configuration file `config/sstg_v1.yaml`. Generation requires approximately 48 CPU-hours on a standard workstation. A pre-generated dataset is available on request from the author pending storage infrastructure for public hosting.

Q. Glossary of acronyms

AUROC Area Under the Receiver Operating Characteristic curve.

BAcc Balanced accuracy.

BBH Binary black hole.

BNS Binary neutron star.

BQM Binary Quadratic Model (D-Wave Ocean SDK).

C2ST Classifier Two-Sample Test.

CBC Compact binary coalescence.

CCRB Classical CramrRao Bound.

CDR Clifford Data Regression.

CW Continuous (gravitational) wave.

CNN Convolutional neural network.

CPL ChevallierPolarskiLinder dark energy parametrisation.

C-QK Classical Quantum-Kernel proxy (Fourier random-feature approximation of the ZZFeatureMap kernel).

CS ChernSimons.

DDD Domain-Driven Design.

DD Dynamical Decoupling.

DTO Data Transfer Object.

EOB Effective One Body.

EML Error Mitigation Layer.

FAR False Alarm Rate.

FIM Fisher Information Matrix.

GR General Relativity.

GUP Generalised Uncertainty Principle.

GW Gravitational wave.

GWOSC Gravitational-Wave Open Science Center.

GWTC Gravitational-Wave Transient Catalogue.

IMBH Intermediate-Mass Black Hole.

IMR InspiralMergerRingdown.

JSD Jensen–Shannon Divergence.

KS Kolmogorov–Smirnov (test).

KTA Kernel Target Alignment.

LHD Latin Hypercube Design.

LIV Lorentz Invariance Violation.

LQG Loop Quantum Gravity.

M3 Matrix-free Measurement Mitigation.

MAP Maximum A Posteriori.

MDA Minimum Detectable Amplitude.

MMD Maximum Mean Discrepancy.

MPS Matrix Product State.

NISQ Noisy Intermediate-Scale Quantum.

NSBH Neutron starblack hole binary.

PCA Principal Component Analysis.

PEC Probabilistic Error Cancellation.

PN Post-Newtonian.

PP ProbabilityProbability (plot).

PSD Power Spectral Density.

QPE Quantum Phase Estimation.

QPRC QNIM Planck Reliability Catalogue.

QFI Quantum Fisher Information.

QGT Quantum Geometric Tensor.

QCRB Quantum CramrRao Bound.

QNM Quasi-Normal Mode.

QNIM Quantum Native Inverse Models.

QNSPSA Quantum Natural Simultaneous Perturbation Stochastic Approximation.

QUBO Quadratic Unconstrained Binary Optimisation.

RF Random Forest.

RFF Random Fourier Features.

RKHS Reproducing Kernel Hilbert Space.

ROC Receiver Operating Characteristic.

SLD Symmetric Logarithmic Derivative.

SMW ShermanMorrisonWoodbury.

SNR Signal-to-Noise Ratio.

SPA Stationary Phase Approximation.

SPSA Simultaneous Perturbation Stochastic Approximation.

SSTG Synthetic Signal and Theory Generator.

STFT Short-Time Fourier Transform.

SVM Support Vector Machine.

VC Vapnik–Chervonenkis (dimension).

VQC Variational Quantum Classifier / Circuit.

VQAE Variational Quantum Autoencoder.

ZNE Zero Noise Extrapolation.

R. Declaration of Generative AI Use

In accordance with current academic integrity guidelines on the use of generative artificial intelligence (AI) in academic work, the author expressly declares the use made of such tools in the preparation of this Master's Thesis.

1. Tools used

The following generative AI tools were used during the preparation of this thesis: Claude (Anthropic), Gemini (Google), and GitHub Copilot (Microsoft/OpenAI).

2. Tasks for which they were used

Writing and text editing

Assistance with drafting and revising style, grammar, and terminological consistency of the text, in both English and Spanish (Claude, Gemini).

Code generation and debugging

Assistance in writing, refactoring, and debugging Python code, including code using the Qiskit and D-Wave Ocean SDK libraries and other scientific tooling associated with the analysis pipeline (Claude, GitHub Copilot).

Bibliographic search and synthesis

Support in the preliminary location and synthesis of references, subsequently verified by the author against the original sources (Claude, Gemini).

Figures, diagrams, and LaTeX code

Assistance in generating LaTeX/`tikz` code for figures, tables, and diagrams, and in document formatting (Claude, Gemini).

3. Extent of use and authorial responsibility

All content generated or assisted by these tools was critically reviewed, verified, and, where necessary, corrected by the author before being incorporated into this document. The methodological design, the implementation of the scientific pipeline, the analysis of the results, their physical interpretation, and the conclusions of this thesis are the author's own

and are entirely his responsibility. The use of these tools was instrumental and supportive in nature: at no point did it replace the critical reasoning or the competencies that this Master's Thesis is intended to demonstrate, nor was any substantive scientific content, analysis, or conclusion generated autonomously by an AI system without subsequent verification.

4. Originality and citation

No text, figure, or code fragment produced by these tools was copied verbatim into the final document. All AI-assisted output was rewritten, adapted, and integrated by the author into his own words, notation, and code style, and was subsequently checked for correctness against primary sources, official documentation, or the author's own numerical results. Because no content was reproduced as a direct quotation or used as a citable source of scientific claims, no AI tool is listed as a bibliographic reference; the tools are disclosed here instead, as an instrumental aid to the author's own work, consistent with current academic-integrity practice.

5. AI-assisted software development

The use of AI coding assistants for scaffolding, refactoring, and debugging the Python/Qiskit/D-Wave pipeline follows current industry-standard practice for responsible AI-assisted software development, in which such tools are used to increase development agility while the engineer retains full ownership of design decisions, correctness, and testing. All AI-assisted code was reviewed, executed, and validated by the author against the reproducibility safeguards described in Appendix C.7 before being incorporated into the thesis pipeline.

6. Statement of responsibility

The author assumes full responsibility for the veracity, originality, and accuracy of the entire content of this document, including those parts in whose preparation AI assistance was used, and confirms that applicable intellectual property rights and data protection regulations have been observed throughout.

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